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(54) **PARVOVIRUS ANTIBODIES FOR
VETERINARY USE**

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62/880,650, filed on Jul. 30, 2019.

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CPC **C07K 16/081** (2013.01); **A61P 31/20**
(2018.01); **C07K 2317/24** (2013.01); **C07K**
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C07K 2317/76 (2013.01); **C07K 2317/92**
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(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,063,905 A 5/2000 Capra et al.
6,372,426 B1 4/2002 Zens
7,604,801 B2 10/2009 Smith-Norowitz et al.
8,052,971 B2 11/2011 Meyer
8,512,981 B2 8/2013 Hermens et al.
8,790,651 B2 7/2014 Bammert et al.
8,828,400 B2 9/2014 Leuchs et al.
9,556,261 B2 1/2017 Gnauer et al.
10,080,798 B2 9/2018 David et al.

10,092,647 B2 10/2018 Feugier et al.
2007/0248612 A1 10/2007 Wilson
2009/0162831 A1 6/2009 Delwart et al.
2014/0044713 A1 2/2014 Lau et al.
2014/0170187 A1 6/2014 Settembre et al.
2015/0086558 A1 3/2015 Chang et al.
2016/0046723 A1 2/2016 Reyes et al.
2016/0213778 A1 7/2016 Feugier et al.
2017/0158756 A1 6/2017 Bergeron et al.
2018/0009869 A1 1/2018 Lu et al.

FOREIGN PATENT DOCUMENTS

EP 914831 B1 4/2003
EP 1009756 B1 9/2004
EP 2384761 B1 9/2013
EP 2332986 B1 10/2013
EP 2566491 B1 7/2015
EP 2736921 B1 6/2018
EP 3019524 B1 4/2020
JP 2005006583 A 1/2005
WO 2004105792 A1 12/2004
WO 2012164372 A1 12/2012
WO 2016210097 A1 12/2016
WO 2018/102795 A2 6/2018
WO 2019035010 A1 2/2019
WO 2020256419 A1 12/2020
WO 2021022067 A2 2/2021

OTHER PUBLICATIONS

Hafenstein, S, et al., "Structural Comparison of Different Antibodies
Interacting with Parvovirus Capsids", Journal of Virology, Jun.
2009, 83(11): 5556-5566.

Salazar G, et al., "Antibody therapies for the prevention and
treatment of viral infections", npj Vaccines, Jul. 10, 2017, 2:19;
doi:10.1038/s41541-017-0019-3 (12 pages).

Carmichael, L.E., et al., "A Modified Live Canine Parvovirus
Vaccine. II. Immune Response", Cornell Vet., 1983, 73: 13-29.

Nelson, C.D.S., et al., "Different mechanisms of antibody-mediated
neutralization of parvoviruses revealed using the Fab fragments of
monoclonal antibodies", Virology, May 1, 2007; 361(2): 283-293
(NIH Public Access Author Manuscript, 23 pages).

Extended European Search Report issued in corresponding Euro-
pean Patent Application No. 20847060.9 (9 pages), dated May 25,
2023.

(Continued)

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(57)

ABSTRACT

Provided are various embodiments relating to parvovirus
antibodies, including caninized, felinized, and chimeric anti-
bodies, that bind to canine and/or feline parvovirus, for
example, having improved expression characteristics. In
various embodiments, the parvovirus antibodies have
ADCC, ADCP, and/or CDC effector functions. In various
embodiments, such monoclonal parvovirus antibodies can
be used in methods to prevent and/or treat parvoviral infec-
tion in subjects, such as dogs and cats. For example, the
parvovirus antibodies provided may be used to provide
passive immunity against infection with a canine or feline
parvovirus.

28 Claims, 7 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited

OTHER PUBLICATIONS

Ayyar, et al., "Optimizing antibody expression: The nuts and bolts,"
Methods, 116:51-62 (2017).

Li, et al., "Antibody Aggregation: Insights from Sequence and
Structure," Antibodies, 5(19) doi: 10.3390/antib5030019 (2016).

International Search Report and Written Opinion received in PCT/
US2020/044302, dated Dec. 17, 2020, 15 pages.

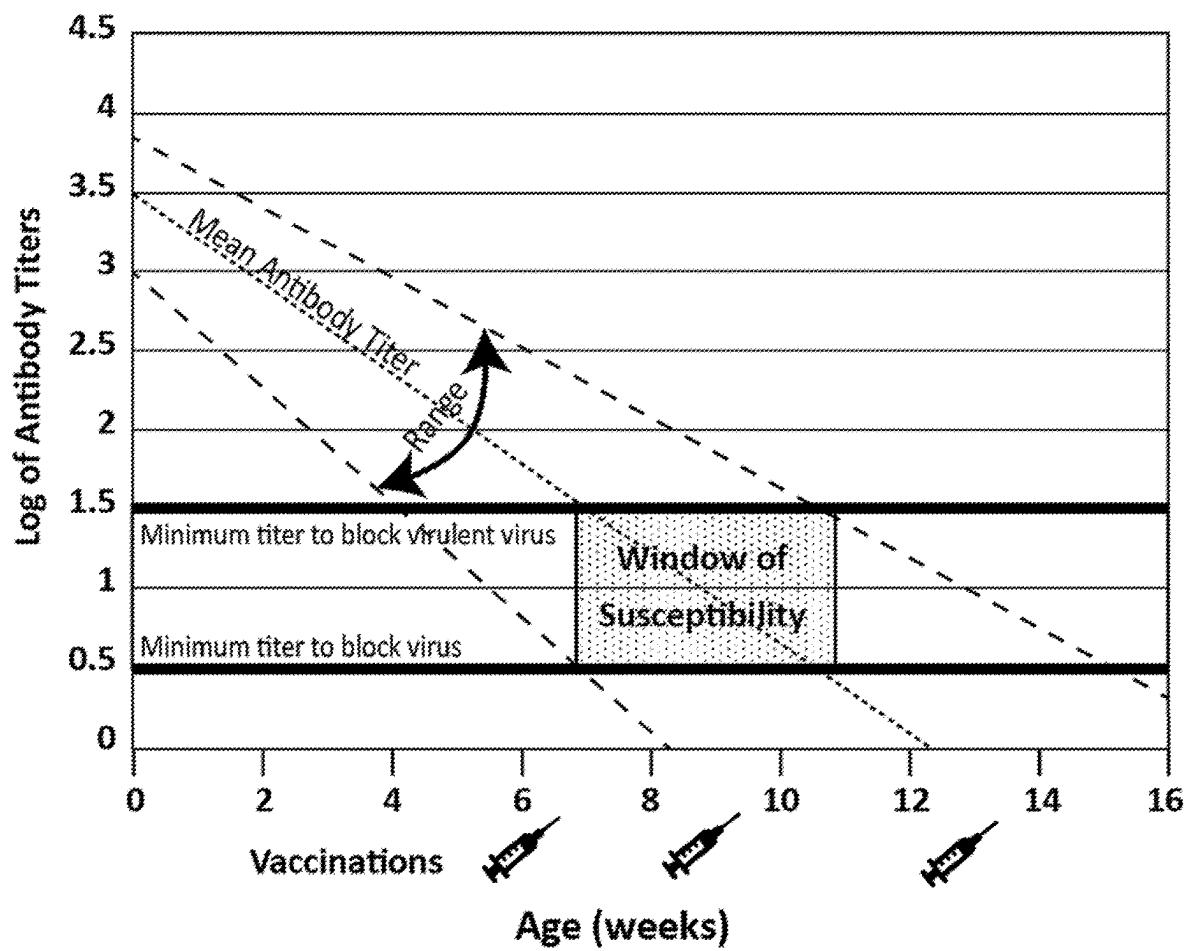


Fig. 1

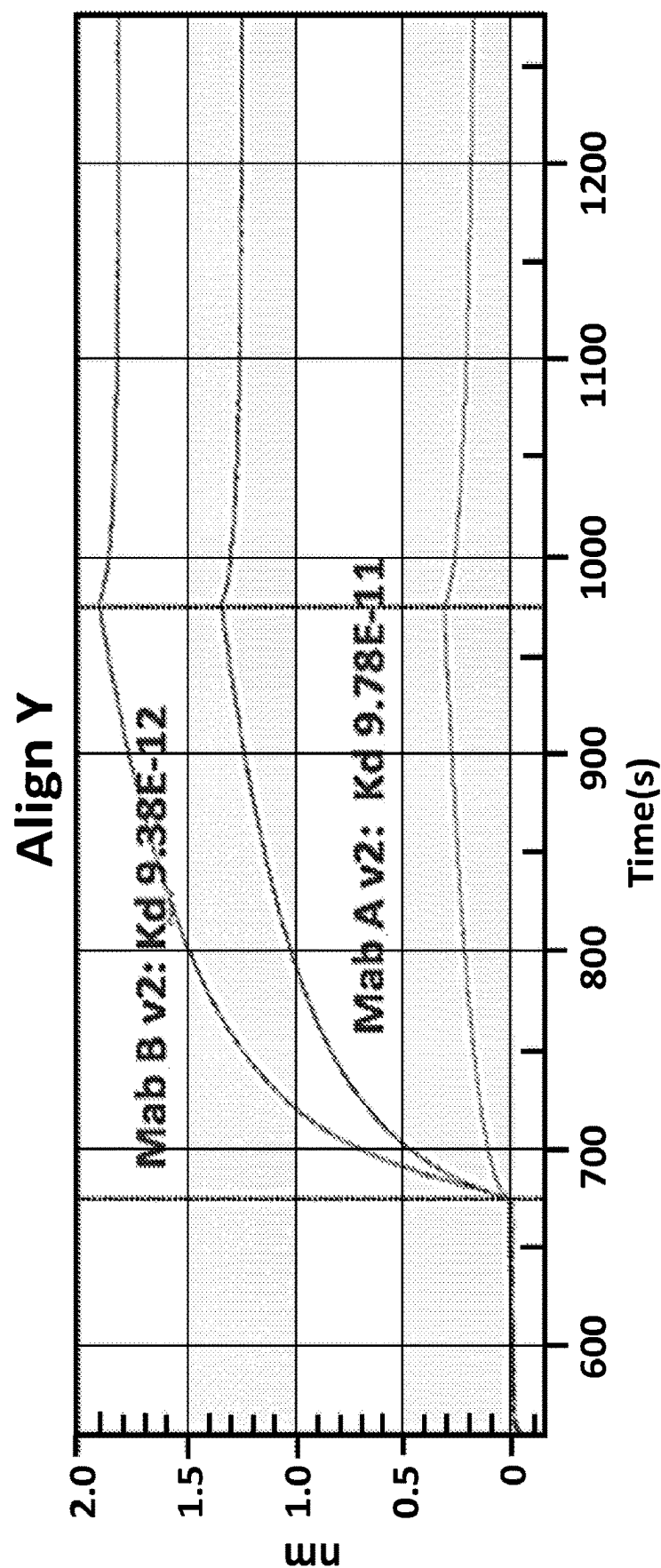


Fig. 2

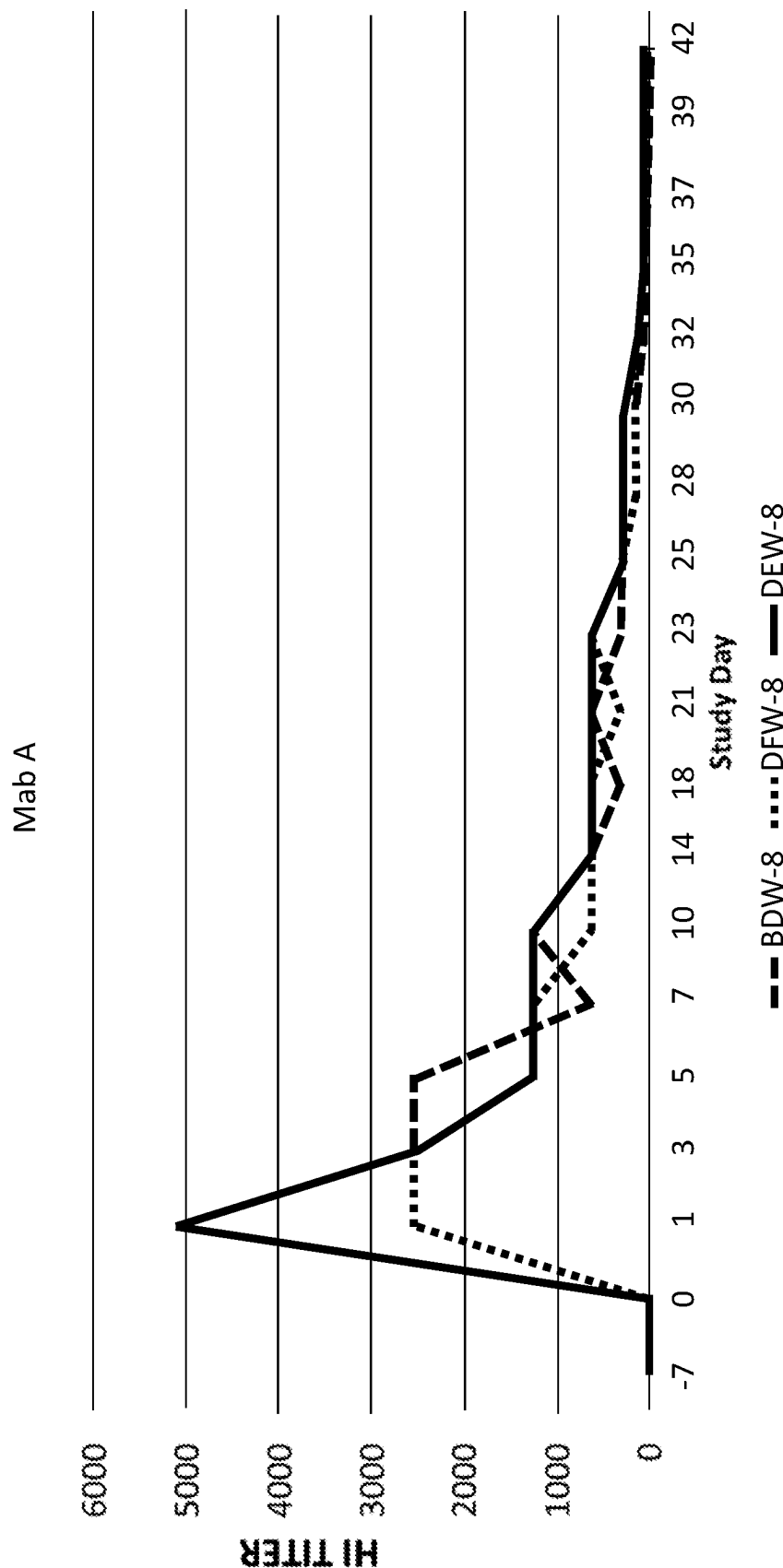


Fig. 3A

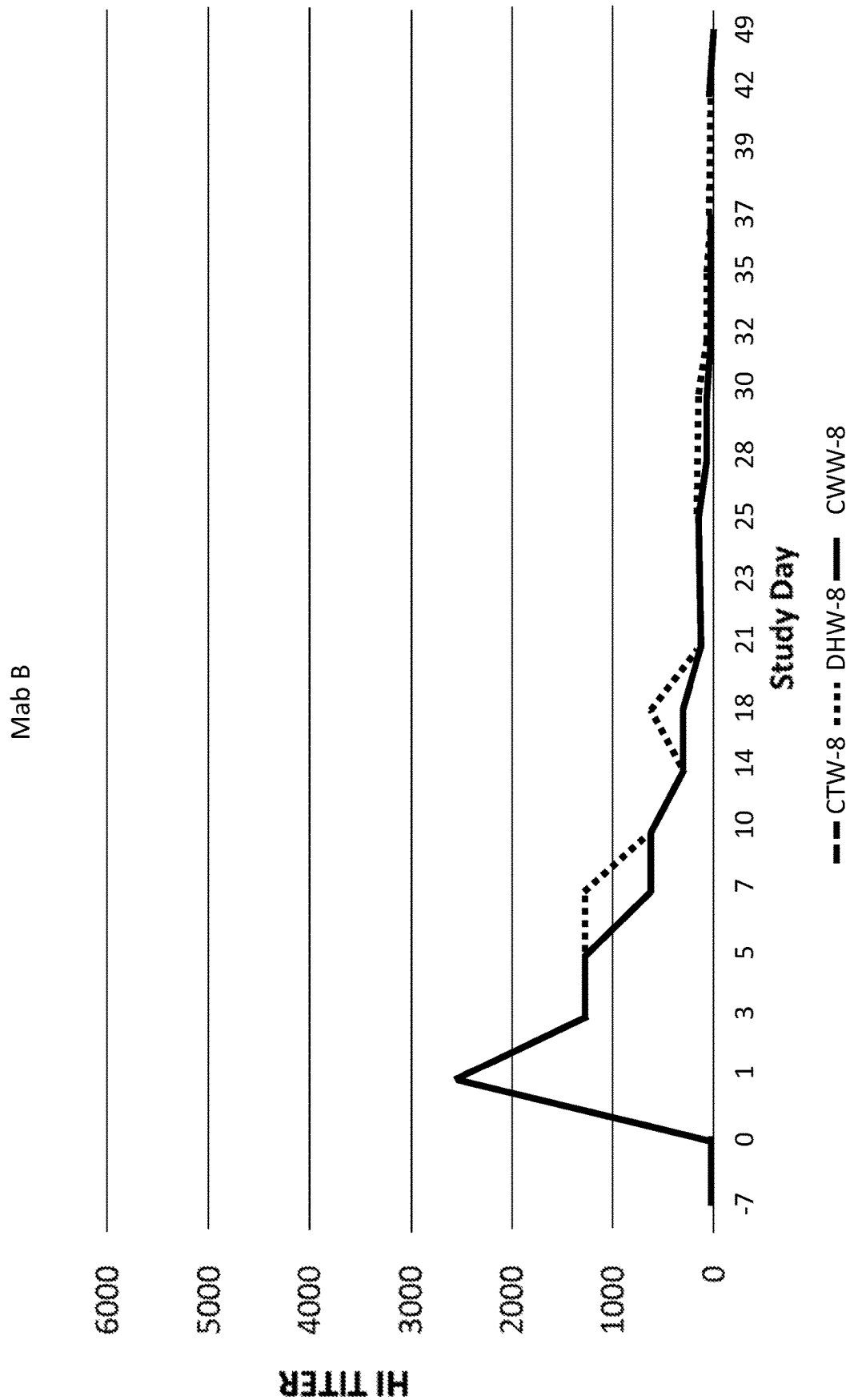


Fig. 3B

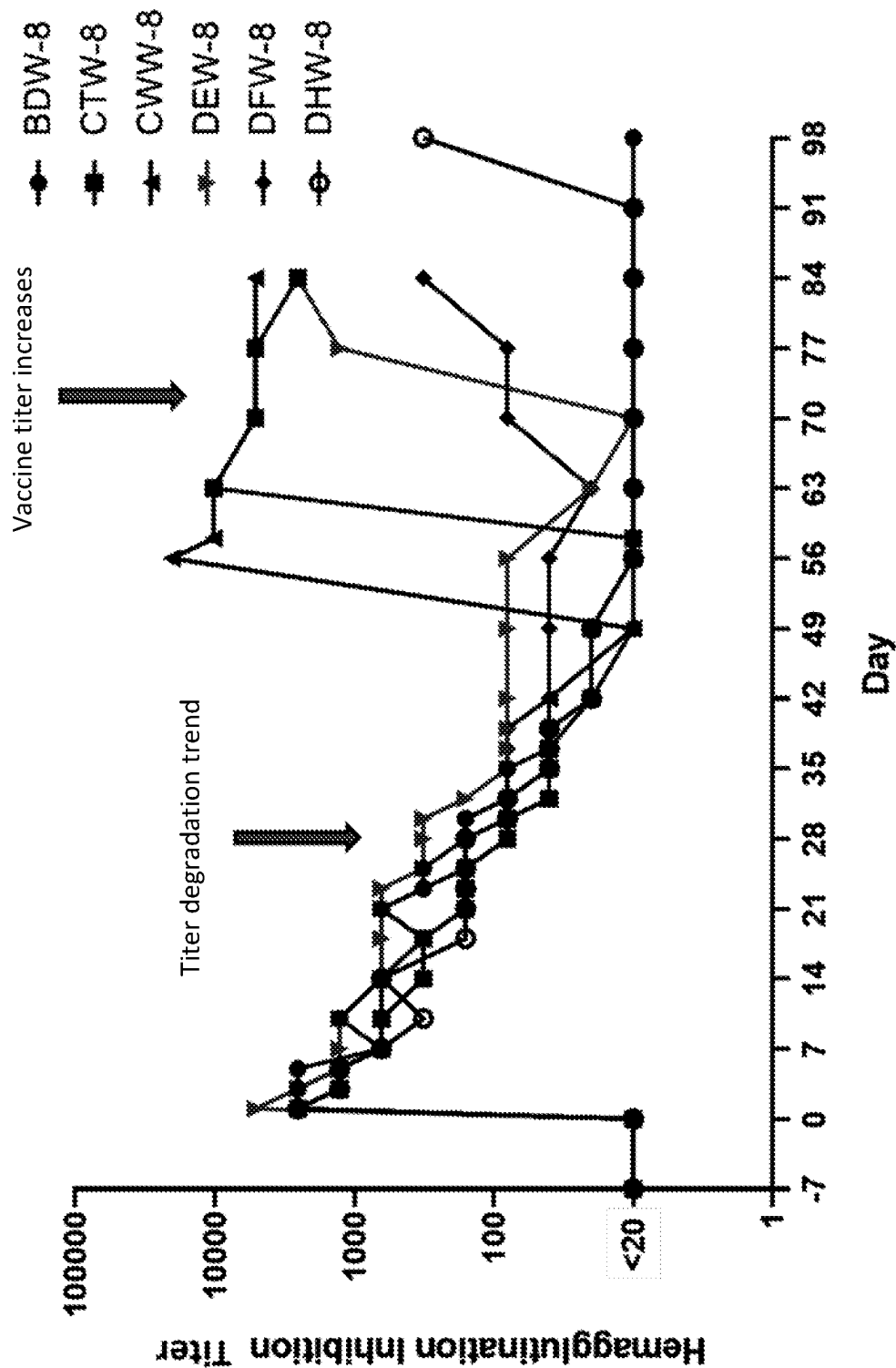
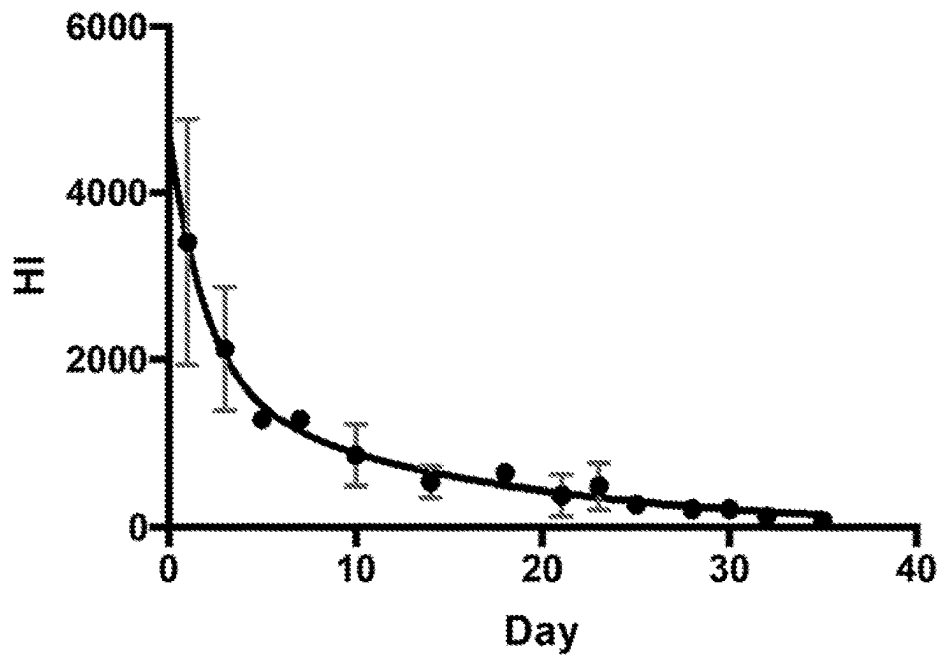
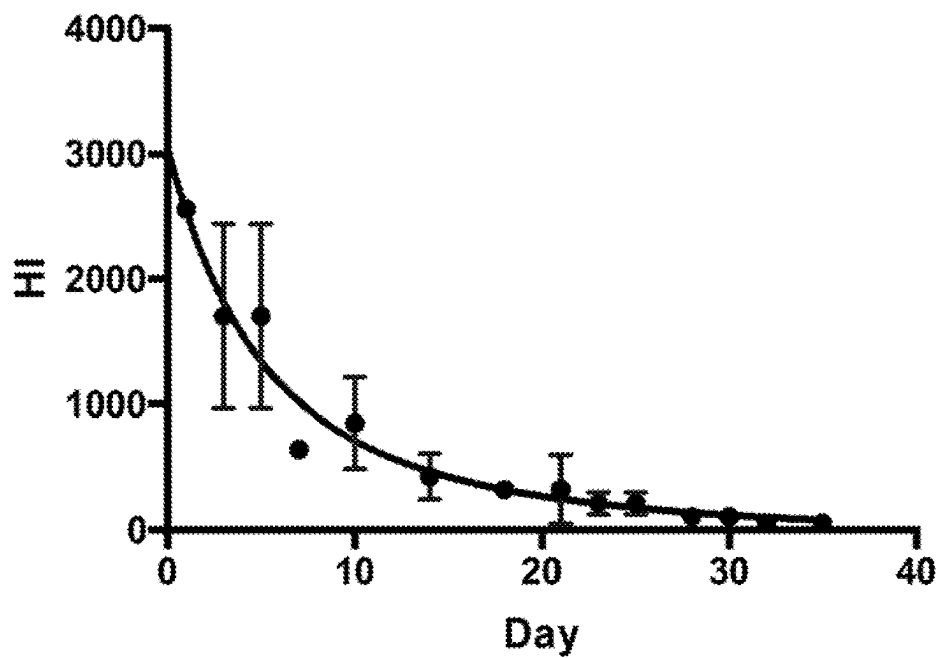


Fig. 3C

Mab A*Fig. 4A***Mab B***Fig. 4B*

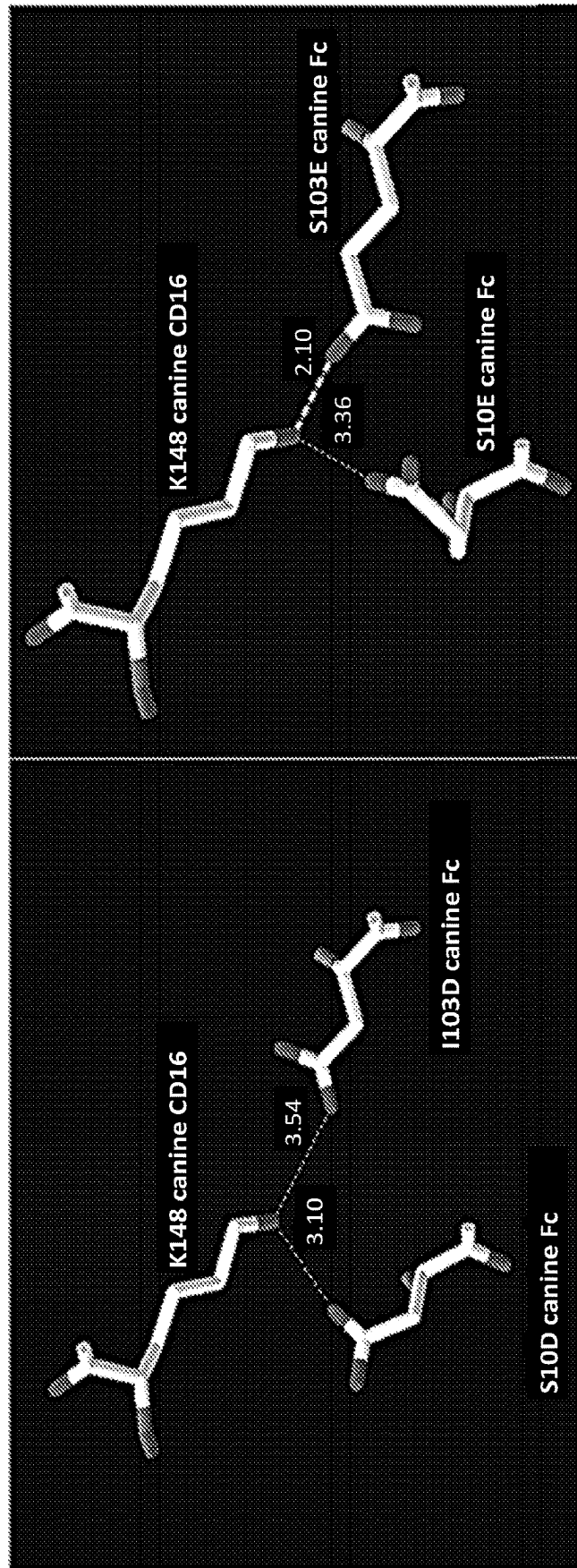


Fig. 5

1

PARVOVIRUS ANTIBODIES FOR VETERINARY USE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a national stage entry of International Patent Application No. PCT/US2020/044302, filed Jul. 30, 2020, which claims the benefit of priority of U.S. Provisional Application No. 62/880,650, filed Jul. 30, 2019, U.S. Provisional Application No. 62/968,970 filed Jan. 31, 2020, and U.S. Provisional Application No. 63/030,123 filed May 26, 2020, each of which is incorporated by reference herein in its entirety for any purpose.

FIELD OF THE INVENTION

This invention relates to isolated canine parvovirus antibodies, for example, with improved recombinant production and neutralizing binding to canine parvovirus and/or feline parvovirus, and methods of using the same, for example, for providing passive immunity against infection with a canine or feline parvovirus and/or treating parvoviral infection in companion animals, such as dogs and cats.

BACKGROUND

Canine Parvovirus (CPV) is the most important enteric virus infecting dogs worldwide¹. The CPV virion is a non-enveloped, DNA virus. There are multiple variants of the original CPV, including CPV-2a, CPV-2b, and CPV-2c. The variants of CPV-2 differ from one another by only a few amino acids². Quite persistent in the environment, CPV may remain infective for months. CPV spreads via direct and indirect contact³. Upon contact with the oral mucosa, CPV begins to replicate in local lymphatics and spreads systemically. With a 4 to 14 day incubation period, CPV targets rapidly dividing cells of the body including bone marrow, lymphocytes and intestinal crypt epithelial cells.

Clinical disease manifests as fever and depression followed by vomiting, diarrhea (profuse and bloody), lymphopenia, dehydration and quite often secondary septicemia and death. Mortality may exceed 70% in puppies³. Infection is most common in puppies after weaning as the maternal antibodies begin to wane. Vaccines are readily available and effectively protect against all variants of CPV-2.

Although properly timed vaccinations are protective, parvovirus infection remains a problem. The primary population of susceptible canines includes puppies in the vulnerable window where maternally derived antibody has waned to non-protective levels and before vaccination has been given (see FIG. 1). Puppies under 24 weeks of age are at highest risk of CPV disease. Additionally, puppies with failure of passive transfer of maternally derived antibodies is another vulnerable population. Outbreaks continue to occur in naïve kennel and shelter populations as well.

Treatment for CPV infection up to this point has been largely supportive. Treatment typically involves intravenous fluids, antiemetics, and broad-spectrum antibiotics to protect against septicemia. Various other treatments have been tried unsuccessfully including anti-virals and hyperimmune plasma⁴. Hospitalization is typically recommended which can be cost-prohibitive for many owners and can lead to the decision to euthanize. Hospitalizing dogs with CPV infection requires strict isolation protocols. This can prove to be

2

a logistics hardship for veterinary facilities and staff⁵. Thus, there is an unmet medical need for prevention and therapeutic treatment.

There are currently no USDA or FDA approved therapies specifically for CPV. A parvovirus monoclonal antibody described herein would be given as a therapeutic intervention to dogs with an active CPV infection to decrease the severity of or eliminate morbidity and mortality associated with CPV. Additionally, a parvovirus monoclonal antibody described herein would be utilized as a prophylactic treatment for dogs exposed to CPV infected dogs to prevent development of CPV infection.

SUMMARY OF THE INVENTION

Embodiment 1. An isolated antibody that binds to canine parvovirus and/or feline parvovirus, wherein the antibody comprises: (a) (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 4, (ii) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 5, (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 6, and (iv) an HC-FR1 sequence of SEQ ID NO: 7 or SEQ ID NO: 8; or (b) (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 42, (ii) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 43, (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 44, and (iv) an HC-FR1 sequence of SEQ ID NO: 45 or SEQ ID NO: 46.

Embodiment 2. An isolated antibody that binds to canine parvovirus and/or feline parvovirus, wherein the antibody comprises: (a) (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 13, (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 14, (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 15, and (iv) an LC-FR1 sequence of SEQ ID NO: 16; or (b) (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 52, (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 53, (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 54, and (iv) an LC-FR1 sequence of SEQ ID NO: 55 or SEQ ID NO: 56.

Embodiment 3. The isolated antibody of embodiment 1, wherein the antibody of (a) comprises an HC-FR4 sequence of SEQ ID NO: 12.

Embodiment 4. The isolated antibody of embodiment 1, wherein the antibody of (b) comprises an HC-FR2 sequence of SEQ ID NO: 48.

Embodiment 5. The isolated antibody of embodiment 1 or embodiment 4, wherein the antibody of (b) comprises an HC-FR3 sequence of SEQ ID NO: 50.

Embodiment 6. The isolated antibody of embodiment 2, wherein the antibody of (b) comprises an HC-FR3 sequence of SEQ ID NO: 59.

Embodiment 7. The isolated antibody of embodiment 2 or embodiment 6, wherein the antibody of (b) comprises an HC-F4 sequence of SEQ ID NO: 61.

Embodiment 8. An isolated antibody that binds to canine parvovirus and/or feline parvovirus, wherein the antibody is a caninized or a felinized antibody comprising: a) a heavy chain comprising (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 4; (ii) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 5; and (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 6, or b) a heavy chain comprising (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 42; (ii) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 43; and (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 44.

Embodiment 9. An isolated antibody that binds to canine parvovirus and/or feline parvovirus, wherein the antibody is a caninized or a felinized antibody comprising: a) a light chain comprising (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 13; (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 14; and (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 15, or b) a light chain comprising (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 52; (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 53; and (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 54.

Embodiment 10. An isolated antibody that binds to canine parvovirus and/or feline parvovirus, wherein the antibody is a caninized or a felinized antibody comprising: a) a heavy chain comprising (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 4; (ii) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 5; and (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 6, and b) a light chain comprising (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 13; (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 14; and (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 15.

Embodiment 11. An isolated antibody that binds to canine parvovirus and/or feline parvovirus, wherein the antibody is a caninized or a felinized antibody comprising: a) a heavy chain comprising (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 42; (ii) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 43; and (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 44, and b) a light chain comprising (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 52; (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 53; and (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 54.

Embodiment 12. The antibody of any one of the preceding embodiments, wherein the antibody is a chimeric antibody.

Embodiment 13. The antibody of any one of the preceding embodiments, wherein the antibody comprises a canine or feline constant heavy chain region or a canine or feline constant light chain region.

Embodiment 14. The antibody of any one of the preceding embodiments, wherein the antibody comprises:

(a) a canine heavy chain constant region selected from an IgG-A, IgG-B, IgG-C, and IgG-D constant region; or

(b) a feline heavy chain constant region selected from an IgG1, IgG2a, and IgG2b constant region.

Embodiment 15. The antibody of any one of the preceding embodiments, wherein the antibody comprises a wild-type or variant IgG Fc having complement fixation activity.

Embodiment 16. The antibody of any one of the preceding embodiments, wherein the antibody comprises a wild-type or variant IgG Fc having antibody-dependent cellular cytotoxicity (ADCC) activity.

Embodiment 17. The antibody of any one of the preceding embodiments, wherein the antibody comprises a wild-type or variant IgG Fc having antibody-dependent cellular phagocytosis (ADCP) activity.

Embodiment 18. The antibody of any one of the preceding embodiments, wherein the antibody comprises: a) an aspartic acid or a glutamic acid at a position corresponding to position 10 of SEQ ID NO: 91; b) an aspartic acid or a glutamic acid at position 10 of SEQ ID NO: 91; c) an aspartic acid or a glutamic acid at a position corresponding to position 103 of SEQ ID NO: 91; d) an aspartic acid or a glutamic acid at position 103 of SEQ ID NO: 91; e) an

aspartic acid or a glutamic acid at a position corresponding to position 10 and/or position 103 of SEQ ID NO: 91; f) an aspartic acid or a glutamic acid at position 10 and/or position 103 of SEQ ID NO: 91; or g) the amino acid sequence of SEQ ID NO: 92, SEQ ID NO: 93, SEQ ID NO: 94, SEQ ID NO: 95, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 98, or SEQ ID NO: 99.

Embodiment 19. The antibody of any one of the preceding embodiments, wherein the antibody comprises a canine κ light constant region or a feline κ light constant region.

Embodiment 20. The antibody of any one of the preceding embodiments, wherein the antibody comprises a feline κ light constant region without one or more N-glycosylation sites.

Embodiment 21. The antibody of any one of the preceding embodiments, wherein the antibody binds to an epitope comprising the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, and/or SEQ ID NO: 3.

Embodiment 22. The antibody of any one of the preceding embodiments, wherein the antibody binds to canine parvovirus or feline parvovirus with a dissociation constant (Kd) of less than 5×10^{-6} M, less than 1×10^{-6} M, less than 5×10^{-7} M, less than 1×10^{-7} M, less than 5×10^{-8} M, less than 1×10^{-8} M, less than 5×10^{-9} M, less than 1×10^{-9} M, less than 5×10^{-10} M, less than 1×10^{-10} M, less than 5×10^{-11} M, less than 1×10^{-11} M, less than 5×10^{-12} M, less than 1×10^{-12} M, less than 5×10^{-13} M, or less than 1×10^{-13} M as measured by biolayer interferometry.

Embodiment 23. The antibody of any one of the preceding embodiments, wherein the antibody binds to canine parvovirus or feline parvovirus as determined by immunoblot analysis and/or biolayer interferometry.

Embodiment 24. The antibody of any one of the preceding embodiments, wherein the antibody at a concentration of 200 mg/mL has an hemagglutination inhibition value of at least 8000, of at least 16000, of at least 32000.

Embodiment 25. The antibody of any one of the preceding embodiments, wherein the antibody is a monoclonal antibody.

Embodiment 26. The antibody of any one of the preceding embodiments comprising one or more of (a) an (HC-FR1) sequence of SEQ ID NO: 7, 8, 45, or 46; (b) a HC-FR2 sequence of SEQ ID NO: 9, 47, or 48; (c) a HC-FR3 sequence of SEQ ID NO: 10, 49, or 50; (d) a HC-FR4 sequence of SEQ ID NO: 11, 12, or 51; (e) a variable region light chain framework 1 (LC-FR1) sequence of SEQ ID NO: 16, 55, or 56; (f) an LC-FR2 sequence of SEQ ID NO: 17 or 57; (g) an LC-FR3 sequence of SEQ ID NO: 18, 58, or 59; or (h) an LC-FR4 sequence of SEQ ID NO: 19, 60, or 61.

Embodiment 27. The antibody of any one of the preceding embodiments, wherein the antibody comprises: (a) a variable heavy chain sequence of SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 85, SEQ ID NO: 86, SEQ ID NO: 40, SEQ ID NO: 62, SEQ ID NO: 63, or SEQ ID NO: 88; and/or (b) a variable light chain sequence of SEQ ID NO: 22, SEQ ID NO: 87, SEQ ID NO: 41, SEQ ID NO: 64, SEQ ID NO: 65, or SEQ ID NO: 89.

Embodiment 28. The antibody of any one of the preceding embodiments, wherein the antibody comprises: (a) a variable heavy chain sequence of SEQ ID NO: 20 and a variable light chain sequence of SEQ ID NO: 22; (b) a variable heavy chain sequence of SEQ ID NO: 21 and a variable light chain sequence of SEQ ID NO: 22; (c) a variable heavy chain sequence of SEQ ID NO: 85 and a variable light chain sequence of SEQ ID NO: 87; (d) a variable heavy chain sequence of SEQ ID NO: 86 and a variable light chain

5

sequence of SEQ ID NO: 87; (e) a variable heavy chain sequence of SEQ ID NO: 40 and a variable light chain sequence of SEQ ID NO: 41; (f) a variable heavy chain sequence of SEQ ID NO: 62 and a variable light chain sequence of SEQ ID NO: 64 or SEQ ID NO: 65; (g) a variable heavy chain sequence of SEQ ID NO: 63 and a variable light chain sequence of SEQ ID NO: 64 or SEQ ID NO: 65; or (h) a variable heavy chain sequence of SEQ ID NO: 88 and a variable light chain sequence of SEQ ID NO: 89.

Embodiment 29. The antibody of any one of the preceding embodiments, wherein the antibody comprises: (a) (i) a heavy chain sequence of SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 66, SEQ ID NO: 67, or SEQ ID NO: 79; and/or (ii) a light chain sequence of SEQ ID NO: 25, SEQ ID NO: 39, SEQ ID NO: 68, SEQ ID NO: 69, or SEQ ID NO: 80; or (b) (i) a heavy chain sequence of SEQ ID NO: 31, SEQ ID NO: 40, or SEQ ID NO: 74; and/or (ii) a light chain sequence of SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 41, SEQ ID NO: 75, or SEQ ID NO: 76.

Embodiment 30. The antibody of any one of the preceding embodiments, wherein the antibody comprises: (a) a heavy chain sequence of SEQ ID NO: 23 or SEQ ID NO: 24, and a light chain sequence of SEQ ID NO: 25; (b) a heavy chain sequence of SEQ ID NO: 31 and a light chain sequence of SEQ ID NO: 32 or SEQ ID NO: 33; (c) a heavy chain sequence of SEQ ID NO: 37 or SEQ ID NO: 38, and a light chain sequence of SEQ ID NO: 39; (d) a heavy chain sequence of SEQ ID NO: 66 or SEQ ID NO: 67, and a light chain sequence of SEQ ID NO: 68 or SEQ ID NO: 69; (e) a heavy chain sequence of SEQ ID NO: 66 and a light chain sequence of SEQ ID NO: 68; (f) a heavy chain sequence of SEQ ID NO: 67 and a light chain sequence of SEQ ID NO: 69; (g) a heavy chain sequence of SEQ ID NO: 74 and a light chain sequence of SEQ ID NO: 75 or SEQ ID NO: 76; or (h) a heavy chain sequence of SEQ ID NO: 79 and a light chain sequence of SEQ ID NO: 80.

Embodiment 31. The antibody of any one of the preceding embodiments, wherein the antibody comprises a heavy chain sequence of SEQ ID NO: 24 and a light chain sequence of SEQ ID NO: 25.

Embodiment 32. An isolated nucleic acid encoding the antibody of any one of the preceding embodiments.

Embodiment 33. A host cell comprising the nucleic acid of embodiment 31.

Embodiment 34. A method of producing an antibody comprising culturing the host cell of embodiment 32 and isolating the antibody.

Embodiment 35. A pharmaceutical composition comprising the antibody of any one or more of embodiments 1 to 31 and a pharmaceutically acceptable carrier.

Embodiment 36. The pharmaceutical composition of embodiment 35, wherein the pharmaceutically acceptable carrier is phosphate buffered saline.

Embodiment 37. A method of providing passive immunity in a subject against infection with a canine or feline parvovirus comprising administering to the subject a therapeutically effective amount of a monoclonal antibody that binds to the canine or feline parvovirus.

Embodiment 38. The method of embodiments 37, wherein the monoclonal antibody is administered before exposure to the canine or feline parvovirus.

Embodiment 39. The method of embodiment 37 or embodiment 38, wherein the monoclonal antibody is administered after exposure to the canine or feline parvovirus.

6

Embodiment 40. The method of any one of embodiments 37 to 39, wherein the monoclonal antibody is administered after infection with the canine or feline parvovirus.

Embodiment 41. The method of any one of embodiments 37 to 40, wherein the monoclonal antibody is administered after the subject has exhibited at least one symptoms selected from fever, vomiting, diarrhea, lymphopenia, and septicemia.

Embodiment 42. The method of any one of embodiments 37 to 41, wherein the monoclonal antibody is administered after canine or feline parvovirus has been detected in feces, such as determined by a positive cage-side SNAP test.

Embodiment 43. The method of any one of embodiments 37 to 42, wherein the subject has previously been administered a parvovirus vaccine.

Embodiment 44. The method of any one of embodiments 37 to 42, wherein the subject has not previously been administered a parvovirus vaccine.

Embodiment 45. The method of any one of embodiments 37 to 44, wherein the subject is unprotected at birth due to lack of maternally-derived antibodies to a canine or feline parvovirus or failure of passive transfer of antibodies to a canine or feline parvovirus.

Embodiment 46. The method of any one of embodiments 37 to 45, wherein the subject is hand-reared, the subject's mother does not produce milk, or the subject is unable to produce antibodies against parvovirus.

Embodiment 47. The method of any one of embodiments 37 to 46, wherein the subject is living in an environment contaminated with the canine or feline parvovirus.

Embodiment 48. A method of treating a canine or feline parvoviral infection in a subject comprising administering to the subject a therapeutically effective amount of a monoclonal antibody that binds to the canine or feline parvovirus.

Embodiment 49. The method of embodiment 48, wherein the monoclonal antibody is administered after the subject has exhibited at least one symptoms selected from fever, vomiting, diarrhea, lymphopenia, and septicemia.

Embodiment 50. The method of embodiment 48 or embodiment 49, wherein the monoclonal antibody is administered after canine or feline parvovirus has been detected in feces, such as determined by a positive cage-side SNAP test.

Embodiment 51. The method of any one of embodiments 48 to 50, wherein the subject has previously been administered a parvovirus vaccine.

Embodiment 52. The method of any one of embodiments 48 to 50, wherein the subject has not previously been administered a parvovirus vaccine.

Embodiment 53. The method of any one of embodiments 48 to 52, wherein the subject is living in an environment contaminated with the canine or feline parvovirus.

Embodiment 54. The method of any one of embodiments 48 to 53, wherein the subject is a canine or feline.

Embodiment 55. The method of any one of embodiments 48 to 53, wherein the subject is a human.

Embodiment 56. The method of any one of embodiments 37 to 55, wherein the method comprises administering to the subject a therapeutically effective amount of a monoclonal antibody that binds to an epitope comprising the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, and/or SEQ ID NO: 3.

Embodiment 57. The method of any one of embodiments 37 to 56, wherein the method comprises administering to the subject a therapeutically effective amount of a monoclonal antibody comprising: (a) a heavy chain comprising (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 42; (ii) a CDR-H2 comprising the amino acid sequence

of SEQ ID NO: 43; and (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 44, and (b) a light chain comprising (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 52; (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 53; and (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 54.

Embodiment 58. The method of any one of embodiments 37 to 57, wherein the method comprises administering to the subject a therapeutically effective amount of the antibody of any one of embodiments 1 to 31 or the pharmaceutical composition of embodiment 35 or embodiment 36.

Embodiment 59. The method of any one of embodiments 37 to 58, wherein the antibody or the pharmaceutical composition is administered parenterally.

Embodiment 60. The method of any one of embodiments 37 to 59, wherein the antibody or the pharmaceutical composition is administered by an intramuscular route, an intraperitoneal route, an intracerebrospinal route, a subcutaneous route, an intra-arterial route, an intrasynovial route, an intrathecal route, an intravenous, or an inhalation route.

Embodiment 61. The method of any one of embodiments 37 to 60, wherein the antibody or the pharmaceutical composition is administered intravenously.

Embodiment 62. The method of any one of embodiments 37 to 60, wherein the antibody or the pharmaceutical composition is administered subcutaneously.

Embodiment 63. The method of any one of embodiments 37 to 62, wherein the subject is less than 1 week of age, less than 2 weeks of age, less than 3 weeks of age, less than 4 weeks of age, less than 5 weeks of age, less than 6 weeks of age, less than 6 weeks of age, less than 7 weeks of age, less than 8 weeks of age, less than 9 weeks of age, less than 10 weeks of age, less than 11 weeks of age, less than 12 weeks of age, less than 6 months of age, between 0 and 12 weeks of age, between 0 and 10 weeks of age, between 0 and 8 weeks of age, between 0 and 6 weeks of age, between 0 and 4 weeks of age, between 0 and 2 weeks of age, between 4 and 12 weeks of age, between 6 and 12 weeks of age, between 10 and 12 weeks of age, between 4 weeks and 6 months of age, between 2 months and 6 months of age, between 4 months and 6 months of age, between 6 months and 1 year of age, greater than 13 weeks of age, or greater than 1 year of age.

Embodiment 64. The method of any one of embodiments 37 to 63, wherein the subject is 13 weeks of age or older.

Embodiment 65. The method of any one of embodiments 37 to 64, wherein the antibody is administered at an amount in the range of 0.01 mg/kg body weight to 100 mg/kg body weight per dose.

Embodiment 66. The method of any one of embodiments 37 to 65, wherein the antibody is administered at an amount of 5 mg/kg body weight per dose.

Embodiment 67. The method of any one of embodiments 37 to 66, wherein the antibody or the pharmaceutical composition is administered as a single dose.

Embodiment 68. The method of any one of embodiments 37 to 67, wherein the antibody or the pharmaceutical composition is administered repeatedly, such as once per week for at least two or three consecutive weeks.

Embodiment 69. The method of any one of embodiments 37 to 68, the method comprises administering to the subject a therapeutically effective amount of two or more different antibodies of any one of embodiments 1 to 31, wherein the two or more different antibodies are administered simultaneously or sequentially, optionally wherein administration of the two or more different antibodies is separated by one or more days.

Embodiment 70. The method of any one of embodiments 37 to 69, wherein the subject has a hemagglutination inhibition titer of less than 20 as determined by hemagglutination inhibition assay prior to administration of the antibody or the pharmaceutical composition.

Embodiment 71. The method of any one of embodiments 37 to 70, wherein the subject is parvovirus titer negative as determined by hemagglutination inhibition assay prior to administration of the antibody or the pharmaceutical composition.

Embodiment 72. The method of any one of embodiments 37 to 71, wherein the subject survives infection with a canine or feline parvovirus following administration of the antibody or the pharmaceutical composition.

Embodiment 73. The method of reducing parvoviral infection of a cell, the method comprising exposing to the cell the antibody of any one of embodiments 1 to 31 or the pharmaceutical composition of embodiment 35 or embodiment 36 under conditions permissive for binding of the antibody to a parvovirus.

Embodiment 74. The method of embodiment 73, wherein the cell is exposed to the antibody or the pharmaceutical composition in vitro.

Embodiment 75. The method of embodiment 73 or embodiment 74, wherein the cell is a mammalian cell, a human cell, a canine cell, or a feline cell.

Embodiment 76. A method for detecting a parvoviral infection in a sample from a subject comprising contacting the sample with the antibody of any one of embodiments 1 to 31 or the pharmaceutical composition of embodiment 35 or embodiment 36 under conditions permissive for the binding of the antibody to a parvovirus, and detecting whether a complex is formed between the antibody and the parvovirus in the sample.

Embodiment 77. The method of embodiment 76, wherein the sample is a biological sample obtained from a canine, a feline, or a human.

Embodiment 78. A variant IgG Fc polypeptide comprising: a) an aspartic acid or a glutamic acid at a position corresponding to position 10 of SEQ ID NO: 91; b) an aspartic acid or a glutamic acid at position 10 of SEQ ID NO: 91; c) an aspartic acid or a glutamic acid at a position corresponding to position 103 of SEQ ID NO: 91; d) an aspartic acid or a glutamic acid at position 103 of SEQ ID NO: 91; e) an aspartic acid or a glutamic acid at a position corresponding to position 10 and/or position 103 of SEQ ID NO: 91; or f) an aspartic acid or a glutamic acid at position 10 and/or position 103 of SEQ ID NO: 91.

Embodiment 79. A polypeptide comprising the variant IgG Fc polypeptide of embodiment 78.

Embodiment 80. A polypeptide comprising the amino acid sequence of SEQ ID NO: 92, SEQ ID NO: 93, SEQ ID NO: 94, SEQ ID NO: 95, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 98, or SEQ ID NO: 99.

Embodiment 81. An isolated nucleic acid encoding the polypeptide of any one of embodiments 78 to 80.

Embodiment 82. A host cell comprising the nucleic acid of embodiment 81.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram of the window of susceptibility for CPV.

FIG. 2 shows Mab A v2 and Mab B v2 binding to VP2.

FIG. 3A shows the HI assay titers for the Mab A group.

FIG. 3B shows the HI assay titers for the Mab B group.

FIG. 3C shows the HI assay titers for both Mab A and Mab B groups.

FIG. 4A shows two phase decay kinetics of Mab A v2.
 FIG. 4B shows two phase decay kinetics of Mab B v2.
 FIG. 5 shows structural models of variant canine IgG-B
 Fc polypeptide sequences of Example 15 and canine CD16.

DESCRIPTION OF CERTAIN SEQUENCES

Table 1 provides a listing of certain sequences referenced herein.

TABLE 1

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
1	ESENYRRVVVNMDKTAVNGNMALDDIHAEIVTPWSL V	Epitope 1
2	TPWRYYFQWDRTLIPSHGTSGTPTNIYHGTDPDVQ FYTIENS	Epitope 2
3	TPWRYYFQWDRTLIPSHGTSGTP	Minimal epitope 2
4	GFSLSYHVH	Variable heavy chain CDR-H1 of Mab A
5	VMWNDGDT	Variable heavy chain CDR-H2 of Mab A
6	PELPGLTYGVWFPY	Variable heavy chain CDR-H3 of Mab A
7	<u>QVQLKES</u> GPGLVQPSQTLSTCTVS	Variable heavy chain framework region HC-FR1 of Mab A
8	<u>QVQLKES</u> GPGLVAPSQTLSTCTVS	Variable heavy chain framework region HC-FR1 of Mab A variant 2 (v2) Q13A
9	WVRQPPGKGLEWLG	Variable region heavy chain framework region HC-FR2 of Mab A
10	YNLALNSRLSISRDTSKSQVFFKMSSLQTEDTATYYC AR	Variable heavy chain framework region HC-FR3 of Mab A
11	WGQGTLLVTVSS	Variable heavy chain framework region HC-FR4 of Mab A
12	WGQGTLLVTVSA	Variable heavy chain framework region HC-FR4 of Mab A variant 2 (v2) S11A
13	KASQNVDSNVD	Variable light chain CDR-L1 of Mab A
14	KASNRNT	Variable light chain CDR-L2 of Mab A
15	MQSTSYPLTF	Variable light chain CDR-L3 of Mab A
16	<u>DIVMTQT</u> PASMSISVGDRVTMNC	Variable light chain framework region LC-FR1 of Mab A
17	WYQQKTGQSPNLLIY	Variable light chain framework region LC-FR2 of Mab A
18	GVPDRFTGSGGTDFTFITISNMQAEDLAVYYC	Variable light chain framework region LC-FR3 of Mab A
19	GSQTKLEIK	Variable light chain framework region LC-FR4 of Mab A

TABLE 1-continued

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
20	<u>QVQLKESG</u> PGLVQPSQTLSLTCTVSGFSLSSYHVHWV RQPPGKGLEWLGVMWNGDTSYNLALNSRLSISRDT KSQVFFKMSSLQTEDTATYYCARPELPGLTYGVWFPY WGQGT LV TVSS	Variable heavy chain of Mab A (Mab A HC)
21	<u>QVQLKESG</u> PGLV <u>APS</u> QTLSLTCTVSGFSLSSYHVHWV RQPPGKGLEWLGVMWNGDTSYNLALNSRLSISRDT KSQVFFKMSSLQTEDTATYYCARPELPGLTYGVWFPY WGQGT LV TVSA	Variable heavy chain of Mab A variant 2 HC-FR1 Q13A HC-FR4 S11A (Mab A HC v2)
22	<u>DI</u> VMTQT P ASMSISVGD R VTMNCASQNVDSNVDWYQ KQTGQSPNLLIYKASNRNTGVPDRFTGSGSGTDFTFT ISNMQAE DL AVYYCMQSTSYP LT FGSGTKLEIK	Variable light chain of Mab A (Mab A LC)
23	<u>QVQLKESG</u> PGLVQPSQTLSLTCTVSGFSLSSYHVHWV RQPPGKGLEWLGVMWNGDTSYNLALNSRLSISRDT KSQVFFKMSSLQTEDTATYYCARPELPGLTYGVWFPY WGQGT LV TVSSASTTAPSVFPLAPSCGTS SG STVALA CLVSGYFPEPVT SV WNSGSLTSGVHTF PS VLQSSGLY SLSSMVTVPSSRW P SETFTCNVAHPASKTKVDKPVK RENGRVPRPPDCPKCPAPEMLGGPSVFIFPPKPKDTL LIARTPEVTCVVVDLDPEDPEVQISWFDGKQMQTAK TQPREEQFNGTYRVVSVLPIGHQDWLKGKQFTCKVNN KALPSPIERTISKARGQAHPQSPVYVLPSSREELSKNT VSLTCLIKDFFPPDIDVEWQSNQQEPESKYRTTPPQ LDEDGSYFLYSKLSVDKSRWQRGDTFICAVMHEALHN HYTQESLSHSPGK	Chimeric variable heavy chain of Mab A and canine IgG-B (Chimeric A HC IgG-B)
24	<u>QVQLKESG</u> PGLV <u>APS</u> QTLSLTCTVSGFSLSSYHVHWV RQPPGKGLEWLGVMWNGDTSYNLALNSRLSISRDT KSQVFFKMSSLQTEDTATYYCARPELPGLTYGVWFPY WGQGT LV TVSSA AST TAPSVFPLAPSCGTS SG STVALA CLVSGYFPEPVT SV WNSGSLTSGVHTF PS VLQSSGLY SLSSMVTVPSSRW P SETFTCNVAHPASKTKVDKPVK RENGRVPRPPDCPKCPAPEMLGGPSVFIFPPKPKDTL LIARTPEVTCVVVDLDPEDPEVQISWFDGKQMQTAK TQPREEQFNGTYRVVSVLPIGHQDWLKGKQFTCKVNN KALPSPIERTISKARGQAHPQSPVYVLPSSREELSKNT VSLTCLIKDFFPPDIDVEWQSNQQEPESKYRTTPPQ LDEDGSYFLYSKLSVDKSRWQRGDTFICAVMHEALHN HYTQESLSHSPGK	Chimeric variable heavy chain of Mab A v2 and canine IgG-B HC-FR1 Q13A HC-FR4 S11A (Chimeric A HC v2 IgG-B)
25	<u>DI</u> VMTQT P ASMSISVGD R VTMNCASQNVDSNVDWYQ KQTGQSPNLLIYKASNRNTGVPDRFTGSGSGTDFTFT ISNMQAE DL AVYYCMQSTSYP LT FGSGTKLEIKRND QPAVYLFQPSPDQLHTGSASVVC LN SFYPKDINVKW KVDGVIQDTGIQESVTEQDKDSTYLSLSTLTMSSTEY LSHELYSCEITHKSLPSTLIKS FQ RSE Q RV	Chimeric variable light chain of Mab A and canine κ light constant region (Chimeric A LC κ)
26	MAVLG LL LLCLVTFPSCVLS	Heavy chain leader
27	METDTLLLV LL LLWVPGSTG	Light chain leader
28	MAVLG LL LLCLVTFPSCVLS <u>QVQLKESG</u> PGLVQPSQTL SLTCTVSGFSLSSYHVHWVRQPPGKGLEWLGVMWNG DTSYNLALNSRLSISRDTSKSQVFFKMSSLQTEDTAT YYCARPELPGLTYGVWFPYWGQGT LV TVSSASTTAP VFPLAPSCGTS SG STVALACLVSGYFPEPVT SV WNSG SLTSGVHTF PS VLQSSGLYSLSSMVTVPSSRW P SETF TCNVAHPASKTKVDKPVKRENGRVPRPPDCPKCPAP EMLGGPSVFIFPPKPKDILLIARTPEVICVVVDLDPE DPEVQISWFDGKQMQTAKTQPREEQFNGTYRVVSVL PIGHQDWLKGKQFTCKVNNKALPSPIERTISKARGQA HPQSVYVLPSSREELSKNIVSLICLIKDFFPPDIDVE WQSNQQEPESKYRTTPQLDEDGSYFLYSKLSVDKS RWQRGDTFICAVMHEALHNHYTQESLSHSPGK	Chimeric variable heavy chain of Mab A and canine IgG-B with leader sequence (Chimeric A HC IgG-B with leader)
29	MAVLG LL LLCLVTFPSCVLS <u>QVQLKESG</u> PGLV <u>APS</u> QTL SLTCTVSGFSLSSYHVHWVRQPPGKGLEWLGVMWNG DTSYNLALNSRLSISRDTSKSQVFFKMSSLQTEDTAT YYCARPELPGLTYGVWFPYWGQGT LV TVSSA AST TAP VFPLAPSCGTS SG STVALACLVSGYFPEPVT SV WNSG SLTSGVHTF PS VLQSSGLYSLSSMVTVPSSRW P SETF	Chimeric variable heavy chain of Mab A v2 and canine IgG-B with leader sequence HC-FR1 Q13A HC-FR4 S11A

TABLE 1-continued

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
	TCNVAHPASKTKVDKPVPKRENGRVPRPPDCPKCPAP EMLGGPSVFIFFPKPKDILLIARTPEVICVVVDLDPE DPEVQISWFDGKQMQTAKTQPREEQFNGTYRVVSVL PIGHQDWLKGKQFTCKVNNKALPSPIERTISKARGQA HQPSVYVLPSSREELSKNIVSLICLIKDFPPPDIDVE WQSNQQEPESKYRTTPQLDEDGSYFLYSKLSVDKS RWQRGDTFICAVMHEALHNHYTQESLSHSPGK	(Chimeric A HC v2 IgG-B with leader)
30	METDTLLLVWLLLVPGSTGDIVMTQTPASMSISVGD RVTMNCASQNVDENVWYQKQTGQSPNLLIYKASNR NTGVDPDRFTGSGSGTDFFTISNMQAEDLAVYYCMQS TSYPLTFGSGTKLEIKRNDAPAVYLFQPSPDQLHTG SASVVCLLNSFYPKDINVKWVDGVIQDTGIQESVTE QDKDSTYLSSTLTMSSTEYLSHELYSCEITHKSLPS TLIKSFQRSECRVD	Chimeric variable light chain of Mab A and canine κ light constant region with leader sequence (Chimeric A LC κ with leader)
31	QVQLKESGPGVLVQPSQTLSTCTVSGFSLSSYHVHWV RQPPGKGLEWLVGMWNGDTSYNLALNSRLSISRDT KSQVFFKMSSLQTEDTATYYCARPELPGITYGVWFPY WGQGLTVTVSSASTTAPSVFPLAPSCGTSGATVALA CLVLGYFPEPVTVSWNSGALTSGVHTFPAVLQASGLY SLSSMVTVPSSRWLSDTFTCNVAHPPSNTKVDKTVRK TDHPGPKPCDCKPCPPPEMLGGPSIIFPPPKPKDTL SISRIVEPTCLVVDLGPDDSDVQITWFVDNTQVYTAK TSPREEQFNSTYRVVSVLPILHQDWLKGKEFKCKVNS KSLPSPIERTISKAKGQPEHPQVYVLPQAQELSRNK VSVTCLIKSFHPPDIAVEWEITGQPEPENNYRTTPQ LDSGTYFVYSKLSVDRSHWQRGNTYTCVSHALHS HHTQKSLTQSPGK	Chimeric variable heavy chain of Mab A and feline IgG-1 (Chimeric A HC IgG-1)
32	DIVMTQTPASMSISVGDRTMNCASQNVDENVWYQ QKTGQSPNLLIYKASNRNTGVDPDRFTGSGSGTDFFT ISNMQAEDLAVYYCMQSTSYPLTFGSGTKLEIKRSDA QPSVFLFQPSLDELHTGSASIVCILNDFYPKEVNVKW KVDGVVQNKGIQESTTEQNSKDYLSSTLTMSSTE YQSHEKFSCEVTHKSLASTLVKSFNRSECRRE	Chimeric variable light chain of Mab A and feline κ light constant region (Chimeric A LC κ)
33	DIVMTQTPASMSISVGDRTMNCASQNVDENVWYQ QKTGQSPNLLIYKASNRNTGVDPDRFTGSGSGTDFFT ISNMQAEDLAVYYCMQSTSYPLTFGSGTKLEIKRSDA QPSVFLFQPSLDELHTGSASIVCILNDFYPKEVNVKW KVDGVVQNKGIQESTTEQNSKDYLSSTLTMSSTE YQSHEKFSCEVTHKSLASTLVKSFNRSECRRE	Chimeric variable light chain of Mab A and feline κ light constant region with no N-linked glycosylation site (Chimeric A LC κ aglycos)
34	MAVLGLLLCLVTFPSCVLSEVQLVESGGDLVKPGGTL RLSCTVSGFSLSSYHVHWVRQPPGKGLEWVAVMWNDG DTSYNLAVKGRFTISRDNKNTLYLQMNSLRAEDTAV YYCARPELPGITYGVWFPYWGQGLTVTVSSASTTAPS VFPLAPSCGTSGSTVALACLVSGYFPEPVTVSWNSG SLTSGVHTFPVSVLQSSGLYLSMVTVPSSRWPSSETF TCNVAHPASKTKVDKPVPKRENGRVPRPPDCPKCPAP EMLGGPSVFIFFPKPKDILLIARTPEVICVVVDLDPE DPEVQISWFDGKQMQTAKTQPREEQFNGTYRVVSVL PIGHQDWLKGKQFTCKVNNKALPSPIERTISKARGQA HQPSVYVLPSSREELSKNIVSLICLIKDFPPPDIDVE WQSNQQEPESKYRTTPQLDEDGSYFLYSKLSVDKS RWQRGDTFICAVMHEALHNHYTQESLSHSPGK	Caninized variable heavy chain of Mab A v3 and canine IgG-B with leader sequence (Chimeric A HC v3 IgG-B with leader)
85	QVQLVESGGDLVKPGGTLRLSCTVSGFSLSSYHVHWV RQPPGKGLEWVAVMWNDGTSYNLAVKGRFTISRDN KNTLYLQMNSLRAEDTAVYYCARPELPGITYGVWFPY WGQGLTVTVSS	Caninized variable heavy chain of Mab A v3
35	MAVLGLLLCLVTFPSCVLSEVQLVESGGDLVKPGGTL RLSCTVSGFSLSSYHVHWVRQPPGKGLEWLVGMWNDG DTSYNLAVKGRFTISRDNKNTLYLQMNSLRAEDTAT YYCARPELPGITYGVWFPYWGQGLTVTVSSASTTAPS VFPLAPSCGTSGSTVALACLVSGYFPEPVTVSWNSG SLTSGVHTFPVSVLQSSGLYLSMVTVPSSRWPSSETF TCNVAHPASKTKVDKPVPKRENGRVPRPPDCPKCPAP EMLGGPSVFIFFPKPKDILLIARTPEVICVVVDLDPE DPEVQISWFDGKQMQTAKTQPREEQFNGTYRVVSVL PIGHQDWLKGKQFTCKVNNKALPSPIERTISKARGQA	Caninized variable heavy chain of Mab A v4 and canine IgG-B with leader sequence (Caninized A HC v4 IgG-B with leader)

TABLE 1-continued

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
	HQPSVYVLPSPREELSKNIVSLICLIKDFPPPDIDVE WQSNQQEPESKYRTTPQLDEGSGYFLYSKLSVDKS RWQRGDTFICAVMHEALHNHYTQESLSHSPGK	
86	EVQLVESGGDLVKPGGTLRLSCTVSGFSLSSYHVHWV RQPPGKGLEWLGVMWMDGDTSYNLAVKGRFTISRDN KNTLYFQMNSLRAEDTATYYCARPELPGITYGVWFPY WGQGTLLTVSS	Caninized variable heavy chain of Mab A v4
36	METDTLLLVLLLVWPGSTGEIVMTQSPASLSLSQEE KVTITCKASQNVDSNVDWYQKPGQAPKLLIYKASNR NTGVPSTRFSGSGSDTDFSTISSLEPEDVAVYYCMQS TSYPLTFGQGTKEIKRNDAPAVYLFQPSPDQLHTG SASVVCLLNSFYPKDINVKWKGVIQDTGIQESVTE QDKDSTYSLSTLTMSSTEYLSHELYSCEITHKSLPS TLIKSFQRSECQRVD	Caninized variable light chain of Mab A and canine κ light constant region with leader sequence (Caninized A LC κ with leader)
87	EIVMTQSPASLSLSQEEKVTITCKASQNVDSNVDWYQ KQPGQAPKLLIYKASNRNTGVPSTRFSGSGSDTDFST ISSLEPEDVAVYYCMQSTSYPLTFGQGTKEIK	Caninized variable light chain of Mab A
37	QVQLVESGGDLVKPGGTLRLSCTVSGFSLSSYHVHWV RQPPGKGLEWVAVMWMDGDTSYNLAVKGRFTISRDN KNTLYLQMNLSRAEDTAVYYCARPELPGITYGVWFPY WGQGTLLTVSSASTTAPSVFPLAPSCGTSSTGVALA CLVSGYFPEPVTVSWNSGSLTSGVHTFPSVLQSSGLY SLSSMVTVPSSRWPSSETFTCNVAHPASKTKVDKPVK RENGRVPRPPDCPKCPAPEMLGGPSVFIFPPKPKDTL LIARTPEVTCVVVDLPEDPEVQISWFDGKQMGTAK TQPREEQFNGTYRVVSVLPIGHQDWLKGKQFTCKVNN KALPSPIERTISKARGQAHQPSVYVLPSPREELSKNT VSLTCLIKDFPPPDIDVEWQSNQQEPESKYRTTPQ LDEGSGYFLYSKLSVDKSRWQRGDTFICAVMHEALHN HYTQESLSHSPGK	Caninized variable heavy chain of Mab A v3 and canine IgG-B (Caninized A HC v3 IgG-B)
38	EVQLVESGGDLVKPGGTLRLSCTVSGFSLSSYHVHWV RQPPGKGLEWLGVMWMDGDTSYNLAVKGRFTISRDN KNTLYFQMNSLRAEDTATYYCARPELPGITYGVWFPY WGQGTLLTVSSASTTAPSVFPLAPSCGTSSTGVALA CLVSGYFPEPVTVSWNSGSLTSGVHTFPSVLQSSGLY SLSSMVTVPSSRWPSSETFTCNVAHPASKTKVDKPVK RENGRVPRPPDCPKCPAPEMLGGPSVFIFPPKPKDTL LIARTPEVTCVVVDLPEDPEVQISWFDGKQMGTAK TQPREEQFNGTYRVVSVLPIGHQDWLKGKQFTCKVNN KALPSPIERTISKARGQAHQPSVYVLPSPREELSKNT VSLTCLIKDFPPPDIDVEWQSNQQEPESKYRTTPQ LDEGSGYFLYSKLSVDKSRWQRGDTFICAVMHEALHN HYTQESLSHSPGK	Caninized variable heavy chain of Mab A v4 and canine IgG-B (Caninized A HC v4 IgG-B)
39	EIVMTQSPASLSLSQEEKVTITCKASQNVDSNVDWYQ KQPGQAPKLLIYKASNRNTGVPSTRFSGSGSDTDFST ISSLEPEDVAVYYCMQSTSYPLTFGQGTKEIKRND QPAVYLFQPSPDQLHTGSASVVCLLNSFYPKDINVKW KVDGVIQDTGIQESVTEQDKDSTYSLSTLTMSSTEY LSHELYSCEITHKSLPSTLIKSFQRSECQRVD	Caninized variable light chain of Mab A and canine κ light constant region (Caninized A LC κ)
40	QLTLRESGPGLVKPSQSLSLTCTVSGFSLSSYHVHWI RQRPRGLEWLGVMWMDGDTSYAFQGRISITADTAQN QFSLQLSSMTTDDTAVYYCARPELPGITYGVWFPYWG QGALVTVS	Felinized variable heavy chain of Mab A (Felinized A HC)
41	AITMTQSPGSLAGSPGQVMTMCKASQNVDSNVD WYQKPGQHPKLLIYKASNRNTGVPDRFSGSGSG TDFTLTISNLAQEDVASYYCMQSTSYPLTFGQGT KLEIK	Felinized variable light chain of Mab A (Felinized A LC)
42	GFSLTSGVGS	Variable heavy chain CDR-H1 of Mab B
43	TMWMDGTD	Variable heavy chain CDR-H2 of Mab B
44	SQLPGYNLRGWVY	Variable heavy chain CDR-H3 of Mab B

TABLE 1-continued

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
45	QVQLKESGPGLVQPSQTLSTCTVS	Variable heavy chain framework region HC-FR1 of Mab B
46	QVQLKESGPGLVQPSSETLSLTCTVS	Variable heavy chain framework region HC-FR1 of Mab B variant 2 (v2) Q16E
47	WVRQPPGKGLEWIG	Variable region heavy chain framework region HC-FR2 of Mab B
48	WVRQPPGKGLEWMG	Variable region heavy chain framework region HC-FR2 of Mab B variant 2 (v2) I13M
49	YHSALRSRLSISRDSKSVLLKMNSLQTEDTAMYFC AR	Variable heavy chain framework region HC-FR3 of Mab B
50	YNSALRSKLSISRDTSKSVFLKMNSLQTEDTAIYYC AR	Variable region heavy chain framework region HC-FR3 of Mab B variant 2 (v2) H2N R8K S15T L21F M34I F36Y
51	WGQGTLLVIVS	Variable heavy chain framework region HC-FR4 of Mab B
52	KASHNINKNLE	Variable light chain CDR-L1 of Mab B
53	YANNLQT	Variable light chain CDR-L2 of Mab B
54	YQYNSGHTF	Variable light chain CDR-L3 of Mab B
55	DIVMTQTPSLLSASVGDRVTLNC	Variable light chain framework region LC-FR1 of Mab B
56	DIQMTQSPPVLSASVGDRVTLSC	Variable light chain framework region LC-FR1 of Mab B variant 2 (v2) V3Q T7S S9P L10V N22S
57	WYQQKLGEPKLLIY	Variable light chain framework region LC-FR2 of Mab B
58	GISSRFSGSGSDYTLTISSLQPEDVATYYC	Variable light chain framework region LC-FR3 of Mab B
59	GIPSRFSGSGSDYTLTISSLQPEDVATYYC	Variable light chain framework region LC-FR3 of Mab B variant 2 (v2) S3P
60	GAGTKLELK	Variable light chain framework region LC-FR4 of Mab B

TABLE 1-continued

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
61	GAGTKLEIK	Variable light chain framework region LC-FR4 of Mab B variant 2 (v2) L8I
62	QVQLKESGPGLVQPSQTLSTCTVSGFSLTSYGVS RWPPGKGLEWIGTMWNGDGDYHSALRSRLSISRDSS KSQVLLKMNSLQTEDTAMYFCARSQPLPGYNLRGWFVY WGQGTLLVIVS	Variable heavy chain of Mab B (Mab B HC)
63	QVQLKESGPGLVQPSQTLSTCTVSGFSLTSYGVS RWPPGKGLEWIGTMWNGDGDYHSALRSRLSISRDSS KSQVLLKMNSLQTEDTAMYFCARSQPLPGYNLRGWFVY WGQGTLLVIVS	Variable heavy chain of Mab B variant 2 HC-FR1 Q16E HC-FR2 I13M HC-FR3 H2N HC-FR3 R8K HC-FR3 S15T HC-FR3 L21F HC-FR3 M34I HC-FR3 F36Y (Mab B HC v2)
64	DIVMTQTPSLLSASVGDRLVILNCKASHNINKNLEWYQ QKLGEAPKLLIYYANNLQGTGSSRFSGSGSGTDYTLT ISSLPQEDVATYYCYQYNSGHTFGAGTKLELK	Variable light chain of Mab B (Mab B LC)
65	DIQMTQSPPVLSASVGDRLVILNCKASHNINKNLEWYQ QKLGEAPKLLIYYANNLQGTGIPSRFSGSGSGTDYTLT ISSLPQEDVATYYCYQYNSGHTFGAGTKLEIK	Variable light chain of Mab B variant 2 LC-FR1 V3Q LC-FR1 T7S LC-FR1 S9P LC-FR1 L10V LC-FR1 N22S LC-FR3 S3P LC-FR4 L8I (Mab B LC v2)
66	QVQLKESGPGLVQPSQTLSTCTVSGFSLTSYGVS RWPPGKGLEWIGTMWNGDGDYHSALRSRLSISRDSS KSQVLLKMNSLQTEDTAMYFCARSQPLPGYNLRGWFVY WGQGTLLVIVSASTTAPSVFPLAPSCGSTSGSTVALAC LVSGYFPEPVTVSWNSGSLTSGVHTFPPSVLQSSGLYS LSSMVTVPSSRWPSSETFTCNVAHPASKTKVDKPVPKR ENGRVPRPPDCPKCPAPEMLGGPSVFIAPPKPKDTLL IARTPEVTCVVVDLDPEDPEVQISWFDGKQMGTAKT QPREEQFNGTYRVVSVLPIGHQDWLKGKQFTCKVNNK ALPSPPIERTISKARGQAHQPSVYVLPSSREELSKNTV SLTCLIKDFPPPIDVWQSNQGEPESESKYRTTPPQL DEDGSYFLYSKLSVDKSRWQRGDTFICAVMHEALHNN YTQESLSHSPGK	Chimeric variable heavy chain of Mab B and canine IgG-B (Chimeric B HC IgG-B)
67	QVQLKESGPGLVQPSQTLSTCTVSGFSLTSYGVS RWPPGKGLEWIGTMWNGDGDYHSALRSRLSISRDSS KSQVLLKMNSLQTEDTAMYFCARSQPLPGYNLRGWFVY WGQGTLLVIVSASTTAPSVFPLAPSCGSTSGSTVALA CLVSGYFPEPVTVSWNSGSLTSGVHTFPPSVLQSSGLY SLSSMVTVPSSRWPSSETFTCNVAHPASKTKVDKPVPK RENGRVPRPPDCPKCPAPEMLGGPSVFIAPPKPKDTL IARTPEVTCVVVDLDPEDPEVQISWFDGKQMGTAK TQPREEQFNGTYRVVSVLPIGHQDWLKGKQFTCKVNN KALPSPPIERTISKARGQAHQPSVYVLPSSREELSKNT VSLTCLIKDFPPPIDVWQSNQGEPESESKYRTTPPQ LDEDGSYFLYSKLSVDKSRWQRGDTFICAVMHEALHNN HYTQESLSHSPGK	Chimeric variable heavy chain of Mab B v2 and canine IgG-B HC-FR1 Q16E HC-FR2 I13M HC-FR3 H2N HC-FR3 R8K HC-FR3 S15T HC-FR3 L21F HC-FR3 M34I HC-FR3 F36Y (Chimeric B HC v2 IgG-B)
68	DIVMTQTPSLLSASVGDRLVILNCKASHNINKNLEWYQ QKLGEAPKLLIYYANNLQGTGSSRFSGSGSGTDYTLT ISSLPQEDVATYYCYQYNSGHTFGAGTKLELKRNDQA PAVYLQFQSPDQLHTGSASVCLLNSFYPKDINVKKK VDGVIQDTGIQESVTEQDKDSTYLSLSTLTMSSTEYL SHELVSCEITHKSLPSTLIKSFQRSECQRVD	Chimeric variable light chain of Mab B and canine κ light constant region (Chimeric B LC κ)
69	DIQMTQSPPVLSASVGDRLVILNCKASHNINKNLEWYQ QKLGEAPKLLIYYANNLQGTGIPSRFSGSGSGTDYTLT ISSLPQEDVATYYCYQYNSGHTFGAGTKLEIKRNDQA	Chimeric variable light chain of Mab B v2 and canine κ light constant

TABLE 1-continued

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
	PAVYLFQPSPDQLHTGSASVCLLNSFYPKDINVKWK VDGVIQDTGIESVTEQDKDSTYLSSTLTMSSTEYL SHELYSCEITHKSLPSTLIKSFQRSECQRVD	region LC-FR1 V3Q LC-FR1 T7S LC-FR1 S9P LC-FR1 L10V LC-FR1 N22S LC-FR3 S3P LC-FR4 L8I (Chimeric B LC v2 κ)
70	MAVLGLLLCLVTFPSCVLSQVQLKESGPGLVQPSQTL SLTCTVSGFSLTSYGVSQVVRQPPGKGLEWIGTMWNDG DTDYHSALRSRLSISRDSKQVLLKMNSLQTEDTAM YFCARSQLPGYNLRGWFVYWGQGLTVIVSASTTAPSV FPLAPSCGSTSGSTVALACLVSGYFPEPVTVSWNSGS LTSGVHTFPVSVLQSSGLYSLSSMVTVPSSRWPSSETFT CNVAHPASKTKVDKPVPKRENGRVPRPPDCPKCPAPE MLGGPSVFIFFPKPKDTLLIARTPEVTCVVVDLDPED PEVQISWFDGKQMQTAKTQPREEQFNGTYRVVSVLP IGHQDWLKGKQFTCKVNNKALPSPIERTISKARGQAH QPSVYVLPSPREELSKNIVSLICLIKDFPPPDIDVEW QSNQGQEPESKYRTTPQLDEDEGSYFLYSKLSVDKSR WQRGDTFICAVMHEALHNHYTQESLSHSPGK	Chimeric variable heavy chain of Mab B and canine IgG-B with leader sequence (Canine chimeric B HC with leader)
71	MAVLGLLLCLVTFPSCVLSQVQLKESGPGLVQPSSETL SLTCTVSGFSLTSYGVSQVVRQPPGKGLEWIGTMWNDG DTDYNSALRSKLSISRDSKQVFLKMNSLQTEDTAI YYCARSQLPGYNLRGWFVYWGQGLTVIVSAASTTAPS VFPLAPSCGSTSGSTVALACLVSGYFPEPVTVSWNSG SLTSGVHTFPVSVLQSSGLYSLSSMVTVPSSRWPSSETF TCNVAHPASKTKVDKPVPKRENGRVPRPPDCPKCPAP EMLGGPSVFIFFPKPKDTLLIARTPEVTCVVVDLDPED DPEVQISWFDGKQMQTAKTQPREEQFNGTYRVVSVL PIGHQDWLKGKQFTCKVNNKALPSPIERTISKARGQA HQPSVYVLPSPREELSKNIVSLICLIKDFPPPDIDVE WQSNQGQEPESKYRTTPQLDEDEGSYFLYSKLSVDKS RWQRGDTFICAVMHEALHNHYTQESLSHSPGK	Chimeric variable heavy chain of Mab B v2 and canine IgG-B with leader sequence HC-FR1 Q16E HC-FR2 I13M HC-FR3 H2N HC-FR3 R8K HC-FR3 S15T HC-FR3 L21F HC-FR3 M34I HC-FR3 F36Y (Canine chimeric B HC v2 IgG-B with leader)
72	METDTLLLVWVLLWVPGSTGDIQMTQTPSLLSASVGD RVTNLCKASHNINKNLEWYQKLGAPKLLIYYANNL QTGISRFRSGSGSGTDYTLTISLQPEDVATYYCYQY NSGHFTFGAGTKLEIKRNDQAQPAVYLFQPSPDQLHIGS ASVVCLLNSFYPKDINVKWKVDGVIQDTGIESVTEQ DKDSTYLSSTLTMSSTEYLSHELYSCEITHKSLPST LIKSFQRSECQRVD	Chimeric variable light chain of Mab B and canine κ light constant region with leader sequence (Canine chimeric B LC κ with leader)
73	METDTLLLVWVLLWVPGSTGDIQMTQSPVLSASVGD RVTLSCKASHNINKNLEWYQKLGAPKLLIYYANNL QTGISRFRSGSGSGTDYTLTISLQPEDVATYYCYQY NSGHFTFGAGTKLEIKRNDQAQPAVYLFQPSPDQLHIGS ASVVCLLNSFYPKDINVKWKVDGVIQDTGIESVTEQ DKDSTYLSSTLTMSSTEYLSHELYSCEITHKSLPST LIKSFQRSECQRVD	Chimeric variable light chain of Mab B v2 and canine κ light constant region with leader sequence LC-FR1 V3Q LC-FR1 T7S LC-FR1 S9P LC-FR1 L10V LC-FR1 N22S LC-FR3 S3P LC-FR4 L8I (Canine chimeric B LC v2 κ with leader)
74	QVQLKESGPGLVQPSQTLSTCTVSGFSLTSYGVSQV RQPPGKGLEWIGTMWNDGDTDYHSALRSRLSISRDS KSQVLLKMNSLQTEDTAMFYFCARSQLPGYNLRGWFVY WGQGLTVIVSASTTAPSVFPLAPSCGSTSGATVALAC LVLYGFPEPVTVSWNSGALTSQVHTFPVAVLQASGLYS LSSMVTVPSSRWLSDTFTCNVAHPSPNTKVDKTVRKT DHPPGPKPCDCKPCPPPEMLGGPSIFIFPPKPKDTLS ISRTPEVTCVVDLGPDDSDVQITWFDVNTQVYTAKT SPREEQFNSTYRVVSVLPILHQDWLKGKEFKCKVNSK SLPSPIERTISKAKGPHEPQVYVLPQAQEELSRNKV SVTCLIKSFHPPDIAVEWEITGQPEPENNYRTTPPQL DSDGTIFYVYSKLSVDRSHWQRGNTYTCVSVSHEALHSH HTQKSLIQSPGK	Chimeric variable heavy chain of Mab B and feline IgG-1 (Feline chimeric B HC IgG- 1)

TABLE 1-continued

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
75	DIVMTQTPSLLSASVGDRLVILNCKASHNINKNLEWYQ QKLGEAPKLLIYYANNLQTGISSRFSGSGSGTDYTLT ISSLQPEDVATYYCYQYNSGHTFGAGTKLELKRSDAQ PSVFLFQPSLDELHTGSASIVCILNDFYPKEVNVKWK VDGVVQNKGIQESTTEQNSKDYSLSSSTLTMSSTEY QSHEKFSCEVTHKSLASTLVKSFNRSECQRE	Chimeric variable light chain of Mab B and feline κ light constant region (Feline chimeric B LC κ)
76	DIVMTQTPSLLSASVGDRLVILNCKASHNINKNLEWYQ QKLGEAPKLLIYYANNLQTGISSRFSGSGSGTDYTLT ISSLQPEDVATYYCYQYNSGHTFGAGTKLELKRSDAQ PSVFLFQPSLDELHTGSASIVCILNDFYPKEVNVKWK VDGVVQNKGIQESTTEQNSKDYSLSSSTLTMSSTEY QSHEKFSCEVTHKSLASTLVKSFQRSECQRE	Chimeric variable light chain of Mab B and feline κ light constant region with no N-linked glycosylation site (Feline chimeric B LC κ aglycos)
88	EVQLVESGGDLVKKPGGTLRLSCTVSIGFSLTSYGVS WVRSQSPGKLEWIGTMWNGDGDYHSAVKGQLSISRDT SKQVFLQMNSLRAEDTAMYYCARSQLPGYNLRGWFVY WGQGTLLTVSS	Caninized variable heavy chain of Mab B v3
77	MAVLGLLLCLVTFPSCVLSEVQLVESGGDLVKKPGGTL RLSCTVSIGFSLTSYGVS WVRSQSPGKLEWIGTMWNGD DTDYHSAVKGQLSISRDTSKQVFLQMNSLRAEDTAM YYCARSQLPGYNLRGWFVYWGQGTLLTVSSASTTAPS VFPLAPSCGSTSGSTVALACLVSGYFPEPVTVSWSG SLTSGVHTFPPSVLQSSGLYSLSMVTVPSSRWPS ETFTCNVAHPASKTKVDKPVKRENGRVPRPDCPKCPAP EMLGGPSVFI FPPKPKDILLIARTPEVICVVDLDPE DPEVQISWFDGKQMQTAKTQPREEQFNGTYRVVSVL PIGHQDWLKGKQFTCKVNNKALPSPIERTISKARGQA HQPSVYVLPSSREELSKNIVSLICLIKDFPPPDIDVE WQSNQGQEPESKYRTPPQLDEDDGSYFLYSKLSVDKS RWQRGDTFICAVMHEALHNHYTQESLSHSPGK	Caninized variable heavy chain of Mab B v3 and canine IgG-B with leader sequence (Caninized B HC v3 IgG-B with leader)
78	METDTLLLVLLWVPGSTGEIVLTQSPASLAVSLGQ RATISCKASHNINKNLEWYQKPGQPPKLLIYYANNL QTGVPARFSGSGSGTDYSLNIHPMEEDDTAMYYCYQY NSGHTFGGGTKLEIKRNDAPAVYLFQPSPDQLHTGS ASVVCLLNSFYPKDINVKWKVDGVIQDTGIQESVTEQ DKDSTYSLSSSTLTMSSTEYLSHELYSCEITHKSLPST LIKSFQRSECQRV	Caninized variable light chain of Mab B v3 and canine κ light constant region with leader sequence (Caninized B LC κ with leader)
79	EVQLVESGGDLVKKPGGTLRLSCTVSIGFSLTSYGVS WVRSQSPGKLEWIGTMWNGDGDYHSAVKGQLSISRDT SKQVFLQMNSLRAEDTAMYYCARSQLPGYNLRGWFVY WGQGTLLTVSSASTTAPSVFPLAPSCGSTSGSTVALA CLVSGYFPEPVTVSWSNGSLTSGVHTFPPSVLQSSGLY SLSSMVTVPSSRWPS ETFTCNVAHPASKTKVDKPVK RENGRVPRPDCPKCPAPEMLGGPSVFI FPPKPKDILLIARTPEVICVVDLDPE DPEVQISWFDGKQMQTAKTQPREEQFNGTYRVVSVL PIGHQDWLKGKQFTCKVNNKALPSPIERTISKARGQA HQPSVYVLPSSREELSKNTVSLTCLIKDFPPPDIDVE WQSNQGQEPESKYRTPPQLDEDDGSYFLYSKLSVDKS RWQRGDTFICAVMHEALHNHYTQESLSHSPGK	Caninized variable heavy chain of Mab B v3 and canine IgG-B (Caninized B HC v3 IgG-B)
89	EIVLTQSPASLAVSLGQRATISCKASHNINKNLEWYQ QKPGQPPKLLIYYANNLQTGVPARFSGSGSGTDYSLN IHPMEEDDTAMYYCYQYNSGHTFGGGTKLEIK	Caninized variable light chain of Mab B v3
80	EIVLTQSPASLAVSLGQRATISCKASHNINKNLEWYQ QKPGQPPKLLIYYANNLQTGVPARFSGSGSGTDYSLN IHPMEEDDTAMYYCYQYNSGHTFGGGTKLEIKRNDAP AVYLFQPSPDQLHTGSASVVCLLNSFYPKDINVKWK VDGVIQDTGIQESVTEQDKDSTYSLSSSTLTMSSTEY LSHELYSCEITHKSLPSTLIKSFQRSECQRV	Caninized variable light chain of Mab B v3 and canine κ light constant region (Caninized B LC v3 κ)
81	GPGLVQPSQTLSTCTVSIGFSLSSYHVHWRQPPGKG LEWLGVMWNGDGTSYNLALNSRLSISRDTSKSQVFFK MSLQTEDTATYYCARPELPGLTYGVWFPYWGQGTLLV TVS	Mab E variable HC
82	ASMSISVGDRLTMNCKASQNVDSNVDWYQKGTGQSPN LLIYKASNRNTGVPDRFTGSGSGTDFTFTISNMQAED LAVYYCMQSTSYPLIFGSGTKLEIKRA	Mab E variable LC

TABLE 1-continued

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
83	GPGLVQPSQTLTSLTCTVSGFSLTSYGVSWVRQPPGKG LEWIGTMWNDGDTDYHSALRSRLSISRSSKSQVLLK MNSLQTEDTAMYFCARSQLPGYNLRGWFVYWGQGTLLV IVS	Mab F variable HC
84	SLLSASVGDVRTLNCKASHNINKNLEWYQOKLGEAPK LLIYYANNLQGTGSSRFSGSGSGTDYTLTISSLQPED VATYYCYQYNSGHTFGAGTKLELKRA	Mab F variable LC
90	MWQLVSSALLLVLSAGTQAADVPKAVVLEPKW NRVLTMDSVTLKCCQGDHLLRDNYTWLHNGRPISN QISTYIIKNASIKNSGEYRCQTDQSKLSDPVQLE VHTGWLLLVQVRLVFQEGELIQLKCHSWKNTPVR NVQYFQNGRGKKFFYNSEYHI PAATSEHNGSYF CRGIIGKKNESSEAVNIIQGSSLPSTSLLSHW PQGGSGSHHHHH	Exemplary canine CD16 with linker, and poly-His
91	PAPEMPLGGPSVFIFFPKPKDTLLIARTPEVTCVV VDLDPEDPEVQISWFDGKQMQTAKTQPREEQFN GTYRVVSVLPIGHQDWLKGKQFTCKVNNKALPSP IERTISKARGQAHQPSVYVLPSPREELS KNTVSL TCLIKDFFPPDIDVEWQSNQGQEPESKYRTTPPQ LDEGGSYFLYSKLSVDKSRWQRGDTFICAVMHEA LHNHYTQESLSHSPGK	Exemplary wild-type canine IgG-B Fc
92	PAPEMPLGGPDVFIFFPKPKDTLLIARTPEVTCVV VDLDPEDPEVQISWFDGKQMQTAKTQPREEQFN GTYRVVSVLPIGHQDWLKGKQFTCKVNNKALPSP IERTISKARGQAHQPSVYVLPSPREELS KNTVSL TCLIKDFFPPDIDVEWQSNQGQEPESKYRTTPPQ LDEGGSYFLYSKLSVDKSRWQRGDTFICAVMHEA LHNHYTQESLSHSPGK	Exemplary variant canine IgG-B Fc CD16 enhancing binding mutant S10D
93	PAPEMPLGGPEVFIFFPKPKDTLLIARTPEVTCVV VDLDPEDPEVQISWFDGKQMQTAKTQPREEQFN GTYRVVSVLPIGHQDWLKGKQFTCKVNNKALPSP IERTISKARGQAHQPSVYVLPSPREELS KNTVSL TCLIKDFFPPDIDVEWQSNQGQEPESKYRTTPPQ LDEGGSYFLYSKLSVDKSRWQRGDTFICAVMHEA LHNHYTQESLSHSPGK	Exemplary variant canine IgG-B Fc CD16 enhancing binding mutant S10E
94	PAPEMPLGGPSVFIFFPKPKDTLLIARTPEVTCVV VDLDPEDPEVQISWFDGKQMQTAKTQPREEQFN GTYRVVSVLPIGHQDWLKGKQFTCKVNNKALPSP DERTISKARGQAHQPSVYVLPSPREELS KNTVSL TCLIKDFFPPDIDVEWQSNQGQEPESKYRTTPPQ LDEGGSYFLYSKLSVDKSRWQRGDTFICAVMHEA LHNHYTQESLSHSPGK	Exemplary variant canine IgG-B Fc CD16 enhancing binding mutant I103D
95	PAPEMPLGGPSVFIFFPKPKDTLLIARTPEVTCVV VDLDPEDPEVQISWFDGKQMQTAKTQPREEQFN GTYRVVSVLPIGHQDWLKGKQFTCKVNNKALPSP ERTISKARGQAHQPSVYVLPSPREELS KNTVSL TCLIKDFFPPDIDVEWQSNQGQEPESKYRTTPPQ LDEGGSYFLYSKLSVDKSRWQRGDTFICAVMHEA LHNHYTQESLSHSPGK	Exemplary variant canine IgG-B Fc CD16 enhancing binding mutant I103E
96	PAPEMPLGGPDVFIFFPKPKDTLLIARTPEVTCVV VDLDPEDPEVQISWFDGKQMQTAKTQPREEQFN GTYRVVSVLPIGHQDWLKGKQFTCKVNNKALPSP DERTISKARGQAHQPSVYVLPSPREELS KNTVSL TCLIKDFFPPDIDVEWQSNQGQEPESKYRTTPPQ LDEGGSYFLYSKLSVDKSRWQRGDTFICAVMHEA LHNHYTQESLSHSPGK	Exemplary variant canine IgG-B Fc CD16 enhancing binding mutant S10D I103D
97	PAPEMPLGGPDVFIFFPKPKDTLLIARTPEVTCVV VDLDPEDPEVQISWFDGKQMQTAKTQPREEQFN GTYRVVSVLPIGHQDWLKGKQFTCKVNNKALPSP ERTISKARGQAHQPSVYVLPSPREELS KNTVSL TCLIKDFFPPDIDVEWQSNQGQEPESKYRTTPPQ LDEGGSYFLYSKLSVDKSRWQRGDTFICAVMHEA LHNHYTQESLSHSPGK	Exemplary variant canine IgG-B Fc CD16 enhancing binding mutant S10D I103E
98	PAPEMPLGGPEVFIFFPKPKDTLLIARTPEVTCVV VDLDPEDPEVQISWFDGKQMQTAKTQPREEQFN	Exemplary variant canine IgG-B Fc

TABLE 1-continued

Description of Certain Sequences		
SEQ ID NO:	SEQUENCE	DESCRIPTION
	GTYRVVSVLPIGHQDWLKGKQFTCKVNNKALPSP	CD16 enhancing binding
	DEERTISKARGQAHQPSVYVLPSPREELS KNTVSL	mutant
	TCLIKDFFPPDIDVEWQNGQEPESKYRTTPPQ	S10E
	LDEGGSYFLYSKLSVDKSRWQRGDTFICAVMHEA	I103D
	LHNHYTQESLSHSPGK	
99	PAPEMLGGPEVFIFPPKPKD TLLIARTPEVTCVV	Exemplary variant canine
	VDLDPEDPEVQISWFDGKQMQTAKTQPREEQFN	IgG-B Fc
	GTYRVVSVLPIGHQDWLKGKQFTCKVNNKALPSP	CD16 enhancing binding
	DEERTISKARGQAHQPSVYVLPSPREELS KNTVSL	mutant
	TCLIKDFFPPDIDVEWQNGQEPESKYRTTPPQ	S10E
	LDEGGSYFLYSKLSVDKSRWQRGDTFICAVMHEA	I103E
	LHNHYTQESLSHSPGK	

DESCRIPTION OF CERTAIN EMBODIMENTS

Antibodies that bind canine parvovirus and/or feline parvovirus are provided. Antibody heavy chains and light chains that are capable of forming antibodies that bind canine parvovirus and/or feline parvovirus are also provided. In addition, antibodies, heavy chains, and light chains comprising one or more particular complementary determining regions (CDRs) are provided. Polynucleotides encoding antibodies to canine parvovirus and/or feline parvovirus are provided. Methods of producing or purifying antibodies to canine parvovirus and/or feline parvovirus are also provided. Methods of providing passive immunity against infection with a canine or feline parvovirus and/or treatment of parvoviral infection using antibodies to canine parvovirus and/or feline parvovirus are provided.

For the convenience of the reader, the following definitions of terms used herein are provided.

As used herein, numerical terms such as Kd are calculated based upon scientific measurements and, thus, are subject to appropriate measurement error. In some instances, a numerical term may include numerical values that are rounded to the nearest significant figure.

As used herein, “a” or “an” means “at least one” or “one or more” unless otherwise specified. As used herein, the term “or” means “and/or” unless specified otherwise. In the context of a multiple dependent claim, the use of “or” when referring back to other claims refers to those claims in the alternative only.

Exemplary Parvovirus Antibodies

Novel antibodies directed against parvovirus are provided, for example antibodies that bind to canine parvovirus and/or feline parvovirus. Parvovirus antibodies provided herein include, but are not limited to, monoclonal antibodies, chimeric antibodies, caninized antibodies, and felinized antibodies.

Also provided herein are amino acid sequences of monoclonal parvovirus antibodies. For example, variable heavy chain CDRs (SEQ ID NOs: 4-6), variable light chain CDRs (SEQ ID NOs: 13-15), variable region heavy chain framework sequences (SEQ ID NOs: 7 and 8-11), and variable region light chain framework sequences (SEQ ID NOs: 16-19) for Mab A are provided. Also provided are variable heavy chain CDRs (SEQ ID NOs: 42-44), variable light chain CDRs (SEQ ID NOs: 52-54), variable region heavy chain framework sequences (SEQ ID NOs: 45, 47, 49, and 51), and variable region light chain framework sequences (SEQ ID NOs: 55, 57, 58, and 60) for Mab B. The amino

acid sequences of the variable heavy chain and variable light chain of Mab A are provided (SEQ ID NOs: 20 and 22, respectively). The amino acid sequences of the variable heavy chain and variable light chain of Mab B are provided (SEQ ID NOs: 62 and 64, respectively).

In addition, the amino acid sequences of the CDRs, framework sequences, variable heavy and light chains of variant Mab A and Mab B antibodies are provided. The variable heavy chain CDRs (SEQ ID NOs 4-6), variable heavy chain framework regions (SEQ ID NOs: 8-10, and 12), the variable heavy chain (SEQ ID NO: 21), the variable light chain CDRs (SEQ ID NOs: 13-15), the variable light chain framework regions (SEQ ID NOs: 16-19), and the variable light chain (SEQ ID NO: 22) of Mab A variant 2 (v2) are provided. The variable heavy chain CDRs (SEQ ID NOs 42-44), variable heavy chain framework regions (SEQ ID NOs: 46, 48, 50, and 51), the variable heavy chain (SEQ ID NO: 63), the variable light chain CDRs (SEQ ID NOs: 52-54), the variable light chain framework regions (SEQ ID NOs: 56, 57, 59, and 61), and the variable light chain (SEQ ID NO: 65) of Mab B variant 2 (v2) are provided.

Also provided herein are chimeric canine, chimeric feline, caninized, and felinized antibodies derived from Mab A, Mab B, Mab A v2, and Mab B v2. In some embodiments, amino acid sequences of caninized and felinized Mab A, Mab B, Mab A v2, and Mab B v2 are provided, such as SEQ ID NOs: 34-41, 88, 77-80, 88, and 89. In some embodiments, amino acid sequences of chimeric antibodies derived from Mab A, Mab B, Mab A v2, and Mab B v2 are provided, such as SEQ ID NOs: 23-25, 28-33, and 66-76.

The term “antibody” herein is used in the broadest sense and encompasses various antibody structures, including but not limited to monoclonal antibodies, polyclonal antibodies, multispecific antibodies (for example, bispecific (such as Bi-specific T-cell engagers) and trispecific antibodies), and antibody fragments (such as Fab, F(ab')₂, ScFv, minibody, diabody, triabody, and tetraabody) so long as they exhibit the desired antigen-binding activity. Canine, feline, and equine species have different varieties (classes) of antibodies that are shared by many mammals.

The term antibody includes, but is not limited to, fragments that are capable of binding to an antigen, such as Fv, single-chain Fv (scFv), Fab, Fab', di-scFv, sdAb (single domain antibody) and (Fab')₂ (including a chemically linked F(ab')₂). Papain digestion of antibodies produces two identical antigen-binding fragments, called “Fab” fragments, each with a single antigen-binding site, and a residual “Fc” fragment, whose name reflects its ability to crystallize

readily. Pepsin treatment yields an F(ab')₂ fragment that has two antigen combining sites and is still capable of cross-linking antigen. The term antibody also includes, but is not limited to, chimeric antibodies, humanized antibodies, and antibodies of various species such as mouse, human, cynomolgus monkey, canine, feline, equine, etc. Furthermore, for all antibody constructs provided herein, variants having the sequences from other organisms are also contemplated. Thus, if a murine version of an antibody is disclosed, one of skill in the art will appreciate how to transform the murine sequence based antibody into a cat, dog, horse, etc. sequence. Antibody fragments also include either orientation of single chain scFvs, tandem di-scFv, diabodies, tandem tri-sdcFv, minibodies, etc. Antibody fragments also include nanobodies (sdAb, an antibody having a single, monomeric domain, such as a pair of variable domains of heavy chains, without a light chain). An antibody fragment can be referred to as being a specific species in some embodiments (for example, mouse scFv or a canine scFv). This denotes the sequences of at least part of the non-CDR regions, rather than the source of the construct. In some embodiments, the antibodies comprise a label or are conjugated to a second moiety.

The terms "label" and "detectable label" mean a moiety attached to an antibody or its analyte to render a reaction (for example, binding) between the members of the specific binding pair, detectable. The labeled member of the specific binding pair is referred to as "detectably labeled." Thus, the term "labeled binding protein" refers to a protein with a label incorporated that provides for the identification of the binding protein. In some embodiments, the label is a detectable marker that can produce a signal that is detectable by visual or instrumental means, for example, incorporation of a radiolabeled amino acid or attachment to a polypeptide of biotinyl moieties that can be detected by marked avidin (for example, streptavidin containing a fluorescent marker or enzymatic activity that can be detected by optical or colorimetric methods). Examples of labels for polypeptides include, but are not limited to, the following: radioisotopes or radionuclides (for example, ³H, ¹⁴C, ³⁵S, ⁹⁰Y, ⁹⁹Tc, ¹¹¹In, ¹²⁵I, ¹³¹I, ¹⁷⁷Lu, ¹⁶⁶Ho, or ¹⁵³Sm); chromogens, fluorescent labels (for example, FITC, rhodamine, lanthanide phosphors), enzymatic labels (for example, horseradish peroxidase, luciferase, alkaline phosphatase); chemiluminescent markers; biotinyl groups; predetermined polypeptide epitopes recognized by a secondary reporter (for example, leucine zipper pair sequences, binding sites for secondary antibodies, metal binding domains, epitope tags); and magnetic agents, such as gadolinium chelates. Representative examples of labels commonly employed for immunoassays include moieties that produce light, for example, acridinium compounds, and moieties that produce fluorescence, for example, fluorescein. In this regard, the moiety itself may not be detectably labeled but may become detectable upon reaction with yet another moiety.

The term "monoclonal antibody" refers to an antibody of a substantially homogeneous population of antibodies, that is, the individual antibodies comprising the population are identical except for possible naturally-occurring mutations that may be present in minor amounts. Monoclonal antibodies are highly specific, being directed against a single antigenic site. Furthermore, in contrast to polyclonal antibody preparations, which typically include different antibodies directed against different determinants (epitopes), each monoclonal antibody is directed against a single determinant on the antigen. Thus, a sample of monoclonal antibodies can bind to the same epitope on the antigen. The

modifier "monoclonal" indicates the character of the antibody as being obtained from a substantially homogeneous population of antibodies, and is not to be construed as requiring production of the antibody by any particular method. For example, the monoclonal antibodies may be made by the hybridoma method first described by Kohler and Milstein, 1975, *Nature* 256:495, or may be made by recombinant DNA methods such as described in U.S. Pat. No. 4,816,567. The monoclonal antibodies may also be isolated from phage libraries generated using the techniques described in McCafferty et al., 1990, *Nature* 348:552-554, for example.

In some embodiments, the monoclonal antibody is Mab A, Mab A v2, Mab B, or Mab B v2.

"Amino acid sequence," means a sequence of amino acid residues in a peptide or protein. The terms "polypeptide" and "protein" are used interchangeably to refer to a polymer of amino acid residues, and are not limited to a minimum length. Such polymers of amino acid residues may contain natural or non-natural amino acid residues, and include, but are not limited to, peptides, oligopeptides, dimers, trimers, and multimers of amino acid residues. Both full-length proteins and fragments thereof are encompassed by the definition. The terms also include post-expression modifications of the polypeptide, for example, glycosylation, sialylation, acetylation, phosphorylation, and the like. Furthermore, for purposes of the present disclosure, a "polypeptide" refers to a protein which includes modifications, such as deletions, additions, and substitutions (generally conservative in nature), to the native sequence, as long as the protein maintains the desired activity. These modifications may be deliberate, as through site-directed mutagenesis, or may be accidental, such as through mutations of hosts which produce the proteins or errors due to PCR amplification.

"Parvovirus" as used herein refers to any naturally occurring parvovirus or parvovirus variant, and includes canine parvovirus (CPV), such as CPV-2a, CPV-2b, and CPV-2c, and feline parvovirus (panleukopenia virus).

As used herein, the term "epitope" refers to a site on a target molecule (for example, an antigen, such as a protein, nucleic acid, carbohydrate or lipid) to which an antigen-binding molecule (for example, an antibody, antibody fragment, or scaffold protein containing antibody binding regions) binds. Epitopes often include a chemically active surface grouping of molecules such as amino acids, polypeptides or sugar side chains and have specific three dimensional structural characteristics as well as specific charge characteristics. Epitopes can be formed both from contiguous or juxtaposed noncontiguous residues (for example, amino acids, nucleotides, sugars, lipid moiety) of the target molecule. Epitopes formed from contiguous residues (for example, amino acids, nucleotides, sugars, lipid moiety) typically are retained on exposure to denaturing solvents whereas epitopes formed by tertiary folding typically are lost on treatment with denaturing solvents. An epitope may include but is not limited to at least 3, at least 5 or 8-10 residues (for example, amino acids or nucleotides). In some examples an epitope is less than 20 residues (for example, amino acids or nucleotides) in length, less than 15 residues or less than 12 residues. Two antibodies may bind the same epitope within an antigen if they exhibit competitive binding for the antigen. In some embodiments, an epitope can be identified by a certain minimal distance to a CDR residue on the antigen-binding molecule. In some embodiments, an epitope can be identified by the above distance, and further limited to those residues involved in a bond (for example, a hydrogen bond) between an antibody residue and an antigen

residue. An epitope can be identified by various scans as well, for example an alanine or arginine scan can indicate one or more residues that the antigen-binding molecule can interact with. Unless explicitly denoted, a set of residues as an epitope does not exclude other residues from being part of the epitope for a particular antibody. Rather, the presence of such a set designates a minimal series (or set of species) of epitopes. Thus, in some embodiments, a set of residues identified as an epitope designates a minimal epitope of relevance for the antigen, rather than an exclusive list of residues for an epitope on an antigen.

In some embodiments, the epitope comprises the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, and/or SEQ ID NO: 3.

The term “CDR” means a complementarity determining region as defined by at least one manner of identification to one of skill in the art. In some embodiments, CDRs can be defined in accordance with any of the Chothia numbering schemes, the Kabat numbering scheme, a combination of Kabat and Chothia, the AbM definition, the contact definition, or a combination of the Kabat, Chothia, AbM, or contact definitions. The various CDRs within an antibody can be designated by their appropriate number and chain type, including, without limitation as CDR-H1, CDR-H2, CDR-H3, CDR-L1, CDR-L2, and CDR-L3. The term “CDR” is used herein to also encompass a “hypervariable region” or HVR, including hypervariable loops.

In some embodiments, a parvovirus antibody comprises a heavy chain comprising (a) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 4; (b) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 5; or (c) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 6. In some embodiments, a parvovirus antibody comprises a light chain comprising (a) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 13; (b) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 14; or (c) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 15.

In some embodiments, an antibody comprises a heavy chain comprising (a) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 42; (b) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 43; or (c) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 44. In some embodiments, a parvovirus antibody comprises a light chain comprising (a) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 52; (b) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 53; or (c) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 54.

In some embodiments, a parvovirus antibody comprises a heavy chain comprising (a) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 4 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-H1 is substituted by a different amino acid; (b) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 5 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-H2 is substituted by a different amino acid; and/or (c) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 6 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-H3 is substituted by a different amino acid. In some embodiments, a parvovirus antibody comprises a light chain comprising (a) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 13 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-L1 is substituted by a different amino acid; (b) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 14 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-L2 is substituted by a different amino acid;

and/or (c) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 15 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-L3 is substituted by a different amino acid.

In some embodiments, an antibody comprises a heavy chain comprising (a) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 42 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-H1 is substituted by a different amino acid; (b) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 43 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-H2 is substituted by a different amino acid; and/or (c) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 44 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-H3 is substituted by a different amino acid. In some embodiments, a parvovirus antibody comprises a light chain comprising (a) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 52 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-L1 is substituted by a different amino acid; (b) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 53 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-L2 is substituted by a different amino acid; and/or (c) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 54 or a variant thereof wherein 1, 2, or 3 amino acids of the CDR-L3 is substituted by a different amino acid.

The term “variable region” as used herein refers to a region comprising at least three CDRs. In some embodiments, the variable region includes the three CDRs and at least one framework region (“FR”). The terms “heavy chain variable region” or “variable heavy chain” are used interchangeably to refer to a region comprising at least three heavy chain CDRs. The terms “light chain variable region” or “variable light chain” are used interchangeably to refer to a region comprising at least three light chain CDRs. In some embodiments, the variable heavy chain or variable light chain comprises at least one framework region. In some embodiments, an antibody comprises at least one heavy chain framework region selected from HC-FR1, HC-FR2, HC-FR3, and HC-FR4. In some embodiments, an antibody comprises at least one light chain framework region selected from LC-FR1, LC-FR2, LC-FR3, and LC-FR4. The framework regions may be juxtaposed between light chain CDRs or between heavy chain CDRs. For example, an antibody may comprise a variable heavy chain having the following structure: (HC-FR1)-(CDR-H1)-(HC-FR2)-(CDR-H2)-(HC-FR3)-(CDR-H3)-(HC-FR4). An antibody may comprise a variable heavy chain having the following structure: (CDR-H1)-(HC-FR2)-(CDR-H2)-(HC-FR3)-(CDR-H3). An antibody may also comprise a variable light chain having the following structure: (LC-FR1)-(CDR-L1)-(LC-FR2)-(CDR-L2)-(LC-FR3)-(CDR-L3)-(LC-FR4). An antibody may also comprise a variable light chain having the following structure: (CDR-L1)-(LC-FR2)-(CDR-L2)-(LC-FR3)-(CDR-L3).

In some embodiments, a parvovirus antibody comprises one or more of (a) a variable region heavy chain framework 1 (HC-FR1) sequence of SEQ ID NO: 7 or SEQ ID NO: 8, (b) a HC-FR2 sequence of SEQ ID NO: 9, (c) a HC-FR3 sequence of SEQ ID NO: 10, (d) a HC-FR4 sequence of SEQ ID NO: 11 or SEQ ID NO: 12, (e) a variable region light chain framework 1 (LC-FR1) sequence of SEQ ID NO: 16, (f) an LC-FR2 sequence of SEQ ID NO: 17, (g) an LC-FR3 sequence of SEQ ID NO: 18, or (h) an LC-FR4 sequence of SEQ ID NO: 19.

In some embodiments, a parvovirus antibody comprises one or more of (a) a variable region heavy chain framework 1 (HC-FR1) sequence of SEQ ID NO: 45 or SEQ ID NO: 46,

(b) a HC-FR2 sequence of SEQ ID NO: 47 or SEQ ID NO: 48, (c) a HC-FR3 sequence of SEQ ID NO: 49 or SEQ ID NO: 50, (d) a HC-FR4 sequence of SEQ ID NO: 51, (e) a variable region light chain framework 1 (LC-FR1) sequence of SEQ ID NO: 55 or SEQ ID NO: 56, (f) an LC-FR2 sequence of SEQ ID NO: 57, (g) an LC-FR3 sequence of SEQ ID NO: 58 or SEQ ID NO: 59, or (h) an LC-FR4 sequence of SEQ ID NO: 60.

In some embodiments, a parvovirus antibody comprises a variable light chain sequence of SEQ ID NO: 22, SEQ ID NO: 87, SEQ ID NO: 41, SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 89, SEQ ID NO: 82, or SEQ ID NO: 84. In some embodiments, a parvovirus antibody comprises a variable heavy chain sequence of SEQ ID NO: 20, SEQ ID NO: 21, SEQ ID NO: 85, SEQ ID NO: 86, SEQ ID NO: 40, SEQ ID NO: 62, SEQ ID NO: 63, SEQ ID NO: 88, SEQ ID NO: 81, or SEQ ID NO: 83.

In some embodiments, a parvovirus antibody comprises a variable heavy chain sequence of SEQ ID NO: 20 and a variable light chain sequence of SEQ ID NO: 22. In some embodiments, a parvovirus antibody comprises a variable heavy chain sequence of SEQ ID NO: 21 and a variable light chain sequence of SEQ ID NO: 22. In some embodiments, a parvovirus antibody comprises a variable heavy chain sequence of SEQ ID NO: 85 and a variable light chain sequence of SEQ ID NO: 87. In some embodiments, a parvovirus antibody comprises a variable heavy chain sequence of SEQ ID NO: 86 and a variable light chain sequence of SEQ ID NO: 87. In some embodiments, a parvovirus antibody comprises a variable heavy chain sequence of SEQ ID NO: 40 and a variable light chain sequence of SEQ ID NO: 41. In some embodiments, a parvovirus antibody comprises a variable heavy chain sequence of SEQ ID NO: 62 and a variable light chain sequence of SEQ ID NO: 64 or SEQ ID NO: 65. In some embodiments, a parvovirus antibody comprises a variable heavy chain sequence of SEQ ID NO: 63 and a variable light chain sequence of SEQ ID NO: 64 or SEQ ID NO: 65. In some embodiments, a parvovirus antibody comprises a variable heavy chain sequence of SEQ ID NO: 88 and a variable light chain sequence of SEQ ID NO: 89.

The term "constant region" as used herein refers to a region comprising at least three constant domains. The terms "heavy chain constant region" or "constant heavy chain" are used interchangeably to refer to a region comprising at least three heavy chain constant domains, CH1, CH2, and CH3. Nonlimiting exemplary heavy chain constant regions include γ , δ , α , ϵ , and μ . Each heavy chain constant region corresponds to an antibody isotype. For example, an antibody comprising a γ constant region is an IgG antibody, an antibody comprising a δ constant region is an IgD antibody, an antibody comprising an α constant region is an IgA antibody, an antibody comprising a μ constant region is an IgM antibody, and an antibody comprising an ϵ constant region is an IgE antibody. Certain isotypes can be further subdivided into subclasses. For example, IgG antibodies include, but are not limited to, IgG1 (comprising a γ_1 constant region), IgG2 (comprising a γ_2 constant region), IgG3 (comprising a γ_3 constant region), and IgG4 (comprising a γ_4 constant region) antibodies; IgA antibodies include, but are not limited to, IgA1 (comprising an α_1 constant region) and IgA2 (comprising an α_2 constant region) antibodies; and IgM antibodies include, but are not limited to IgM1 and IgM2. The terms "light chain constant region" or "constant light chain" are used interchangeably to refer to a region comprising a light chain constant domain, CL. Non-limiting exemplary light chain constant regions include λ

and κ . Non-function-altering deletions and alterations within the domains are encompassed within the scope of the term "constant region" unless designated otherwise. Canine and feline have antibody classes such as IgG, IgA, IgD, IgE, and IgM. Within the canine IgG antibody class are IgG-A, IgG-B, IgG-C, and IgG-D. Within the feline IgG antibody class are IgG1, IgG2a, and IgG2b.

The term "chimeric antibody" or "chimeric" refers to an antibody in which a portion of the heavy chain or light chain is derived from a particular source or species, while at least a part of the remainder of the heavy chain or light chain is derived from a different source or species. In some embodiments, a chimeric antibody refers to an antibody comprising at least one variable region from a first species (such as mouse, rat, cynomolgus monkey, etc.) and at least one constant region from a second species (such as human, dog, cat, equine, etc.). In some embodiments, a chimeric antibody comprises at least one mouse variable region and at least one canine constant region. In some embodiments, a chimeric antibody comprises at least one mouse variable region and at least one feline constant region. In some embodiments, all of the variable regions of a chimeric antibody are from a first species and all of the constant regions of the chimeric antibody are from a second species. In some embodiments, a chimeric antibody comprises a constant heavy chain region or constant light chain region from a companion animal. In some embodiments, a chimeric antibody comprises a mouse variable heavy and light chains and a companion animal constant heavy and light chains. For example, a chimeric antibody may comprise a mouse variable heavy and light chains and a canine constant heavy and light chains; a chimeric antibody may comprise a mouse variable heavy and light chains and a feline constant heavy and light chains; or a chimeric antibody may comprise a mouse variable heavy and light chains and an equine constant heavy and light chains.

In some embodiments, a parvovirus antibody comprises a chimeric antibody comprising: (a) (i) a heavy chain amino acid sequence of SEQ ID NO: 23; (ii) a light chain amino acid sequence of SEQ ID NO: 25; or (iii) a heavy chain amino acid sequence as in (i) and a light chain sequence as in (ii); (b) (i) a heavy chain amino acid sequence of SEQ ID NO: 24; (ii) a light chain amino acid sequence of SEQ ID NO: 25; or (iii) a heavy chain amino acid sequence as in (i) and a light chain sequence as in (ii); (c) (i) a heavy chain amino acid sequence of SEQ ID NO: 31; (ii) a light chain amino acid sequence of SEQ ID NO: 32; or (iii) a heavy chain amino acid sequence as in (i) and a light chain sequence as in (ii); (d) (i) a heavy chain amino acid sequence of SEQ ID NO: 31; (ii) a light chain amino acid sequence of SEQ ID NO: 33; or (iii) a heavy chain amino acid sequence as in (i) and a light chain sequence as in (ii); (e) (i) a heavy chain amino acid sequence of SEQ ID NO: 66; (ii) a light chain amino acid sequence of SEQ ID NO: 68; or (iii) a heavy chain amino acid sequence as in (i) and a light chain sequence as in (ii); (f) (i) a heavy chain amino acid sequence of SEQ ID NO: 67; (ii) a light chain amino acid sequence of SEQ ID NO: 69; or (iii) a heavy chain amino acid sequence as in (i) and a light chain sequence as in (ii); (g) (i) a heavy chain amino acid sequence of SEQ ID NO: 74; (ii) a light chain amino acid sequence of SEQ ID NO: 75; or (iii) a heavy chain amino acid sequence as in (i) and a light chain sequence as in (ii); or (h) (i) a heavy chain amino acid sequence of SEQ ID NO: 74; (ii) a light chain amino acid sequence of SEQ ID NO: 75; or (iii) a heavy chain amino acid sequence as in (i) and a light chain sequence as in (ii).

A “canine chimeric,” “chimeric canine,” or “canine chimeric antibody” refers to a chimeric antibody having at least a portion of a heavy chain or a portion of a light chain derived from a dog. A “feline chimeric,” “chimeric feline,” or “feline chimeric antibody” refers to a chimeric antibody having at least a portion of a heavy chain or a portion of a light chain derived from a cat. In some embodiments, a canine chimeric antibody comprises a mouse or rat variable heavy and light chains and a canine constant heavy and light chains. In some embodiments, a feline chimeric antibody comprises a mouse or rat variable heavy and light chains and a feline constant heavy and light chains.

In some embodiments, a parvovirus antibody comprises a canine heavy chain constant region selected from an IgG-A, IgG-B, IgG-C, and IgG-D constant region.

In some embodiments, a parvovirus antibody comprises a feline heavy chain constant region selected from an IgG1, IgG2a, and IgG2b constant region.

A “caninized antibody” means an antibody in which at least one amino acid in a portion of a non-canine variable region has been replaced with the corresponding amino acid from a canine variable region. In some embodiments, a caninized antibody comprises at least one canine constant region (e.g., a γ constant region, an α constant region, a δ constant region, an ϵ constant region, a μ constant region, or etc.) or fragment thereof. In some embodiments, a caninized antibody is an antibody fragment, such as Fab, scFv, (Fab')₂, etc. The term “caninized” also denotes forms of non-canine (for example, murine) antibodies that are chimeric immunoglobulins, immunoglobulin chains, or fragments thereof (such as Fv, Fab, Fab', F(ab')₂ or other antigen-binding sequences of antibodies) that contain minimal sequence of non-canine immunoglobulin. Caninized antibodies can include canine immunoglobulins (recipient antibody) in which residues from a CDR of the recipient are substituted by residues from a CDR of a non-canine species (donor antibody) such as mouse, rat, or rabbit having the desired specificity, affinity, and capacity. In some instances, Fv framework region (FR) residues of the canine immunoglobulin are replaced by corresponding non-canine residues. Furthermore, the caninized antibody can comprise residues that are found neither in the recipient antibody nor in the imported CDR or framework sequences, but are included to further refine and optimize antibody performance.

In some embodiments, at least one amino acid residue in a portion of a rat or a mouse variable heavy chain or a rat or a mouse variable light chain has been replaced with the corresponding amino acid from a canine variable region. In some embodiments, the modified chain is fused to a canine constant heavy chain or a canine constant light chain.

A “felinized antibody” means an antibody in which at least one amino acid in a portion of a non-feline variable region has been replaced with the corresponding amino acid from a feline variable region. In some embodiments, a felinized antibody comprises at least one feline constant region (e.g., a γ constant region, an α constant region, a δ constant region, an ϵ constant region, a μ constant region, or etc.) or fragment thereof. In some embodiments, a felinized antibody is an antibody fragment, such as Fab, scFv, (Fab')₂, etc. The term “felinized” also denotes forms of non-feline (for example, murine) antibodies that are chimeric immunoglobulins, immunoglobulin chains, or fragments thereof (such as Fv, Fab, Fab', F(ab')₂ or other antigen-binding sequences of antibodies) that contain minimal sequence of non-feline immunoglobulin. Felinized antibodies can include feline immunoglobulins (recipient antibody) in which residues from a CDR of the recipient are substituted

by residues from a CDR of a non-feline species (donor antibody) such as mouse, rat, or rabbit having the desired specificity, affinity, and capacity. In some instances, Fv framework region (FR) residues of the feline immunoglobulin are replaced by corresponding non-feline residues. Furthermore, the felinized antibody can comprise residues that are found neither in the recipient antibody nor in the imported CDR or framework sequences, but are included to further refine and optimize antibody performance.

In some embodiments, at least one amino acid residue in a portion of a mouse variable heavy chain or a mouse variable light chain has been replaced with the corresponding amino acid from a feline variable region. In some embodiments, the modified chain is fused to a feline constant heavy chain or a canine constant light chain.

A “fragment crystallizable polypeptide” or “Fc polypeptide” is the portion of an antibody molecule that interacts with effector molecules and cells. It comprises the C-terminal portions of the immunoglobulin heavy chains. As used herein, an Fc polypeptide includes fragments of the Fc domain having one or more biological activities of an entire Fc polypeptide. An “effector function” of the Fc polypeptide is an action or activity performed in whole or in part by any antibody in response to a stimulus and may include complement fixation and/or ADCC (antibody-dependent cellular cytotoxicity) induction and/or ADCP (antibody-dependent cellular phagocytosis).

In some embodiments, a biological activity of an Fc polypeptide is the ability to bind FcRn. In some embodiments, a biological activity of an Fc polypeptide is the ability to bind C1q. In some embodiments, a biological activity of an Fc polypeptide is the ability to bind CD16. In some embodiments, a biological activity of an Fc polypeptide is the ability to bind protein A.

The term “IgX Fc” means the Fc region is derived from a particular antibody isotype (e.g., IgG, IgA, IgD, IgE, IgM, etc.), where “X” denotes the antibody isotype. Thus, “IgG Fc” denotes the Fc region of a γ chain, “IgA Fc” denotes the Fc region of an α chain, “IgD Fc” denotes the Fc region of a δ chain, “IgE Fc” denotes the Fc region of an ϵ chain, “IgM Fc” denotes the Fc region of a μ chain, etc. In some embodiments, the IgG Fc region comprises CH1, hinge, CH2, CH3, and CL1. “IgX-N-Fc” denotes that the Fc region is derived from a particular subclass of antibody isotype (such as canine IgG subclass A, B, C, or D; or feline IgG subclass 1, 2a, or 2b), where “N” denotes the subclass. In some embodiments, IgX Fc or IgX-N-Fc regions are derived from a companion animal, such as a dog or a cat. In some embodiments, IgG Fc regions are isolated from canine γ heavy chains, such as IgG-A, IgG-B, IgG-C, or IgG-D. In some instances, IgG Fc regions are isolated from feline γ heavy chains, such as IgG1, IgG2a, or IgG2b. Antibodies comprising an Fc region of IgG-A, IgG-B, IgG-C, or IgG-D may provide for higher expression levels in recombination production systems.

The terms “IgX Fc” and “IgX Fc polypeptide” include wild-type IgX Fc polypeptides and variant IgX Fc polypeptides, unless indicated otherwise.

In some embodiments, a variant IgG Fc polypeptide comprises a variant IgG Fc polypeptide of a companion animal species. In some embodiments, a variant IgG Fc polypeptide comprises a variant canine IgG Fc polypeptide or a feline IgG Fc polypeptide. In some embodiments, a variant IgG Fc polypeptide (e.g., a variant canine IgG-A Fc polypeptide, a variant canine IgG-C Fc polypeptide, or a variant canine IgG-D Fc polypeptide, variant feline IgG1a Fc polypeptide, variant feline IgG1b Fc polypeptide, or

variant feline IgG2 Fc polypeptide) has an activity that the reference (e.g., wild-type) polypeptide substantially lacks.

An antibody may be modified to extend or shorten its half-life. In some embodiments involving a higher dose of antibody, a shorter half-life may be desirable for acute treatment. In some embodiments involving a lower dose of antibody, a longer half-life may be desirable for prolonged treatment. For example, as discussed below, mutations in IgG Fc that affect FcRn interactions may be introduced.

In some embodiments, a parvovirus antibody comprises a wild-type or variant IgG Fc having complement fixation activity (or complement-dependent cytotoxicity (CDC)). In some embodiments, a parvovirus antibody comprises a wild-type or variant IgG Fc having antibody-dependent cellular cytotoxicity (ADCC) activity. In some embodiments, a parvovirus antibody comprises a wild-type or variant IgG Fc having antibody-dependent cellular phagocytosis (ADCP) activity. In some embodiments, a parvovirus antibody comprises a wild-type or variant IgG Fc having complement fixation activity and/or ADCC activity and/or ADCP activity. IgG Fc polypeptides may be modified to have an effector function or to have an enhanced effector function.

In some embodiments, a parvovirus antibody comprises a wild-type or variant IgG Fc that binds to canine FcRn at low pH. In some embodiments, a parvovirus comprises a wild-type or variant IgG Fc that binds to C1q. In some embodiments, a parvovirus comprises a wild-type or variant IgG Fc that binds to CD16. In some embodiments, a parvovirus comprises a variant IgG Fc comprising one or more afucosylated glycan.

In some embodiments, a variant IgG Fc (e.g., a variant canine IgG Fc polypeptide or a variant feline IgG Fc polypeptide) has modified FcRn binding affinity compared to a reference polypeptide. In some embodiments, a variant IgG Fc has increased FcRn binding affinity at an acidic pH (e.g., at a pH in the range of from about 5.0 to about 6.5, such as at a pH of about 5.0, a pH of about 5.5, a pH of about 6.0, or a pH of about 6.5) compared to a reference polypeptide. Exemplary variant IgG Fc polypeptides having increased FcRn binding affinity are disclosed in WO 2020/082048, which is incorporated by reference herein in its entirety.

In some embodiments, a variant IgG Fc (e.g., a variant canine IgG Fc polypeptide or a variant feline IgG Fc polypeptide) has modified C1q binding affinity compared to a reference polypeptide. In some embodiments, a variant IgG Fc has increased C1q binding affinity compared to a reference polypeptide. Exemplary variant IgG Fc polypeptides having increased C1q binding affinity are disclosed in WO 2020/139984 (e.g., Example 2), which is incorporated by reference herein in its entirety.

In some embodiments, a variant IgG Fc (e.g., a variant canine IgG Fc polypeptide or a variant feline IgG Fc polypeptide) has modified CD16 binding affinity compared to a reference polypeptide. In some embodiments, a variant IgG Fc has increased CD16 binding affinity compared to a reference polypeptide. Exemplary variant IgG Fc polypeptides having increased CD16 binding affinity are disclosed in WO 2020/139984 (e.g., Example 2), which is incorporated by reference herein in its entirety.

In some embodiments, a variant canine IgG Fc has enhanced CD16 binding affinity compared to a reference polypeptide. In some embodiments a variant IgG Fc comprises a) an aspartic acid or a glutamic acid at a position corresponding to position 10 of SEQ ID NO: 91; b) an aspartic acid or a glutamic acid at position 10 of SEQ ID

NO: 91; c) an aspartic acid or a glutamic acid at a position corresponding to position 103 of SEQ ID NO: 91; d) an aspartic acid or a glutamic acid at position 103 of SEQ ID NO: 91; e) an aspartic acid or a glutamic acid at a position corresponding to position 10 and/or position 103 of SEQ ID NO: 91; f) an aspartic acid or a glutamic acid at position 10 and/or position 103 of SEQ ID NO: 91. In some embodiments a variant IgG Fc comprises the amino acid sequence of SEQ ID NO: 92, SEQ ID NO: 93, SEQ ID NO: 94, SEQ ID NO: 95, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 98, or SEQ ID NO: 99.

In some embodiments, a variant IgG Fc (e.g., a variant canine IgG Fc polypeptide or a variant feline IgG Fc polypeptide) has modified Protein A binding affinity compared to a reference polypeptide. In some embodiments, a variant IgG Fc has increased Protein A binding affinity compared to a reference polypeptide. Exemplary variant IgG Fc polypeptides having increased Protein A binding affinity are disclosed in WO 2020/139984 (e.g., Example 2), which is incorporated by reference herein in its entirety.

The term "affinity" means the strength of the sum total of noncovalent interactions between a single binding site of a molecule (for example, an antibody) and its binding partner (for example, an antigen). The affinity of a molecule X for its partner Y can generally be represented by the dissociation constant (K_D). Affinity can be measured by common methods known in the art, such as, for example, immunoblot, ELISA KD, KinEx A, biolayer interferometry (BLI), or surface plasmon resonance devices.

The terms " K_D ," " K_d ," "Kd" or "Kd value" as used interchangeably to refer to the equilibrium dissociation constant of an antibody-antigen interaction. In some embodiments, the K_d of the antibody is measured by using biolayer interferometry assays using a biosensor, such as an Octet® System (Pall ForteBio LLC, Fremont, CA) according to the supplier's instructions. Briefly, biotinylated antigen is bound to the sensor tip and the association of antibody is monitored for ninety seconds and the dissociation is monitored for 600 seconds. The buffer for dilutions and binding steps is 20 mM phosphate, 150 mM NaCl, pH 7.2. A buffer only blank curve is subtracted to correct for any drift. The data are fit to a 2:1 binding model using ForteBio data analysis software to determine association rate constant (k_{on}), dissociation rate constant (k_{off}), and the K_d . The equilibrium dissociation constant (K_d) is calculated as the ratio of k_{off}/k_{on} . The term " k_{on} " refers to the rate constant for association of an antibody to an antigen and the term " k_{off} " refers to the rate constant for dissociation of an antibody from the antibody/antigen complex.

The term "binds" to an antigen or epitope is a term that is well understood in the art, and methods to determine such binding are also well known in the art. A molecule is said to exhibit "binding" if it reacts, associates with, or has affinity for a particular cell or substance and the reaction, association, or affinity is detectable by one or more methods known in the art, such as, for example, immunoblot, ELISA KD, KinEx A, biolayer interferometry (BLI), surface plasmon resonance devices, or etc.

"Surface plasmon resonance" denotes an optical phenomenon that allows for the analysis of real-time biospecific interactions by detection of alterations in protein concentrations within a biosensor matrix, for example using the BiAcCore™ system (BiAcCore International AB, a GE Healthcare company, Uppsala, Sweden and Piscataway, N.J.). For further descriptions, see Jonsson et al. (1993) *Ann. Biol. Clin.* 51: 19-26.

“Biolayer interferometry” refers to an optical analytical technique that analyzes the interference pattern of light reflected from a layer of immobilized protein on a biosensor tip and an internal reference layer. Changes in the number of molecules bound to the biosensor tip cause shifts in the interference pattern that can be measured in real-time. A nonlimiting exemplary device for biolayer interferometry is an Octet® system (Pall ForteBio LLC). See, e.g., Abdiche et al., 2008, *Anal. Biochem.* 377: 209-277.

In some embodiments, a parvovirus antibody binds to a canine parvovirus or a feline parvovirus with a dissociation constant (Kd) of less than 5×10^{-6} M, less than 1×10^{-6} M, less than 5×10^{-7} M, less than 1×10^{-7} M, less than 5×10^{-8} M, less than 1×10^{-8} M, less than 5×10^{-9} M, less than 1×10^{-9} M, less than 5×10^{-10} M, less than 1×10^{-10} M, less than 5×10^{-11} M, less than 1×10^{-11} M, or less than 1×10^{-12} M, as measured by biolayer interferometry. In some embodiments, a parvovirus antibody binds to a canine parvovirus or a feline parvovirus with a Kd of between 5×10^{-6} M and 1×10^{-6} M, between 5×10^{-6} M and 5×10^{-7} M, between 5×10^{-6} M and 1×10^{-7} M, between 5×10^{-6} M and 5×10^{-8} M, 5×10^{-6} M and 1×10^{-8} M, between 5×10^{-6} M and 5×10^{-9} M, between 5×10^{-6} M and 1×10^{-9} M, between 5×10^{-6} M and 5×10^{-10} M, between 5×10^{-6} M and 1×10^{-10} M, between 5×10^{-6} M and 5×10^{-11} M, between 5×10^{-6} M and 1×10^{-11} M, between 5×10^{-6} M and 5×10^{-12} M, between 5×10^{-6} M and 1×10^{-12} M, between 1×10^{-6} M and 1×10^{-7} M, between 1×10^{-6} M and 5×10^{-8} M, 1×10^{-6} M and 1×10^{-8} M, between 1×10^{-6} M and 5×10^{-9} M, between 1×10^{-7} M and 1×10^{-9} M, between 1×10^{-7} M and 5×10^{-10} M, between 1×10^{-7} M and 1×10^{-10} M, between 1×10^{-6} M and 5×10^{-11} M, between 1×10^{-6} M and 1×10^{-11} M, between 1×10^{-6} M and 5×10^{-12} M, between 1×10^{-6} M and 1×10^{-12} M, between 5×10^{-7} M and 5×10^{-8} M, 5×10^{-7} M and 1×10^{-8} M, between 5×10^{-7} M and 5×10^{-9} M, between 5×10^{-7} M and 1×10^{-9} M, between 5×10^{-7} M and 5×10^{-10} M, between 5×10^{-7} M and 1×10^{-10} M, between 5×10^{-7} M and 5×10^{-11} M, between 5×10^{-7} M and 1×10^{-11} M, between 5×10^{-7} M and 5×10^{-12} M, between 5×10^{-7} M and 1×10^{-12} M, between 1×10^{-7} M and 5×10^{-8} M, 1×10^{-7} M and 1×10^{-8} M, between 1×10^{-7} M and 5×10^{-9} M, between 1×10^{-7} M and 1×10^{-9} M, between 1×10^{-7} M and 5×10^{-10} M, between 1×10^{-7} M and 1×10^{-10} M, between 1×10^{-7} M and 5×10^{-11} M, between 1×10^{-7} M and 1×10^{-11} M, between 1×10^{-7} M and 5×10^{-12} M, between 1×10^{-7} M and 1×10^{-12} M, between 5×10^{-8} M and 5×10^{-9} M, between 5×10^{-8} M and 1×10^{-9} M, between 5×10^{-8} M and 5×10^{-10} M, between 5×10^{-8} M and 1×10^{-10} M, between 5×10^{-8} M and 5×10^{-11} M, between 5×10^{-8} M and 1×10^{-11} M, between 5×10^{-8} M and 5×10^{-12} M, between 5×10^{-8} M and 1×10^{-12} M, between 1×10^{-8} M and 1×10^{-9} M, between 1×10^{-8} M and 5×10^{-10} M, between 1×10^{-8} M and 1×10^{-10} M, between 1×10^{-8} M and 5×10^{-11} M, between 1×10^{-8} M and 1×10^{-11} M, between 1×10^{-8} M and 5×10^{-12} M, between 1×10^{-8} M and 1×10^{-12} M, between 5×10^{-9} M and 1×10^{-9} M, between 5×10^{-9} M and 5×10^{-10} M, between 5×10^{-9} M and 1×10^{-10} M, between 5×10^{-9} M and 5×10^{-11} M, between 5×10^{-9} M and 1×10^{-11} M, between 5×10^{-9} M and 5×10^{-12} M, between 5×10^{-9} M and 1×10^{-12} M, between 1×10^{-9} M and 5×10^{-10} M, between 1×10^{-9} M and 1×10^{-10} M, between 1×10^{-9} M and 5×10^{-11} M, between 1×10^{-9} M and 1×10^{-11} M, between 1×10^{-9} M and 5×10^{-12} M, between 1×10^{-9} M and 1×10^{-12} M, between 5×10^{-10} M and 1×10^{-10} M, between 5×10^{-10} M and 5×10^{-11} M, between 5×10^{-10} M and 1×10^{-11} M, between 5×10^{-10} M and 5×10^{-12} M, between 5×10^{-10} M and 1×10^{-12} M, between 1×10^{-10} M and 1×10^{-11} M, between 1×10^{-10} M and 5×10^{-12} M, between 1×10^{-10} M and 1×10^{-12} M,

between 5×10^{-11} M and 1×10^{-12} M, between 5×10^{-11} M and 5×10^{-12} M, between 5×10^{-11} M and 1×10^{-12} M, between 1×10^{-11} M and 5×10^{-12} M, or between 1×10^{-11} M and 1×10^{-12} M, as measured by biolayer interferometry. In some embodiments, a parvovirus antibody binds to a canine parvovirus or a feline parvovirus, as determined by immunoblot analysis.

“Wild-type” refers to a non-mutated version of a polypeptide that occurs in nature, or a fragment thereof. A wild-type polypeptide may be produced recombinantly.

A “variant” means a biologically active polypeptide having at least about 50% amino acid sequence identity with the native sequence polypeptide after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity. Such variants include, for instance, polypeptides wherein one or more amino acid residues are added, deleted, at the N- or C-terminus of the polypeptide.

In some embodiments, a variant has at least 1, 2, 3, 4, or 5 amino acids substituted by a different amino acid.

In some embodiments, a variant has at least about 50% sequence identity with the reference nucleic acid molecule or polypeptide after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity. Such variants include, for instance, polypeptides wherein one or more amino acid residues are added, deleted, at the N- or C-terminus of the polypeptide. In some embodiments, a variant has at least about 50% sequence identity, at least about 60% sequence identity, at least about 65% sequence identity, at least about 70% sequence identity, at least about 75% sequence identity, at least about 80% sequence identity, at least about 85% sequence identity, at least about 90% sequence identity, at least about 95% sequence identity, at least about 97% sequence identity, at least about 98% sequence identity, or at least about 99% sequence identity with the sequence of the reference nucleic acid or polypeptide.

As used herein, “position corresponding to position n,” wherein n is any number, refers to an amino acid position of a subject polypeptide that aligns with position n of a reference polypeptide after aligning the amino acid sequences of the subject and reference polypeptides and introducing gaps. Alignment for purposes of whether a position of a subject polypeptide corresponds with position n of a reference polypeptide can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as BLAST, BLAST-2, CLUSTAL OMEGA, ALIGN, or MEGALIGN™ (DNASTAR) software. Those skilled in the art can determine appropriate parameters for alignment, including any parameters needed to achieve maximal alignment over the full length of two sequences being compared. In some embodiments, the subject polypeptide and the reference polypeptide are of different lengths.

A “point mutation” is a mutation that involves a single amino acid residue. The mutation may be the loss of an amino acid, substitution of one amino acid residue for another, or the insertion of an additional amino acid residue.

An “amino acid substitution” refers to the replacement of one amino acid in a polypeptide with another amino acid. In some embodiments, an amino acid substitution is a conservative substitution. Nonlimiting exemplary conservative amino acid substitutions are shown in Table 2. Amino acid substitutions may be introduced into a molecule of interest

and the products screened for a desired activity, for example, retained/improved antigen binding, decreased immunogenicity, or improved ADCC or CDC or enhanced pharmacokinetics.

TABLE 2

Original Residue	Exemplary Substitutions
Ala (A)	Val; Leu; Ile
Arg (R)	Lys; Gln; Asn
Asn (N)	Gln; His; Asp; Lys; Arg
Asp (D)	Glu; Asn
Cys (C)	Ser; Ala
Gln (Q)	Asn; Glu
Glu (E)	Asp; Gln
Gly (G)	Ala
His (H)	Asn; Gln; Lys; Arg
Ile (I)	Leu; Val; Met; Ala; Phe; Norleucine
Leu (L)	Norleucine; Ile; Val; Met; Ala; Phe
Lys (K)	Arg; Gln; Asn
Met (M)	Leu; Phe; Ile
Phe (F)	Trp; Leu; Val; Ile; Ala; Tyr
Pro (P)	Ala
Ser (S)	Thr
Thr (T)	Val; Ser
Trp (W)	Tyr; Phe
Tyr (Y)	Trp; Phe; Thr; Ser
Val (V)	Ile; Leu; Met; Phe; Ala; Norleucine

Amino acids may be grouped according to common side-chain properties:

- (1) hydrophobic: Norleucine, Met, Ala, Val, Leu, Ile;
- (2) neutral hydrophilic: Cys, Ser, Thr, Asn, Gln;
- (3) acidic: Asp, Glu;
- (4) basic: His, Lys, Arg;
- (5) residues that influence chain orientation: Gly, Pro;
- (6) aromatic: Trp, Tyr, Phe.

Non-conservative substitutions will entail exchanging a member of one of these classes with another class.

The term "vector" is used to describe a polynucleotide that can be engineered to contain a cloned polynucleotide or polynucleotides that can be propagated in a host cell. A vector can include one or more of the following elements: an origin of replication, one or more regulatory sequences (such as, for example, promoters or enhancers) that regulate the expression of the polypeptide of interest, or one or more selectable marker genes (such as, for example, antibiotic resistance genes and genes that can be used in colorimetric assays, for example, β -galactosidase). The term "expression vector" refers to a vector that is used to express a polypeptide of interest in a host cell.

A "host cell" refers to a cell that may be or has been a recipient of a vector or isolated polynucleotide. Host cells may be prokaryotic cells or eukaryotic cells. Exemplary eukaryotic cells include mammalian cells, such as primate or non-primate animal cells; fungal cells, such as yeast; plant cells; and insect cells. Nonlimiting exemplary mammalian cells include, but are not limited to, NSO cells, PER.C6 cells (Crucell), 293 cells, and CHO cells, and their derivatives, such as 293-6E, DG44, CHO-S, and CHO-K cells. Host cells include progeny of a single host cell, and the progeny may not necessarily be completely identical (in morphology or in genomic DNA complement) to the original parent cell due to natural, accidental, or deliberate mutation. A host cell includes cells transfected in vivo with a polynucleotide(s) encoding an amino acid sequence(s) provided herein.

The term "isolated" as used herein refers to a molecule that has been separated from at least some of the components with which it is typically found in nature or produced. For

example, a polypeptide is referred to as "isolated" when it is separated from at least some of the components of the cell in which it was produced. Where a polypeptide is secreted by a cell after expression, physically separating the supernatant containing the polypeptide from the cell that produced it is considered to be "isolating" the polypeptide. Similarly, a polynucleotide is referred to as "isolated" when it is not part of the larger polynucleotide (such as, for example, genomic DNA or mitochondrial DNA, in the case of a DNA polynucleotide) in which it is typically found in nature, or is separated from at least some of the components of the cell in which it was produced, for example, in the case of an RNA polynucleotide. Thus, a DNA polynucleotide that is contained in a vector inside a host cell may be referred to as "isolated." In some embodiments, the parvovirus antibody is purified using chromatography, such as size exclusion chromatography, ion exchange chromatography, protein A column chromatography, hydrophobic interaction chromatography, and CHT chromatography.

To "reduce" or "inhibit" means to decrease, reduce, or arrest an activity, function, or amount as compared to a reference. In some embodiments, by "reduce" or "inhibit" is meant the ability to cause an overall decrease of 20% or greater. In some embodiments, by "reduce" or "inhibit" is meant the ability to cause an overall decrease of 50% or greater. In some embodiments, by "reduce" or "inhibit" is meant the ability to cause an overall decrease of 75%, 85%, 90%, 95%, or greater. In some embodiments, the amount noted above is inhibited or decreased over a period of time, relative to a control dose (such as a placebo) over the same period of time. A "reference" as used herein, refers to any sample, standard, or level that is used for comparison purposes. A reference may be obtained from a healthy or non-diseased sample. In some examples, a reference is obtained from a non-diseased or non-treated sample of a companion animal. In some examples, a reference is obtained from one or more healthy animals of a particular species, which are not the animal being tested or treated.

The term "substantially reduced," as used herein, denotes a sufficiently high degree of reduction between a numeric value and a reference numeric value such that one of skill in the art would consider the difference between the two values to be of statistical significance within the context of the biological characteristic measured by said values. In some embodiments, the substantially reduced numeric values is reduced by greater than about any one of 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 70%, 80%, 90%, or 100% compared to the reference value.

In some embodiments, an parvovirus antibody may reduce parvovirus titers in a canine or a feline by at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, or 100% compared to parvovirus titers in the absence of the antibody, as measured by a Hemagglutination Inhibition (HI) assay or Virus Neutralization (VN) assay. In some embodiments, the reduction in parvovirus titer is between 10% and 15%, between 10% and 20%, between 10% and 25%, between 10% and 30%, between 10% and 35%, between 10% and 40%, between 10% and 45%, between 10% and 50%, between 10% and 60%, between 10% and 70%, between 10% and 80%, between 10% and 90%, between 10% and 100%, between 15% and 20%, between 15% and 25%, between 15% and 30%, between 15% and 35%, between 15% and 40%, between 15% and 45%, between 15% and 50%, between 15% and 60%, between 15% and 70%, between 15% and 80%, between 15% and 90%, between

15% and 100%, between 20% and 25%, between 20% and 30%, between 20% and 35%, between 20% and 40%, between 20% and 45%, between 20% and 50%, between 20% and 60%, between 20% and 70%, between 20% and 80%, between 20% and 90%, between 20% and 100%, between 25% and 30%, between 25% and 35%, between 25% and 40%, between 25% and 45%, between 25% and 50%, between 25% and 60%, between 25% and 70%, between 25% and 80%, between 25% and 90%, between 25% and 100%, between 30% and 35%, between 30% and 40%, between 30% and 45%, between 30% and 50%, between 30% and 60%, between 30% and 70%, between 30% and 80%, between 30% and 90%, between 30% and 100%, between 35% and 40%, between 35% and 45%, between 35% and 50%, between 35% and 60%, between 35% and 70%, between 35% and 80%, between 35% and 90%, between 35% and 100%, between 40% and 45%, between 40% and 50%, between 40% and 60%, between 40% and 70%, between 40% and 80%, between 40% and 90%, between 40% and 100%, between 45% and 50%, between 45% and 60%, between 45% and 70%, between 45% and 80%, between 45% and 90%, between 45% and 100%, between 50% and 60%, between 50% and 70%, between 50% and 80%, between 50% and 90%, between 50% and 100%, between 60% and 70%, between 60% and 80%, between 60% and 90%, between 60% and 100%, between 70% and 80%, between 70% and 90%, between 70% and 100%, between 80% and 90%, between 80% and 100%, or between 90% and 100%.

Exemplary Pharmaceutical Compositions

The terms “pharmaceutical formulation” and “pharmaceutical composition” refer to a preparation which is in such form as to permit the biological activity of the active ingredient(s) to be effective, and which contains no additional components that are unacceptably toxic to a subject to which the formulation would be administered.

A “pharmaceutically acceptable carrier” refers to a non-toxic solid, semisolid, or liquid filler, diluent, encapsulating material, formulation auxiliary, or carrier conventional in the art for use with a therapeutic agent that together comprise a “pharmaceutical composition” for administration to a subject. A pharmaceutically acceptable carrier is non-toxic to recipients at the dosages and concentrations employed and is compatible with other ingredients of the formulation. The pharmaceutically acceptable carrier is appropriate for the formulation employed. Examples of pharmaceutically acceptable carriers include alumina; aluminum stearate; lecithin; serum proteins, such as human serum albumin, canine or other animal albumin; buffers such as phosphate, citrate, tromethamine or HEPES buffers; glycine; sorbic acid; potassium sorbate; partial glyceride mixtures of saturated vegetable fatty acids; water; salts or electrolytes, such as protamine sulfate, disodium hydrogen phosphate, potassium hydrogen phosphate, sodium chloride, zinc salts, colloidal silica, or magnesium trisilicate; polyvinyl pyrrolidone, cellulose-based substances; polyethylene glycol; sucrose; mannitol; or amino acids including, but not limited to, arginine.

The pharmaceutical composition can be stored in lyophilized form. Thus, in some embodiments, the preparation process includes a lyophilization step. The lyophilized composition may then be reformulated, typically as an aqueous composition suitable for parenteral administration, prior to administration to the dog, cat, or horse. In other embodiments, particularly where the antibody is highly stable to thermal and oxidative denaturation, the pharmaceutical composition can be stored as a liquid, i.e., as an aqueous

composition, which may be administered directly, or with appropriate dilution, to the dog, cat, or horse. A lyophilized composition can be reconstituted with sterile Water for Injection (WFI). Anti-bacterial agents (e.g., bacteriostatic reagents, such as benzyl alcohol, may be included. Thus, the invention provides pharmaceutical compositions in solid or liquid form.

The pH of the pharmaceutical compositions may be in the range of from about pH 5 to about pH 8, when administered. The compositions of the invention are sterile if they are to be used for therapeutic purposes. Sterility can be achieved by any of several means known in the art, including by filtration through sterile filtration membranes (e.g., 0.2 micron membranes). Sterility may be maintained with or without anti-bacterial agents.

In some embodiments, the pharmaceutically acceptable carrier or the pharmaceutical composition has a pH of from 5.0 to 6.2, from 5.0 to 6.0, or from 5.3 to 5.7. In some embodiments, the pharmaceutical carrier is phosphate buffered saline, pH 7.2. In some embodiments, the pharmaceutical carrier is 50 mM NaCitrate pH 7, 150 mM NaCl.

In some embodiments, the pharmaceutically acceptable carrier or a pharmaceutical composition comprises an anti-bacterial agent.

Exemplary Uses of Antibodies and Pharmaceutical Compositions

The antibodies or pharmaceutical compositions comprising the antibodies of the invention may be useful for providing passive immunity against infection with parvovirus and/or treating a parvoviral infection. As used herein, a “parvoviral infection” means a condition associated with, caused by, or characterized by, a parvoviral infection. Such conditions include, but are not limited to, infections confirmed by cage-side ELISA tests, hemagglutination assay (HA), histopathology, virus isolation or virus titers, or PCR. Infections with parvovirus often include fever, vomiting, diarrhea, lymphopenia, dehydration, and/or secondary septicemia.

As used herein, “treatment” is an approach for obtaining beneficial or desired clinical results. “Treatment” as used herein, covers any administration or application of a therapeutic for disease in a subject, such as a mammal, including a human and a companion animal (e.g., a canine or feline). For purposes of this disclosure, beneficial or desired clinical results include, but are not limited to, any one or more of: alleviation of one or more symptoms, prevention of mortality, diminishment of extent and severity of disease, preventing or delaying spread of disease, eliminating or shortening duration of viral shedding, preventing or delaying recurrence of disease, preventing or decreasing viral cytopathic effects, delay or slowing of disease progression, amelioration of the disease state, inhibiting the disease or progression of the disease, resolution of clinical signs of disease, inhibiting or slowing the disease or its progression, arresting its development, and remission (whether partial or total). Also encompassed by “treatment” is a reduction of pathological consequence of a proliferative disease. The methods provided herein contemplate any one or more of these aspects of treatment. In-line with the above, the term treatment does not require one-hundred percent removal of all aspects of the disorder.

In some embodiments, a parvovirus antibody or pharmaceutical compositions comprising it can be utilized in accordance with the methods herein to provide passive immunity against infection parvovirus and/or treat parvoviral infections. In some embodiments, a parvovirus antibody or pharmaceutical compositions is administered to subject,

such as a companion animal (e.g., a canine or a feline) or a human to provide passive immunity against infection with parvovirus and/or treat an a parvoviral infection.

A "therapeutically effective amount" of a substance/molecule, agonist or antagonist may vary according to factors such as the type of disease to be treated, the disease state, the immune status of the individual subject, the virulent viral load encountered, the severity and extent of viremia, the severity and course of the disease, the type of therapeutic purpose, any previous therapy, the clinical history, the response to prior treatment, the maternally-derived antibody passive transfer status, the previous immunization status of the individual animal, the discretion of the attending veterinarian, age, sex, and weight of the subject, and the ability of the substance/molecule, agonist or antagonist to elicit a desired response in the subject. A therapeutically effective amount is also one in which any toxic or detrimental effects of the substance/molecule, agonist or antagonist are outweighed by the therapeutically beneficial effects. A therapeutically effective amount may be delivered in one or more administrations. A therapeutically effective amount refers to an amount effective, at dosages and for periods of time necessary, to achieve the desired therapeutic or prophylactic result.

In some embodiments, a parvovirus antibody or pharmaceutical composition comprising a parvovirus antibody is administered parenterally, by subcutaneous administration, intravenous infusion, or intramuscular injection. In some embodiments, a parvovirus antibody or pharmaceutical composition comprising a parvovirus antibody is administered as a single dose or multiple dose bolus injection. In some embodiments, a parvovirus antibody or pharmaceutical composition comprising a parvovirus antibody is administered by an intramuscular, an intravenous, an intraperitoneal, an intracerebrospinal, a subcutaneous, an intra-arterial, an intrasynovial, an intrathecal, or an inhalation route.

Parvovirus antibodies described herein may be administered in an amount in the range of 0.01 mg/kg body weight to 100 mg/kg body weight per dose. In some embodiments, parvovirus antibodies may be administered in an amount in the range of 0.5 mg/kg body weight to 50 mg/kg body weight per dose. In some embodiments, parvovirus antibodies may be administered in an amount in the range of 0.1 mg/kg body weight to 10 mg/kg body weight per dose. In some embodiments, parvovirus antibodies may be administered in an amount in the range of 0.1 mg/kg body weight to 100 mg/kg body weight per dose. In some embodiments, parvovirus antibodies may be administered in an amount in the range of 1 mg/kg body weight to 10 mg/kg body weight per dose. In some embodiments, parvovirus antibodies may be administered in an amount in the range of 0.5 mg/kg body weight to 100 mg/kg body weight, in the range of 1 mg/kg body weight to 100 mg/kg body weight, in the range of 5 mg/kg body weight to 100 mg/kg body weight, in the range of 10 mg/kg body weight to 100 mg/kg body weight, in the range of 20 mg/kg body weight to 100 mg/kg body weight, in the range of 50 mg/kg body weight to 100 mg/kg body weight, in the range of 1 mg/kg body weight to 10 mg/kg body weight, in the range of 5 mg/kg body weight to 10 mg/kg body weight, in the range of 0.5 mg/kg body weight to 10 mg/kg body weight, in the range of 0.01 mg/kg body weight to 0.5 mg/kg body weight, in the range of 0.01 mg/kg body weight to 0.1 mg/kg body weight, or in the range of 5 mg/kg body weight to 50 mg/kg body weight. In some embodiments, parvovirus antibodies may be administered in an amount of 0.5 mg/kg body weight.

A parvovirus antibody or a pharmaceutical composition comprising a parvovirus antibody can be administered to a subject, such as a human or companion animal (e.g., a canine or a feline) as a single dose, at one time or over a series of treatments. For example, a parvovirus antibody or a pharmaceutical composition comprising a parvovirus antibody may be administered at least once, more than once, at least twice, at least three times, at least four times, or at least five times.

In some embodiments, the dose is administered once per week for at least two or three consecutive weeks, and in some embodiments, this cycle of treatment is repeated two or more times, optionally interspersed with one or more weeks of no treatment. In other embodiments, the therapeutically effective dose is administered once per day for two to five consecutive days, and in some embodiments, this cycle of treatment is repeated two or more times, optionally interspersed with one or more days or weeks of no treatment.

In some embodiments, the dose is administered to a subject, such as a human or companion animal (e.g., a canine or a feline) less than 1 week of age, less than 2 weeks of age, less than 3 weeks of age, less than 4 weeks of age, less than 5 weeks of age, less than 6 weeks of age, less than 6 weeks of age, less than 7 weeks of age, less than 8 weeks of age, less than 9 weeks of age, less than 10 weeks of age, less than 11 weeks of age, less than 12 weeks of age, less than 6 months of age, between 0 and 12 weeks of age, between 0 and 10 weeks of age, between 0 and 8 weeks of age, between 0 and 6 weeks of age, between 0 and 4 weeks of age, between 0 and 2 weeks of age, between 4 and 12 weeks of age, between 6 and 12 weeks of age, between 10 and 12 weeks of age, between 4 weeks and 6 months of age, between 2 months and 6 months of age, between 4 months and 6 months of age, between 6 months and 1 year of age, greater than 13 weeks of age, or greater than 1 year of age.

It may be advantageous to deliver the parvovirus antibody or a nucleic acid encoding the parvovirus antibody to an infant subject prenatally and/or postnatally to provide passive immunity against parvovirus infection. In some embodiments, the parvovirus antibody is administered to a pregnant or nursing maternal subject, such as a human or companion animal (e.g., a canine or a feline). In some embodiments, the parvovirus antibody is administered to the placenta of a pregnant subject.

In some embodiments, a method of providing passive immunity in an infant subject against infection with a canine or feline parvovirus comprises administering to a pregnant or nursing maternal subject a therapeutically effective amount of a monoclonal antibody that binds to the canine or feline parvovirus. In some embodiments the parvovirus antibody is administered to the placenta of a pregnant subject. In some embodiments, the parvovirus antibody is administered to a nursing subject.

Provided herein are methods of using the parvovirus antibodies, polypeptides and polynucleotides for detection, diagnosis and monitoring of a parvoviral infection. Provided herein are methods of determining whether a subject will respond to parvovirus antibody therapy. In some embodiments, the method comprises virus serum neutralization. In some embodiments, the method comprises detecting whether the subject has cells that express parvovirus using a parvovirus antibody. In some embodiments, the method of detection comprises contacting the sample with an antibody, polypeptide, or polynucleotide and determining whether the level of binding differs from that of a reference or comparison sample (such as a control). In some embodiments, the

47

method may be useful to determine whether the antibodies or polypeptides described herein are an appropriate treatment for the subject.

In some embodiments, the sample is a biological sample. The term "biological sample" means a quantity of a substance from a living thing or formerly living thing. In some embodiments, the biological sample is a swab containing cellular debris, cell or cell/tissue lysate. In some embodiments, the biological sample includes, but is not limited to, blood, (for example, whole blood), plasma, serum, urine, synovial fluid, lymphatic tissue and epithelial cells.

In some embodiments, the cells or cell/tissue lysate are contacted with a parvovirus antibody and the binding between the antibody and the cell is determined. When the test cells show binding activity as compared to a reference cell of the same tissue type, it may indicate that the subject would benefit from treatment with a parvovirus antibody. In some embodiments, the test cells are from tissue of a subject, such as a human or companion animal (e.g., a canine or a feline).

Various methods known in the art for detecting specific antibody-antigen binding can be used. Exemplary immunoassays which can be conducted include fluorescence polarization immunoassay (FPIA), fluorescence immunoassay (FIA), enzyme immunoassay (EIA), nephelometric inhibition immunoassay (NIA), enzyme linked immunosorbent assay (ELISA), and radioimmunoassay (RIA). An indicator moiety, or label group, can be attached to the subject antibodies and is selected so as to meet the needs of various uses of the method which are often dictated by the availability of assay equipment and compatible immunoassay procedures. Appropriate labels include, without limitation, radionuclides (for example ^{125}I , ^{131}I , ^{35}S , ^3H , or ^{32}P), enzymes (for example, alkaline phosphatase, horseradish peroxidase, luciferase, or β -galactosidase), fluorescent moieties or proteins (for example, fluorescein, rhodamine, phycoerythrin, GFP, or BFP), or luminescent moieties (for example, QdotTM nanoparticles supplied by the Quantum Dot Corporation, Palo Alto, Calif.). General techniques to be used in performing the various immunoassays noted above are known to those of ordinary skill in the art.

For purposes of diagnosis, the polypeptide including antibodies can be labeled with a detectable moiety including but not limited to radioisotopes, fluorescent labels, and various enzyme-substrate labels known in the art. Methods of conjugating labels to an antibody are known in the art. In some embodiments, the parvovirus antibodies need not be labeled, and the presence thereof can be detected using a second labeled antibody which binds to the first parvovirus antibody. In some embodiments, the parvovirus antibody can be employed in any known assay method, such as competitive binding assays, direct and indirect sandwich assays, and immunoprecipitation assays. Zola, *Monoclonal Antibodies: A Manual of Techniques*, pp. 147-158 (CRC Press, Inc. 1987). The parvovirus antibodies and polypeptides can also be used for in vivo diagnostic assays, such as in vivo imaging. Generally, the antibody or the polypeptide is labeled with a radionuclide (such as ^{111}In , ^{99}Tc , ^{14}C , ^{131}I , ^{125}I , ^3H , or any other radionuclide label, including those outlined herein) so that the cells or tissue of interest can be localized using immunoscintigraphy. The antibody may also be used as staining reagent in pathology using techniques well known in the art.

In some embodiments, a first antibody is used for a diagnostic and a second antibody is used as a therapeutic. In some embodiments, the first and second antibodies are

48

different. In some embodiments, the first and second antibodies can both bind to the antigen at the same time, by binding to separate epitopes.

The following examples illustrate particular aspects of the disclosure and are not intended in any way to limit the disclosure.

EXAMPLES

Example 1

Preparation of Chimeric Canine Antibodies Mab A and Mab B

Structural analysis of canine parvovirus and feline parvovirus complexed with antibody fragments from eight different neutralizing monoclonal antibodies were reported in Hafenstein S., et al, *J Virol.* 2009 June, 83(11):5556-66. Incomplete amino acid sequence information was provided for rat Mab E and Mab F Fab fragments, which exhibited canine parvovirus neutralization in an in vitro assay. The incomplete amino acid sequences for the variable heavy chain and variable light chain of Mab E are SEQ ID NOS. 81 and 82, respectively. The incomplete amino acid sequences for the variable heavy chain and variable light chain of Mab F are SEQ ID NOS. 83 and 84, respectively.

Three-dimensional protein structure analysis was performed to construct a redesigned first framework region for both heavy and light chains of Mab E and Mab F. The amino acid sequences of the heavy and light chain of redesigned monoclonal antibodies, designated Mab A and Mab B, are SEQ ID NO: 20 (Mab A heavy chain), SEQ ID NO: 22 (Mab A light chain), SEQ ID NO: 62 (Mab B heavy chain), and SEQ ID NO: 64 (Mab B light chain).

DNA sequences encoding a chimeric antibody were designed for fusions of Mab A VH (SEQ ID NO: 20) and Mab B VH (SEQ ID NO: 62) to canine constant IgG-B heavy chain and Mab A VL (SEQ ID NO: 22) and Mab B LC (SEQ ID NO: 64) to canine constant κ light chain resulting in SEQ ID NO: 23 (Chimeric A HC IgG-B), SEQ ID NO: 66 (Chimeric B HC IgG-B), SEQ ID NO: 25 (Chimeric A LC κ), and SEQ ID NO: 68 (Chimeric B LC κ). The heavy chain and light chain nucleotide sequences were synthesized chemically and inserted into an expression vector suitable for transfection into a mammalian host cell. Following transfection of heavy and light chain vector pairs into cells and culture, antibodies were affinity purified from the culture media using CaptivA[®] Protein A Affinity Resin (Repligen). The purified Chimeric Canine Mab A and Mab B antibodies were confirmed by SDS-PAGE analysis (data not shown).

Chimeric antibodies may also be designed and prepared based on fusions of Mab A VH (SEQ ID NO: 20) and Mab B VH (SEQ ID NO: 62) to canine constant IgG-A, IgG-C, and IgG-D heavy chains.

Example 2

Chimeric Canine Antibodies Mab A and Mab B Activity

Both purified Chimeric Canine Mab A and Mab B (formulated in PBS, pH 7.2 at 200 $\mu\text{g/mL}$) were used for assessing anti-canine parvovirus activity using a hemagglutination inhibition (HI) assay. The assay was performed at Companion Animal Vaccine and Immuno Diagnostic Service Laboratory, University of Wisconsin, essentially as described by Carmichael, et al., *Am J of Vet Res.* 1980 May,

41 (5):784-91. Serial dilutions of samples of Chimeric Canine Mab A and Chimeric Canine Mab B incubated with CPV were prepared. Porcine red blood cells were then added. Canine parvovirus induces porcine red blood cells to agglutinate. Both Chimeric Canine Mab A and Chimeric Canine Mab B prevented agglutination. Both antibodies resulted in a CPV-2b HI of 40960, while irrelevant canine IgG produced HI of <20. Most vaccinated dogs are understood to have antibody HI around 1280.

Furthermore, the activity of the antibodies to prevent live CPV from infecting and destroying cells was analyzed using Virus Serum Neutralization assay.

Example 3

Variant VII and VL Sequences for Enhanced Expression of Chimeric Canine Mab A and Mab B

Chimeric Canine Mab A and Chimeric Canine Mab B consistently expressed poorly in transient CHOS or stable CHOK1 cells. To enhance expression levels, three-dimensional protein structure analysis was performed to identify VH and VL amino acid residues suitable for modification. Multiple constructs with various mutations in VH and VL were generated. The expression level of antibodies having combinations of canine IgG-B with VH and VL variants were tested.

Enhanced expression variants were analyzed for their specific activity by canine parvovirus HI assay. In these experiments, each of the antibody variants tested was formulated in PBS, pH 7.2 at 200 µg/mL. Both Chimeric Canine Mab A (SEQ ID NOs: 23 and 25) and Chimeric Canine Mab B (SEQ ID NOs: 66 and 68) demonstrated a CPV-2b HI of 10240. Enhanced expression variants, Chimeric Canine Mab A v2 (SEQ ID NOs: 24 and 25) and Chimeric Canine Mab B v2 (SEQ ID NOs: 67 and 69), maintained the same HI value of 10240, while other variants displayed lower specific activity.

Example 4

Parvovirus Infection Assay Activity (SVN)

To determine the ability of Chimeric Canine Mab A v2 (SEQ ID NOs: 24 and 25) and Chimeric Canine Mab B v2 (SEQ ID NOs: 67 and 69) to prevent live parvovirus from infecting and destroying cells, a serum virus neutralization (SVN) assay was performed. In this assay, Madin Darby Canine Kidney (MDCK) cells were grown. Serial dilutions of antibody were mixed with a fixed titer of parvovirus. The mixture was then incubated with the cells. After incubation, the cells were examined by microscopy to identify the viral cytopathic effect (CPE). The readout is the final dilution of the antibody sample where it reached limits in preventing parvovirus from causing the cytopathic effect.

Both Chimeric Canine Mab A v2 (1) kept at 4° C. and (2) subjected to 7 freeze-thaw cycles at 200 µg/mL had the same activity: SVN for CPV-2b 32000 and SVN for CPV2c 8000.

Example 5

Stability of Chimeric Canine Mab A v2 and Mab B v2

Stability of Chimeric Canine Mab A v2 (SEQ ID NOs: 24 and 25) at 25 mg/mL and Chimeric Canine Mab B v2 (SEQ ID NOs: 67 and 69) at 20.5 mg/mL was tested. Chimeric

Canine Mab A v2 and Chimeric Canine Mab B v2 have similar specific activity by HI assay at a concentration of about 25 mg/mL and about 20.5 mg/mL, respectively.

Both Chimeric Canine Mab A v2 and Chimeric Canine Mab B v2 produced and purified from CHOK1 stable pool can be readily concentrated to 25 mg/mL in PBS, pH7.2, suggesting that both antibodies have no solubility concerns.

Chimeric Canine Mab A v2 and Chimeric Canine Mab B v2 are monomeric (>99%) at neutral pH and after storage for 10 hours at pH 4.1, as assessed by size-exclusion high performance liquid chromatography (HPLC). In this analysis, monomer and aggregates of antibodies were separated by SEC-HPLC, using Agilent 1100 system, Shodex KW803 column (8 mm×300 mm) with a KW-G guard column, and mobile phase buffer of 2×PBS pH7.2 (270 mM NaCl, 5.4 mM KCl, 8.6 mM Na₂HPO₄, 2.8 mM KPO₄) at a constant flow rate of 0.5 mL/minute. The column was calibrated using Bio Rad markers (Catalog No. 151-1901) composed of Thyroglobulin, γ-globulin, Ovalbumin, Myoglobin and Vitamin B12.

Chimeric Canine Mab A v2 real-time stability in (1) PBS, pH 7.2 at 25 mg/mL and (2) 50 mM NaCitrate pH 7, 150 mM NaCl at 25 mg/mL at various temperatures were evaluated. The HPLC-SEC results at ten months are summarized in Table 3.

TABLE 3

Buffer	Temperature	% HMW	Chimeric Canine Mab A v2 (% monomer)
PBS pH 7.2	−80° C. (control)	<LOQ	100
	−20° C.	<LOQ	100
	4° C.	<LOQ	100
	25° C.	2.1%	97.9
50 mM NaCitrate pH 7, 150 mM NaCl	−80° C. (control)	<LOQ	100
	−20° C.	<LOQ	100

Stability was also tested by subjecting the antibodies to three freeze-thaw cycles with freezing at −70° C. and thawing at room temperature. No loss of activity was observed for either antibody in terms of both CPV-2b HI and CPV-2c HI value.

Stability was further tested by subjecting the antibodies to seven freeze-thaw cycles with freezing at −20° C. and thawing at room temperature. Both antibodies maintained full activity as measured by HI assay.

Thermostability of the two antibodies in different formulations and across a range of pH was analyzed using differential scanning fluorescence (DSF). The melting temperature (T_m) of each antibody at the different conditions were measured. Buffer and 12 µg of antibody were mixed together with 1× Protein thermal shift dye (Applied Biosystem, Catalog No. 4461146). A melting curve was performed with StepOne Real Time PCR System (Applied Biosystem, Catalog No. 4376357). The temperature was increased from 25° C. to 99° C. with a ramp rate of 1% according to the manufacturer's instructions. The data was analyzed by Protein Thermal Shift™ Software v1.0 (Applied Biosystem, Catalog No. 4466038) to determine the T_m, which was calculated as the highest value derived from taking the first derivative of the protein melting curve. For each of the formulations tested, including PBS pH7.2, Chimeric Canine Mab A v2 was significantly more stable than Chimeric Canine Mab B v2.

51

Example 6

VLP Bindings of Chimeric Canine Antibodies Mab A v2 and Mab B v2

Apparent affinity of Chimeric Canine Mab A v2 (SEQ ID NOs: 24 and 25) and Chimeric Canine Mab B v2 (SEQ ID NOs: 67 and 69) with parvovirus VLP was measured. First, biotinylated Chimeric Canine Mab A v2 and Chimeric Canine Mab B v2 were bound to streptavidin sensor tips. Then, binding of recombinant canine parvovirus CPV2b capsid (Advanced ImmunoChemical Inc Cat #8-CPV) to the Mab-bound sensor tip was assessed.

Chimeric Canine Mab A v2 and Chimeric Canine Mab B v2 antibodies each exhibited affinity to canine CPV 2b VLP with kinetics potentially sufficient for therapeutic activity. The binding analysis was performed using an Octet Biosensor as follows. Briefly, Chimeric Canine Mab A v2 (SEQ ID NOs: 24 and 25) and Chimeric Canine Mab B v2 (SEQ ID NOs: 67 and 69) was biotinylated through amine chemistry. The free unreacted biotin was removed by extensive dialysis. Biotinylated Chimeric Canine Mab A v2 and Chimeric Canine Mab B v2 was captured on streptavidin sensor tips. The association of either Mab A v2 or Mab B v2 and canine CPV 2b VLP (40 µg/mL) was monitored for 300 seconds. Dissociation was monitored for 300 seconds. A biotinylated irrelevant canine IgG-B was used as negative control. The data were fit to a 1:1 binding model using ForteBio™ data analysis software to determine the k_{on} , k_{off} , and the K_d . The buffer for dilutions and all binding steps was 20 mM phosphate, 150 mM NaCl, pH 7.2. The K_d of Chimeric Canine Mab A v2 and VLP2b was 9.78×10^{-11} M and of Chimeric Canine Mab B v2 and VLP2b was 9.38×10^{-12} M (See FIG. 2).

Example 7

Caninization of Mab A v2 and Mab B v2

To reduce potential immunogenicity, particularly where repeated administration of antibodies A or B are needed, VH and VL from Mab A v2 and B v2 were subjected to caninization. A number of caninized VH and VL were designed, expressed, and purified. Good expressors were selected for specific activity using CPV2b HI assay. Caninized A HC v3 IgG-B (SEQ ID NO: 37) plus Caninized A LC κ (SEQ ID NO: 39) and Caninized A HC v4 IgG-B (SEQ ID NO: 38) plus Caninized A LC κ (SEQ ID NO: 39) maintained full activity compared to Chimeric A HC IgG-B (SEQ ID NO: 23) plus Chimeric A LC κ (SEQ ID NO: 25) and compared to Chimeric A HC v2 IgG-B (SEQ ID NO: 24) plus Chimeric A LC κ (SEQ ID NO: 25). Caninized B HC v3 IgG-B (SEQ ID NO: 79) plus Caninized B LC v3 κ (SEQ ID NO: 80) maintained full activity compared to Chimeric B HC IgG-B (SEQ ID NO: 66) plus Chimeric B LC κ (SEQ ID NO: 68) and compared to Chimeric B HC v2 IgG-B (SEQ ID NO: 67) plus Chimeric B LC v2 κ (SEQ ID NO: 69).

Example 8

Feline Chimeric Mab A and Mab B

Chimeric Canine Mab A v2 (SEQ ID NOs: 24 and 25) and Chimeric Canine Mab B v2 (SEQ ID NOs: 67 and 69) showed activity in a feline parvovirus HI assay. Chimeric Canine Mab A v2 at 200 m/mL exhibited a feline parvovirus

52

HI of 262144, while Chimeric Canine Mab B v2 at 200 µg/mL exhibited a feline parvovirus HI of 131072.

Thus, Mab A and Mab B may be used in feline Parvovirus (panleukopenia virus) prevention or treatment. Furthermore, chimeric feline Mab A variable heavy chain and feline IgG-1 was designed to pair with chimeric feline Mab A variable light chain and feline kappa (SEQ ID NOs: 31 and 32). To remove potential heterogeneity, a mutation that removes the glycosylation site can be introduced to make chimeric feline Mab A-IgG1 paired with an aglycosylated kappa (e.g., SEQ ID NOs: 31 and 33).

Further, chimeric feline Mab B variable heavy chain and feline IgG-1 was designed to pair with chimeric feline Mab B variable light chain and feline kappa (SEQ ID NOs: 74 and 75). A mutation that removes the glycosylation site can be introduced to make chimeric feline Mab A-IgG1 paired with an aglycosylated kappa (e.g., SEQ ID NOs: 74 and 76).

Example 9

Safety and Pharmacodynamics in Puppies

An in vivo study was performed using Chimeric Canine Mab A (SEQ ID NOs: 23 and 25) and Chimeric Canine Mab B (SEQ ID NOs: 66 and 68) monoclonal antibodies. The objectives of this study were to evaluate safety, effect of Chimeric Canine Mab A and Chimeric Canine Mab B on Canine Parvovirus-2 (CPV-2) titers, virus neutralization, and response to a monovalent modified live vaccine after treatment with the two antibodies in purpose-bred, unvaccinated, healthy puppies. See study design summary in Table 4, below.

TABLE 4

Identity	Anti-Canine Parvovirus Neutralizing Antibodies Chimeric Canine Mab A (SEQ ID NOs: 23 and 25) and Chimeric Canine Mab B (SEQ ID NOs: 66 and 68).
Protocol Number	KB-030-PK-301
Protocol Title	Pilot Pharmacodynamic and Safety Study of Mab A and Mab B in Purpose Bred Puppies
Materials and Methods	The study was open label Non-GLP Single-dose laboratory PD and safety study in unvaccinated, CPV-2 titer negative, healthy beagles Puppies were randomized to one of two equal treatment groups (3 to Mab A; 3 to Mab B) Allocation occurred using a randomization list supplied by Sponsor Concomitant medications were given for coccidia and helminth prophylaxis during the acclimation phase.
Study Population	6 healthy female beagle puppies 10-12 weeks of age Weight at least 1.5 kg Sourced from Ridgland Farms, Wisconsin
Study Dates	Study Initiation/Acclimation: 21 Jan. 2019 TA Administered: 28 Jan. 2019 Study End: 9 May 2019

After randomization when puppies were allocated into two groups (3 Mab A (Dog IDs BDW-8, DEW-8, and DFW-8); 3 Mab B (Dog IDs CTW-8, CWW-8, and DHW-8)). Puppies received the respective antibody in a single intravenous dose (5 mg/kg) via a clipped and prepared cephalic vein on study Day 0. A monovalent CPV vaccine, Nobivac 1-Pv (Merck Animal Health, Kenilworth, NJ) was initially administered to puppies at titers of 80 or lower. A second dose was administered at 21 days after the first dose. A third dose was administered if the puppy had not seroconverted at about 21 days after the second dose.

Puppies were monitored using Hemagglutination Inhibition (HI) and Virus Neutralization (VN) for a total of 100 days. HI quantifies the amount of antibody in a serum sample and VN is utilized to determine the ability of an antibody to prevent live virus from infecting and destroying cells. The puppies were also followed to determine if they seroconverted following CPV vaccination.

All puppies were validated to have HI titers of <20 on Day 0 before intravenous administration of Mab A or B antibodies. Day 1 HI titers after administration of Mab A or B were between 2560 and 5120. Titers were monitored for 100 days to document the degradation of the antibodies and evaluate serum half-life.

On Day 37 when titers from individual puppies began to fall to 1:40, vaccinations began to be administered to assess neutralization of vaccinal CPV-2. Mab A or B antibodies were documented to block immunization at varying levels in the puppies. Puppies were given subsequent doses of vaccination on schedule and response was verified. Vaccinal seroconversion ultimately occurred in 5 of the 6 puppies. The HI assay titers for both groups of animals are reflected in the Table 5, below and in FIG. 3A (Mab A group), FIG. 3B (Mab B group), and FIG. 3C (both groups combined).

Two phase decay kinetics was used to obtain the fast distribution and slow phase decay using HI as function of time. Half-life of Chimeric Canine Mab A was 10.1 days (FIG. 4A) while half-life of Chimeric Canine Mab B was 8.6 days (FIG. 4B). Serum VN assays mirrored the HI titers (See Table 6, below). The half-life roughly matches maternally-derived antibody. (Carmichael, L E, Joubert, J C and Pol-

lock, RVH; "A Modified Live Canine Parvovirus Vaccine. II. Immune Response;" Cornell Vet, Volume 73, Pages 13-29; 1983).

This initial pharmacodynamic study demonstrates that intravenously administered Chimeric Canine Mab A (SEQ ID NOs: 23 and 25) and Chimeric Canine Mab B (SEQ ID NOs: 66 and 68) resulted in initial antibody titers between 2560 and 5120 within 24 hours post-administration. Additionally, VN follows HI titers very closely. The approximate half-life of the antibodies is between 7 and 10 days. The duration of immunity (i.e., the duration in which the HI titer was maintained at 80 or higher), ranged from 30-42 days after one dose of Chimeric Canine Mab A or Chimeric Canine Mab B. Interference with vaccination is likely until the antibody degrades sufficiently to no longer neutralize vaccinal CPV-2. Both Chimeric Canine Mab A and Chimeric Canine Mab B resulted in high titers, were well-tolerated and no adverse events or injection site reactions were reported.

The initial CPV vaccination did not result in seroconversion in 5 out of 6 puppies, potentially due to residual Chimeric Canine Mab A or B antibodies neutralizing the vaccine. Blocking of vaccinal virus replication by both Canine Mab A or Chimeric Canine Mab B was the first in vivo correlation to the in vitro neutralization data. Subsequently, three puppies seroconverted after the second and one after the third vaccination.

Regarding puppy CWW-8, seroconversion on Day 56 did not correlate with the timeframe of vaccination administered on Day 37. It is suspected that vaccinal virus shedding from other puppies in the study exposed CWW-8 and produced the seroconversion once the titers were <20.

TABLE 5

	Study Day														
	Day -7	Day 0	Day 1	Day 3	Day 5	Day 7	Day 10	Day 14	Day 18	Day 21	Day 23	Day 25	Day 28	Day 30	Day 32
Animal ID	Hemagglutination Inhibition Assay/Parvovirus titer														
BDW-8	<20	<20	2560	2560	2560	640	1280	640	320	640	320	320	160	160	80
DEW-8	<20	<20	5120	2560	1280	1280	1280	640	640	640	640	320	320	320	160
DFW-8	<20	<20	2560	2560	1280	1280	640	640	640	320	640	320	160	160	160
CTW-8	<20	<20	2560	1280	1280	640	640	320	320	160	160	160	80	80	40
CWW-8	<20	<20	2560	1280	1280	640	640	320	320	160	160	160	80	80	40
DHW-8	<20	<20	2560	1280	1280	1280	640	320	640	160	160	160	160	160	80

	Study Day														
	Day 35	Day 37	Day 39	Day 42	Day 49	Day 56	Day 58	Day 63	Day 70	Day 77	Day 84	Day 87	Day 91	Day 98	
Animal ID	Hemagglutination Inhibition Assay/Parvovirus titer														
BDW-8	80	40	40	20 ¹	<20	<20		<20 ²	<20	<20	<20	XX ⁴	<20	<20	
DEW-8	80	80	80	80 ¹	40	40		20 ²	<20	1280 ³	2560				
DFW-8	80	80	80	40 ¹	40	40		20 ²	80 ³	80	320				
CTW-8	40	40 ¹		20	20	<20	<20 ²	10240 ³	5120	5120	2560				
CWW-8	40	40 ¹		40	<20	20480 ³	10240 ²	10240	5120	5120	5120				
DHW-8	80	40	40	20 ¹	20	<20		<20 ²	<20	<20	<20	XX ⁴	<20	320 ³	

^{1st} Monovalent parvo vaccine given^{2nd} Monovalent parvo vaccine given³Vaccinal Seroconversion^{43rd} Monovalent parvo vaccine given

TABLE 6

Animal ID	Study Day										
	Day 0	Day 1	Day 7	Day 14	Day 21	Day 28	Day 35	Day 42	Day 49	Day 56	Day 63
Virus Neutralization Assay											
BDW-8	<2	1024	512	256	128	32	32	16	8	2	2
DEW-8	<2	1024	512	256	128	64	32	32	16	4	1024
DFW-8	<2	1024	512	256	128	16	32	16	16	8	32
CTW-8	<2	512	256	64	64	32	16	4	4	>4096	1024
CWW-8	<2	512	256	128	64	32	16	16	4	>4096	4096
DHW-8	<2	1024	256	128	64	32	16	8	8	4	<2

Underline = 1st Monovalent parvo vaccine givenItalic = 2nd Monovalent parvo vaccine given

Example 10

Prevention and Treatment

This in vivo study utilized MabA v2 antibody. The objective of this study was to validate the dose of live CPV-2 that will be used to challenge dogs in future pivotal studies of MabA v2 and to provide proof of concept for MabA v2 for two indications, both as a prophylactic treatment to prevent CPV-2 infection and as a therapeutic treatment for dogs with CPV-2 infection. See study design summary in Table 7, below.

TABLE 7

Identity	Anti-Canine Parvovirus Neutralizing Antibody Chimeric Canine Mab A v2 (SEQ ID NOs: 24 and 25)
Protocol Number	KB-030-PLC-301
Protocol Title	Evaluation of the Efficacy of Mab A v2 as a Prophylactic and Therapeutic Treatment in Dogs to Prevent Clinical Signs of CPV-2 Infection
Materials and Methods	The study was open label Non-GLP Single-dose laboratory PD and safety study in unvaccinated, CPV-2 titer negative, healthy beagles
Study Population	Puppies were randomized to one of six equal treatment groups 12 healthy beagle puppies Equal numbers of males and females 11 weeks of age Weight at least 1.5 kg
Study Dates	Sourced from Ridgman Farms Inc., Wisconsin Study Day 1: 22 Jun. 2019 CPV-2 Inoculation: 25 Jun. 2019 Study End: 7 Jul. 2019

Twelve dogs were randomized and assigned to one of 6 groups with a male and female in each group. Groups 1-3 were only to receive the live CPV-2 challenge in order to validate the dose required to produce required morbidity and mortality at three different viral logs. Dogs in groups 4 and 5 received Mab A v2 as a prophylactic treatment before inoculation with live CPV-2b. Dogs in group 6 were given Mab A v2 after challenge and confirmed development CPV-2 infection via fecal cage-side ELISA SNAP Test. See Table 8 (below) for CPV-2 inoculation challenge schedule.

TABLE 8

Group	Number of Dogs	Challenge	Study Day of Inoculation	Route	Dose
1	2	CPV-2b	Day 4	IN	10 ² TCID ₅₀
2	2	CPV-2b	Day 4	IN	10 ⁴ TCID ₅₀

15

TABLE 8-continued

Group	Number of Dogs	Challenge	Study Day of Inoculation	Route	Dose
3	2	CPV-2b	Day 4	IN	10 ⁶ TCID ₅₀
4*	2	CPV-2b	Day 4	IN	10 ⁶ TCID ₅₀
5*	2	CPV-2b	Day 4	IN	10 ⁶ TCID ₅₀
6**	2	CPV-2b	Day 4	IN	10 ⁶ TCID ₅₀

*Groups 4 and 5 were treated with Mab on Day 1 per Table 9

25 **Group 6 was treated with Mab after CPV-2 infection was confirmed by detection of virus in feces (via cage-side CPV-2 ELISA) per Table 10

Groups 4 and 5 were given MabA v2 on study Day 1 per Table 9, below.

TABLE 9

Group	Number of Dogs	Treatment	Study Day of Treatment	Route	Dose
4	2	Mab A v2	Day 1	IV	5 mg/kg
5	2	Mab A v2	Day 1	SC	5 mg/kg

Group 6 was given Mab A v2 on study Day 7, after each puppy had developed a positive cage-side CPV-2 ELISA test documenting CPV-2 in the feces confirming parvovirus infection. See Table 10, below.

TABLE 10

Group	Number of Dogs	Treatment	Study Day of Treatment	Route	Dose
6	2	Mab A v2	Upon detection of virus in feces	IV	5 mg/kg

This study provided the first pharmacodynamic data on MabA v2 administered subcutaneously. The intravenously administered dose in group 4 dogs provided instantaneous rise in HI titers as predicted. The subcutaneously administered dose in group 5 dogs produced comparable titers over a 24-hour period. In Table 11, below, it can be seen how the intravenous and sub-cutaneous doses differ in timing at which the peak post-administration of MabA v2 HI titers were achieved.

As shown in Table 12, below, administration of MabA v2 produced circulating antibody titers in the dogs within 24 hours (IV) and 48 hours (SC) after administration. Protective passive antibody titers were maintained for 16 days in face of a high load virulent challenge.

TABLE 11

Treatment			Serum HI Titers								
Group	Treatment	Dog ID	Day 1	Day 2	Day 3	Day 4	Day 6	Day 8	Day 11	Day 15	Day 16
1	None	AA1001	<20	<20	<20	<20	<20	<20	2560	20480	>20480
		AA1101	<20	<20	<20	<20	<20	<20	320	20480	>20480
2	None	AA2001	<20	<20	<20	<20	<20	<20	160	>20480	>20480
		AA2101	<20	<20	<20	<20	<20	<20	10240	NA	NA
3	None	AA3001	<20	<20	<20	<20	<20	<20	no sample	NA	NA
		AA3101	<20	<20	<20	<20	<20	<20	2560	NA	NA
4	Mab A v2 (Day 1, IV)	AA4001	5120	5120	5120	5120	2560	2560	1280	2560	2560
		AA4101	10240	5120	5120	5120	2560	2560	2560	1280	2560
5	Mab A v2 (Day 1, SC)	AA5001	<20	2560	2560	2560	2560	2560	2560	1280	2560
		AA5101	<20	2560	2560	2560	2560	2560	5120	1280	1280
6	Mab A v2 (upon detection of CPV-2 in feces, IV)	AA6001	<20	<20	<20	<20	<20	5120	5120	5120	10240
		AA6101	<20	<20	<20	<20	<20	5120	10240	10240	10240

TABLE 12

Treatment			CPV-2 SVN Antibody Titers								
Group	Treatment	Dog ID	Day 1	Day 2	Day 3	Day 4	Day 6	Day 8	Day 11	Day 15	Day 16
1	None	AA1001	<10	<10	<10	<10	<10	<10	2560	10240	10240
		AA1101	<10	<10	<10	<10	<10	<10	160	20480	5120
2	None	AA2001	<10	<10	<10	<10	<10	<10	160	5120	10240
		AA2101	<10	<10	<10	<10	<10	<10	2560	NA	NA
3	None	AA3001	<10	<10	<10	<10	<10	<10	no sample	NA	NA
		AA3101	<10	<10	<10	<10	<10	<10	2560	NA	NA
4	Mab A v2 (Day 1, IV)	AA4001	2560	2560	1280	1280	1280	640	640	320	320
		AA4101	2560	1280	1280	1280	1280	640	320	320	320
5	Mab A v2 (Day 1, SC)	AA5001	<10	640	640	1280	640	640	640	320	320
		AA5101	<10	640	640	1280	1280	640	640	320	320
6	Mab A v2 (upon detection of CPV-2 in feces, IV)	AA6001	<10	<10	<10	<10	<10	1280	5120	5120	5120
		AA6101	<10	<10	<10	<10	<10	640	5120	5120	5210

NA Not applicable

On Day 1, groups 4 and 5 were given MabA v2 intravenously or subcutaneously, respectively. On Day 4, all 12 dogs in all 6 groups were administered virulent CPV-2 intranasally per the inoculation dose in Table 8. For a total of 14 days, the dogs were monitored every 6 hours to record General Health Observations (GHO), every 12 hours to document clinical scores (via validated system, Mohr et al, 2003), and frequent sampling of blood for hematology, HI, and VN as well as fecal hemagglutination assay (HA). Additionally, all dogs were given a cage-side CPV-2 ELISA test every day after inoculation until tested positive to evaluate when parvovirus infection was present. Table 13 summarizes the clinical signs of CPV-2 observed during the study.

TABLE 13-continued

Group	Dog ID	Diarrhea, blood or mucus in feces	RT $\geq$ 103.4° F.	Lympho- penia $\geq$ 50%	Fecal Viral HA	Early Euthanasia
3	AA3001	-	-	+	+	+ (at study day 10)
	AA3101	+	+	+	+	+ (at study day 11)
4	AA4001	-	-	-	-	-
	AA4101	-	-	-	-	-
5	AA5001	-	-	-	-	-
	AA5101	-	-	-	-	-
6	AA6001	-	-	+	-	-
	AA6101	-	-	+	+	-

TABLE 13

Group	Dog ID	Diarrhea, blood or mucus in feces	RT $\geq$ 103.4° F.	Lympho- penia $\geq$ 50%	Fecal Viral HA	Early Euthanasia
1	AA1001	+	-	-	+	-
	AA1101	-	-	-	-	-
2	AA2001	+	-	-	+	-
	AA2101	+	-	+	+	+ (at study day 11)

Dogs in groups 3 and 6 (those who were challenged with 10⁶ live CPV-2 and not pre-treated with MabA v2) showed positive cage-side CPV-2 ELISA tests on the third or fourth day post-inoculation. Both dogs in group 3 (effectively the control group) developed profound classic clinical signs of parvovirus infection including depression, vomiting, bloody diarrhea, progressive dehydration and were moribund and euthanized by 7 days post-inoculation. Both dogs in group 6 tested positive on cage-side CPV-2 ELISA test 3 days after inoculation with live CPV-2. After confirming the positive cage-side CPV-2 ELISA test they were subsequently treated

with Mab A v2 intravenously at 5 mg/kg. They received no additional adjunctive or supportive care.

While the dogs in group 3 continued to get sicker and eventually became moribund and were humanely euthanized, the dogs in group 6 survived and experienced full clinical recovery during the study period. Group 6 dogs also returned to a negative cage-side CPV-2 ELISA test, indicating that they were no longer shedding viral particles.

All dogs in groups 4 and 5 given Mab A v2 as a prophylactic treatment not only survived, but remained healthy (e.g. never developed a positive cage-side CPV-2 ELISA and never exhibited clinical signs of parvovirus infection).

Anatomic histopathology of Groups 3-6 following humane euthanasia at scheduled study completion was performed, unless euthanasia was performed early for humane reasons. Groups 4 and 5 showed no significant pathologic changes. Group 6 showed lymphoid depletion of the thymus, but the other organs examined were normal. Group 3 showed lymphoid depletion of all lymphoid organs, moderate to severe small intestinal villus atrophy/necrosis with mild to severe crypt necrosis indicative of CPV-2 infection.

This pilot proof of concept study demonstrated Mab A v2 is promising as a prophylactic and therapeutic intervention. All dogs in the prophylactic groups (n=4) never developed parvovirus infection and remained clinically normal. All dogs in the therapeutic group (n=2) which were confirmed to be infected with parvovirus survived and experienced full clinical recovery after administration of Mab A v2 with no other adjunctive or supportive treatments. This study also validated that sub-cutaneous dosing of Mab A v2 for prophylaxis is a viable option. Finally, the product was well-tolerated with no adverse events or injection site reactions. In this proof of concept study, prophylactic administration of Mab A v2 prior to challenge protected 100% of puppies from CPV-2 infection. In addition, 100% of puppies given therapeutic administration of Mab A v2 after confirmed CPV infection survived and recovered.

Example 11

Therapeutic Treatment Pivotal Effectiveness Study

This pivotal study evaluates the effectiveness of Mab A v2 as a treatment of canine parvovirus (CPV) disease. Satisfactory completion of this study and demonstration of efficacy may support a label claim stating, "This product has been shown to be effective for the treatment of dogs 13 weeks of age or older against canine parvovirus (CPV) disease."

This GCP, randomized, blinded, placebo-controlled study is 14 days in length following at least 7-days acclimation period. A total of 28 naïve, CPV-2 titer negative dogs (≤ 13 weeks old on Day 0, of any sex, and weighing ≥ 1.5 kg) are randomized to the study at a 3:1 ratio to two groups (n=21 dogs in Group 1 and n=7 in Group 2) as per Table 14.

During acclimation, the dogs are acclimated to feeding, housing and handling procedures while being administered anthelmintic and antibiotic medications to eliminate common helminth and coccidia infections.

The day of CPV-2b challenge is considered as Day 0. On Day 0 all dogs in Groups 1 and 2 are administered virulent CPV-2b at a dose of approximately 1×10^6 TCID₅₀ intranasally (IN) (approximately 1 mL volume; approximately 0.5 mL per nostril). On the same day as detection of CPV in feces (via positive cage-side CPV SNAP test) each dog is administered Mab A v2 or Control Product (CP; phosphate

buffered saline (PBS)) via intravenous (IV) injection to an accessible vein (e.g., cephalic) catheter using a luer-lock syringe. Mab A v2 is administered in a single 5 mg/kg dose based on body weight. The CP has a pH of 7.2 ± 0.2 . The dose for Mab A v2 and CP is calculated as follows:

$$\frac{\{(\text{Body weight in kg}) \times (5 \text{ mg/kg})\} \pm \{25 \text{ mg/mL}\}}{\text{mg/mL}} = \text{volume in mL to be injected}$$

At the scheduled times outlined in Table 15 individual dog health is closely monitored for 14 days with physical examinations (PEs), general health observations (GHO), measurement of rectal temperature (RT), cage-side CPV SNP test, serum HI, fecal hemagglutination assay (HA), serum biochemistry, and hematology.

The primary efficacy variable is prevention of mortality from CPV infection in Group 1 dogs versus Group 2 dogs. See treatment groups in Table 14 below.

TABLE 14

Group	Number of Dogs	Treatment	Day	Route	Concentration (mg/mL)	Dose
1	21	IVP	Day of Positive CPV SNAP Test	IV	25 mg/mL	5 mg/kg
2	7	CP	Day of Positive CPV SNAP Test	IV	NA	Volume equivalent to Group 1

Randomization. This study may require a two-step randomization. The first step is to determine which 28 dogs are enrolled in the study. Dogs are then randomized at a block size of 4 based on litter if possible, with a 3:1 ratio to Groups 1 and 2, respectively. The randomization is based on random numbers generated by the PLAN procedure in SAS (version 9.4 or later, SAS Institute, Cary NC) using the randomized selection method with a seed number.

After arrival to the Test Facility, dogs are placed into cages in 2 rooms based on the order of arrival, and dogs from the same litter/block stay in the same room if possible. Dogs might get secondary exposure to virus that is shed from infected dogs. To keep a similar level of secondary virus exposure for both treatment groups and to maintain blinding to Test Facility personnel, each room contains both treatment groups. After the 28 dogs are selected, their cages are sequentially numbered from 1-28. Dogs from the same litter have adjacent cage numbers, with litters of 4 taking earlier numbers (e.g., 1-4, 5-8, etc.) per room. One room should contain cage numbers 1-12, and the other room contain cage numbers 13-28. The dogs are enrolled in the order of cage numbers in the randomization list.

Study Blinding. The Investigator and all other Test Facility Personnel (excluding the Dispenser and one designated person who restrains dogs for dosing) do not have knowledge of the actual treatment group assignment at Day 0. Only the Dispenser and the person designated for restraining dogs during dosing at the Test Facility have access to actual treatment group assignment (Mab A v2 or CP) for each dog. The Dispenser and the dog restrainer/holder do not perform any study assessments.

Specifications. A total of at least 30 purpose-bred, unvaccinated, CPV-2 seronegative (HI<20), healthy Beagle dogs are procured. The dogs are confirmed CPV negative via CPV SNAP test upon arrival to the Test Facility. Although the

study enrollment includes 28 dogs, two additional dogs are purchased to ensure dog numbers are met after screening and acclimation. Ideally, the 30 dogs are comprised of 7 litters with 4 dogs per litter, plus 2 dogs from another litter (a partial litter).

Pivotal Potency Determination. The pivotal efficacy serial sets the release potency of subsequent serials, each serial needs to be at, or above potency of the pivotal efficacy serial as compared in the potency assay in order to be considered satisfactory for release. In order to set a Master Reference for the potency assay, the pivotal efficacy serial is tested using the reference monitoring assay(s). The testing is conducted on Day 0±7 days of administration for the first efficacy dose administered to the study dogs.

Observations

Physical Examination (PE). Physical examinations are performed on all dogs by the Test Facility Veterinarian on at least Days -7, -1, Mab A v2/CP dosing day, Day 13 and on an unscheduled basis when necessary. Physical examinations include a comprehensive assessment of all body systems, including hydration status and BW, and are recorded.

Food Consumption. Once daily, qualitative assessment of food consumption is made in conjunction with general health observations (GHOs). All abnormalities in food consumption observed are recorded.

General Health Observations (GHOs) and Measurement of Rectal Temperature (RT). GHOs are conducted by the Test Facility Veterinarian or their designee at least once daily during the acclimation period from Day -7 through Day -1. GHOs are conducted every 12 hours (±1 hour) starting on Day 0 through Day 12. Beginning on Day 0, in conjunction with each GHO, the RT is also measured using a calibrated thermometer in Fahrenheit scale.

Specific GHO information are collected on these four parameters:

- Attitude—recorded as normal, mild to moderate depression, severe depression, or collapsed or moribund
- Appetite—recorded as normal, voluntarily eats small amounts, no interest in food, or not offered
- Vomiting—recorded as absent, mild (once per 12 hours), moderate (2-5 times per 12 hours), or severe (≥6 times per 12 hours)
- Feces—recorded as well-formed or absent, soft or pasty feces, watery non-bloody diarrhea, or water, bloody diarrhea

GHOs may also include other observations in addition to these four parameters. These parameters are documented for each dog at each observation period.

Cage-side CPV SNAP Test. Cage-side CPV ELISA tests are performed using the IDXX CPV SNAP test according to the manufacturer's instructions. Preliminary CPV SNAP tests are performed on all dogs upon arrival on Day -7 from the Vendor to confirm the dogs do not have active CPV infection. Subsequent CPV SNAP tests are evaluated at a consistent time of morning from Day -1 through study termination. In the unlikely event that a dog does not test positive on CPV SNAP test and therefore does not receive Mab A v2/CP, it is excluded from efficacy analysis.

Serum CPV-2 Hemagglutination (HI) Testing. Whole blood samples are collected (approximately 1.5 mL) from all dogs via a jugular, cephalic, or lateral saphenous vein into serum separator tubes and processed to serum on Days -7, 0, 3, 4, 6, 8, 10, and 13 with the Day 0 sample being collected prior to being challenged with virulent CPV-2b. Samples are also collected prior to unscheduled euthanasia. Samples are inventoried and expedited in shipment on dry

ice to the Laboratory Investigator at the earliest opportunity. Samples are stored at frozen conditions (≤-60° C.) until shipment. Data are recorded.

Fecal CPV-2 Hemagglutination Assay (HA) Testing. A bulk fecal sample is collected from each dog on Day 0 through study termination. Bulk feces is collected from each dog into a 50 mL tube, individually frozen, and kept for fecal HA testing to validate CPV SNAP test results and to meet one of the four CPV-2b infection criteria. Fecal samples are homogenized prior to HA testing to maximize accuracy of results and sample size. Procedures for collection, storage and shipping of feces follow the Laboratory Investigator's instructions. Data are recorded.

CPV-2b Infection Criteria. The criteria for CPV infection are in accordance with 9 CFR § 113.317, (c), (3), (i). This regulation defines the four criteria for an active CPV infection. A valid CPV-2b challenge should produce three of the four criteria or mortality in at least four of seven CP dogs. Per 9 CFR § 113.317 definition, the four criteria of parvovirus are as follows:

Body temp ≥103.4° F.

Lymphopenia of ≥50% of pre-challenge normal

Diarrhea, mucus and/or blood in the feces

Viral hemagglutinins at a level of ≥1:64 in a 1:5 dilution of feces (or a test of equal sensitivity)

These four criteria are monitored daily in all dogs via q 12-hour GHOs including rectal temperature (RT), hematology at specified intervals to monitor lymphocyte counts, and CPV SNAP tests which are performed daily with results verified via fecal HA. Post challenge, dogs from either group that die due to CPV infection are considered a mortality from CPV infection. Dogs that are deemed moribund and must be euthanized via the rescue clause are also considered a mortality from CPV infection.

Clinical Pathology (Hematology and Clinical Biochemistry). On Day -7 a baseline complete blood count (CBC) and biochemical profile is evaluated to verify the dogs are healthy with no pre-existing conditions. On Days -1, 0, 3, 4, 5, 6, 7, 8, 10 and 13 blood is collected for CBC to monitor blood lymphocyte counts. The samples should be collected at a consistent time each morning and recorded. Approximately 0.5 mL of blood is collected for hematology and approximately 1 mL of blood is collected for clinical biochemistry via any accessible vein (e.g. jugular, cephalic, or lateral saphenous vein). All samples are analyzed to evaluate CBC data in real-time to determine if lymphopenia has occurred which is one of the four clinical signs consistent with active CPV per 9 CFR 113.317.

Effectiveness Outcomes

Primary Outcome. The primary efficacy variable is CPV-induced mortality. Mortality rate in Mab A v2 and CP groups is calculated. The prevented fraction (PF) and lower 95% confidence limit is calculated to compare the Mab A v2 group to the CP group with Room as a stratified variable using the Cochran-Mantel-Haenszel method (the FREQ procedure in SAS, SAS Institute, Cary NC, version 9.4 or later). In case the PF cannot be generated due to the data (e.g., 100% Mab A v2 survival rate), and results are homogeneous across rooms, the same analysis without Room as a stratified variable is conducted. The primary efficacy endpoint is met if the lower 95% confidence limit of the PF is >0 and the mortality rate in the CP group is higher than the Mab A v2 group. Mortality in Mab A v2/CP dogs is defined as death from CPV infection or dogs which are deemed moribund and must be euthanized via the rescue clause. In the unlikely event that a dog does not test positive on CPV

63

SNAP test and therefore does not receive Mab A v2/CP, it is excluded from efficacy analysis.

Rescue Clause. If an individual dog is deemed moribund by the Test Facility Veterinarian, they can remove the dog from the study via the rescue clause and elect euthanasia. Dogs removed from the study for humane reasons are included in the efficacy assessment as a “mortality” unless the reason for euthanasia is unrelated to CPV.

The schedule of events is provided in Table 15 below.

TABLE 15

	Acclimation							In-Life			
	Study Day										
	Day -7	Day -6	Day -5	Day -4	Day -3	Day -2	Day -1	Day 0	Day 1	Day 2	Day 3
GHOs and RT ¹	X	X	X	X	X	X	X	q 12	q 12	q 12	q 12
PE and BW ²	X						X				
IVP/CP Dose ³									TBD		
Intranasal CVP-2b Challenge								X			
Randomization							X				
Hematology	X						X	X			X
Biochemistry	X										
CPV SNAP Test	X						X	X	X	X	X
CPV-2 HI (serum)	X							X			X
Bulk feces collection for CPV-2 HA								X	X	X	X
(feces) ⁴											
Fenbendazole	X	X	X	X	X						
Sulfadimethoxine	X	X	X	X	X						
Study Termination											

	In-Life Study Day										May also be Unscheduled
	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10	Day 11	Day 12	Day 13	
GHOs and RT ¹	q 12	q 12	q 12	q 12	q 12	q 12	q 12	q 12	q 12		X
PE and BW ²											
IVP/CP Dose ³										X	X
Intranasal CVP-2b Challenge											
Randomization											
Hematology	X	X	X	X	X		X			X	
Biochemistry											
CPV SNAP Test	X	X	X	X	X	X	X	X	X	X	X
CPV-2 HI (serum)	X		X		X		X		X	X	
Bulk feces collection for CPV-2 HA	X	X	X	X	X	X	X	X	X	X	X
(feces) ⁴											
Fenbendazole											
Sulfadimethoxine											
Study Termination										X	

¹GHOs occur at least once daily during Acclimation. On Day 0, GHOs and rectal temperature (RT) are recorded every 12 hours (±1 hour) through study termination.

²PE and BW are also performed on the same day as IVP/CP Dosing.

³IVP/CP Dose are administered upon detection of CPV-2 in individual dog's feces using the CPV SNAP test.

⁴Fecal HA are conducted to verify parvovirus infection and validate CPV SNAP test results.

55

The following is the Canine Parvovirus Challenge Inoculation Procedure:

1. Remove food ~12 hours before the challenge material is administered. Water remains available.
2. Remove the challenge material from the ultracold (−80° C.) freezer and allow to thaw at room temperature approximately 1 hour before use.
3. Thaw five extra aliquots in order to have back-up doses in case of handling error.
4. Retain challenge material on wet ice until use.
5. Draw up 1 mL aliquot challenge material with a syringe one dose at a time.

64

6. Use syringe (without needle) with nasal canula to administer 0.5 mL into each nostril.
7. Restraint is by hand; no sedation is needed.
8. Handler holds the dog's head up with nose slightly elevated during dosing.
9. Allow a few seconds time between each nostril for the dog to swallow and be comfortable.
10. Expect that some challenge material may be expelled from the nostrils.

11. Record Dog ID, date and time on the Animal Challenge Record as dosing is completed.
12. Return food to all dogs immediately after dosing.
13. Return unused challenge material to −80° C. freezer, label as “retention aliquot, date”.
14. Retain aliquot until study end and return to the Laboratory Investigator.

The following is the Bulk Feces Collection Procedure:

1. Select individual fecal pile (or section of pile) that appears to be most abnormal.
2. Using a separate wooden tongue depressor, collect up to several grams of feces from each individual dog—

65

- one collection per dog per day. Highly diarrheic feces may need to be collected using a syringe.
- Place feces into 50 mL conical plastic centrifuge tube. Tube should be no more than half full.
 - Label with Dog ID and date.
 - Store in plastic freezer bags, grouped by date and treatment group.
 - Freeze at -80° C. until shipping to the Laboratory Investigator.
 - At the end of the collection period, ship all collected feces on dry ice to arrive overnight to the Laboratory Investigator.

Example 12

Prophylaxis Pivotal Effectiveness Study

The purpose of this pivotal study is to evaluate the effectiveness of Mab A v2 as a prophylactic therapy to prevent clinic signs of canine parvovirus (CPV) infection.

Satisfactory completion of this study and demonstration of efficacy may support a label claim stating, "This product has been shown to be effective for passive immunity of healthy dogs 13 weeks of age or older against canine parvovirus (CPV) disease".

This GCP, randomized, blinded, placebo-controlled study is 14 days in length following at least a 7-day acclimation period. A total of 25 naïve, CPV-2 seronegative (CPV hemagglutination inhibition (HI) titer <20) dogs are randomized to two groups (n=20 dogs in Group 1 and n=5 in Group 2) as per Table 16. Dogs are ≤ 13 weeks old on Day 0, of any sex, weigh ≥ 1.5 kg upon arrival and are in good health and free from any clinical signs of disease at study start.

During acclimation, the dogs are acclimated to feeding, housing and handling procedures while being administered anthelmintic and antibiotic medications to eliminate common helminth and coccidia infections.

On Day 0, all dogs are administered their assigned Mab A v2 (Group 1) or Control Product [CP; Group 2; phosphate buffered saline (PBS)] via subcutaneous (SC) injection. Mab A v2 is administered in a single 5 mg/kg dose based on body weight. The CP has a pH of 7.2 ± 0.2 administered in a volume equal to Group 1. The dose for Mab A v2 and CP is calculated as follows:

$$\{(\text{Body weight in kg}) \times (5 \text{ mg/kg})\} \pm \{25 \text{ mg/mL}\} = \text{volume in mL to be injected}$$

Local and systemic safety data are collected from Day 0 (post treatment) to Day 2. On Day 3 all dogs from both Groups 1 and 2 are administered virulent CPV-2b at a dose of approximately 1×10^6 TCID₅₀/mL intranasally (IN) (a volume equaling approximately 0.5 mL per nostril).

At scheduled times outlined in Table 17 individual dog health is closely monitored for 14 days with physical examinations (PEs), measurement of rectal temperature (RT), cage-side CPV-2 ELISA IDEXX SNAP test (CPV SNAP test), serum hemagglutination inhibition assay (HI), fecal hemagglutination assay (HA), and hematology.

The primary efficacy variable is prevention of CPV infection in Group 1 dogs as defined by the criteria of 9 CFR § 113.317, (c), (3), (i):

Body temp $\geq 103.4^{\circ}$ F.

Lymphopenia of $\geq 50\%$ of pre-CPV-2b challenge normal Diarrhea, mucus and/or blood in feces

Viral hemagglutinins at a level of $\geq 1:64$ in a 1:5 dilution of feces (or a test of equal sensitivity)

66

Efficacy is demonstrated if at least 80% of Group 2 (CP) dogs have at least three of the four criteria outlined in 9 CFR § 113.317, (c), (3), (i) and if at least 19 Mab A v2 treated dogs remain alive and show ≤ 1 of the four possible clinical signs.

See treatment groups in Table 16 below.

TABLE 16

Group	Number of Dogs	Treatment	Route	Concentration (mg/mL)	Dose
1	20	Mab A v2	SC	25 mg/mL	5 mg/kg
2	5	CP	SC	25 NA	Volume equivalent to Group 1

Randomization. This study may require a two-step randomization. The first step is to determine which 25 dogs are enrolled in the study. Dogs are then randomized at a block size of 5 with a 4:1 ratio to Groups 1 and 2, respectively. The randomization is based on random numbers generated by the PLAN procedure in SAS (version 9.4 or later, SAS Institute, Cary NC) using the randomized selection method with a seed number.

After arrival to the Test Facility, dogs are placed into cages in 2 rooms based on the order of arrival, and dogs from the same litter/block stay in the same room if possible. Dogs might get secondary exposure to virus that is shed from infected dogs. To keep a similar level of secondary virus exposure for both treatment groups and to maintain blinding to Test Facility personnel, each room contains both treatment groups. After the 25 dogs are selected, their cages are sequentially numbered from 1-25. Dogs from the same litter have adjacent cage numbers, with litters of 5 taking earlier numbers (e.g., 1-5, 6-10, etc.) per room. One room should contain cage numbers 1-10, and the other room contain cage numbers 11-25. The dogs are enrolled in the order of cage numbers in the randomization list.

Masking of Study. The Investigator and all other Test Facility Personnel (excluding the Dispenser and one designated person who restrains dogs for dosing) do not have knowledge of the actual treatment group assignment at Day 0. Only the Dispenser and the person designated for restraining dogs during dosing at the Test Facility has access to actual treatment group assignment (Mab A v2 or CP) for each dog. The Dispenser and the dog restrainer/holder do not perform any study assessments.

Specifications. A total of at least 27 purpose-bred, unvaccinated, CPV-2 seronegative (HI <20), healthy Beagle dogs are procured. The dogs are confirmed CPV negative via CPV SNAP test upon arrival to the Test Facility. Although the study enrollment includes 25 dogs, two additional dogs are purchased to ensure dog numbers are met after screening and acclimation. Ideally, the 27 dogs are comprised of 5 litters with 5 dogs per litter, plus 2 dogs from another litter (a partial litter).

Observations

Physical Examination (PE). Physical examinations are performed on all dogs by the Test Facility Veterinarian on Days -7, 0, 3, 13 and on an unscheduled basis when necessary. Physical examinations include a comprehensive assessment of all body systems, including hydration status and BW.

Food Consumption. Once daily, qualitative assessment of food consumption is made in conjunction with general health observations (GHOs). All abnormalities in food consumption observed are recorded.

Injection Site Observations (ISO). ISO is made at 4 hours $\pm$ 1 hour after administration of Mab A v2/CPV for clinical signs of local inflammation (e.g. erythema, heat or swelling) once daily on Day 1 and Day 2. The size of any reactions is recorded. All abnormalities observed are recorded.

General Health Observations (GHOs) and Measurement of Rectal Temperature (RT). GHOs are conducted by the Test Facility Veterinarian or their designee once daily during the acclimation period from Day -7 through Day -1. GHOs are conducted every 12 hours ($\pm$ 1 hour) from Day 0 through Day 12 or on an unscheduled basis. Beginning on Day 0, in conjunction with each GHO, the rectal temperature (RT) is also measured using a calibrated thermometer in Fahrenheit scale. General health observations include, but are not be limited to, observations of general physical appearance, abnormalities of food or water consumption, and/or occurrence of vomiting or diarrhea. The condition of all dogs, including those without abnormal signs, are documented.

Cage-side CPV ELISA (CPV SNAP Test). Cage-side CPV ELISA tests are performed using the IDEXX CPV SNAP test according to the manufacturer's instructions. Preliminary CPV SNAP tests are performed on all dogs upon arrival on Day -7 from the Vendor to confirm the dogs do not have active CPV infection. Subsequent CPV SNAP tests are evaluated at a consistent time of morning on Day 0, Day 3 (prior to inoculation with virulent CPV-2b) and then once daily through study termination. CPV SNAP test data are recorded.

Serum CPV-2 Hemagglutination (HI) Testing. Whole blood samples are collected (~1.5 mL) from all dogs via a jugular, cephalic, or lateral saphenous vein into serum separator tubes and processed to serum at a consistent time of morning on Days -7, 0, 2, 3, 4, 6, 8, 10, and 13 with the Day 3 sample being collected prior to inoculation with virulent CPV-2b. Samples are also collected prior to unscheduled euthanasia. Samples are inventoried and expedited in shipment on dry ice to the Laboratory Investigator at the earliest opportunity. Samples are stored at frozen conditions ($\leq -60^{\circ}$ C.) until shipment. Data are recorded.

Fecal CPV-2 Hemagglutination Assay (HA) Testing. A bulk fecal sample is collected from each dog on Day 0, and Day 3 through study termination at a consistent time of morning. Bulk feces is collected from each dog into a 50 mL tube, individually frozen, and kept for fecal HA testing to validate CPV SNAP test results and to meet one of the four CPV-2b infection criteria. Fecal samples are homogenized prior to HA testing to maximize accuracy of results and sample size. Procedures for collection, storage and shipping of feces follow the Laboratory Investigator's instructions. Data are recorded.

CPV-2b Infection Criteria. The criteria for CPV infection are in accordance with 9 CFR § 113.317, (c), (3), (i). This regulation defines the four criteria for an active CPV infection. A valid CPV-2b challenge should produce three of the four criteria in at least 80% of the challenge control dogs. Per 9 CFR § 113.317 definition, the four criteria of parvovirus are as follows:

Body temp $\geq 103.4^{\circ}$ F.

Lymphopenia of $\geq 50\%$ of pre-challenge normal

Diarrhea, mucus and/or blood in the feces

Viral hemagglutinins at a level of $\geq 1:64$ in a 1:5 dilution of feces (or a test of equal sensitivity)

These four criteria are monitored daily in all dogs via q 12-hour GHOs including rectal temperature (RT), hematology at specified intervals to monitor lymphocyte counts, and CPV SNAP tests which are performed daily with results verified via fecal HA.

Clinical Pathology (Hematology and Clinical Biochemistry). On Day -7 a baseline complete blood count (CBC) and biochemical profile is evaluated to verify the dogs are healthy with no pre-existing conditions. On Days 0, 3, 5, 6, 7, 8, 9, 10 and 13 blood is collected for CBC to monitor blood lymphocyte counts. The samples are collected at a consistent time each morning and recorded. Approximately 0.5 mL of blood is collected for hematology and approximately 1 mL of blood is collected for clinical biochemistry via any accessible vein (e.g. jugular, cephalic, or lateral saphenous vein). All samples are analyzed to evaluate CBC data in real-time to determine if lymphopenia has occurred which is one of the four clinical signs consistent with active CPV per 9 CFR 113.317.

Effectiveness Outcomes

Primary Outcome. The primary efficacy variable is prevention of CPV infection as defined by the criteria of 9 CFR § 113.317, (c), (3), (i):

Body temp $\geq 103.4^{\circ}$ F.

Lymphopenia of $\geq 50\%$ of pre-CPV-2b challenge normal (Average of Days -7, 0 and 3

pre-challenge)

Diarrhea, mucus and/or blood in feces

Viral hemagglutinins at a level of $\geq 1:64$ in a 1:5 dilution of feces (or a test of equal sensitivity) Efficacy is established if 80% of control dogs have three of the four criteria outlined in 9 CFR 113.317 and if at least 19 Mab A v2 treated dogs remain alive and show ≤ 1 of the four possible clinical signs.

Second Outcomes. Secondary outcomes such as CPV SNAP test results, serum HI titers, or other parameters may be evaluated.

Safety Outcomes. Dogs from this pivotal study are applied to the required number of dogs treated in the Mab A v2 Pivotal Field Safety Study. Safety evaluation is through collation of all local and systemic adverse events (AEs) from Day 0 (post-treatment) through Day 3. Specifically, dogs are monitored for injection site reactions such as erythema, heat and swellings (via daily ISO) and systemic reactions, including anaphylaxis post administration of Mab A v2. The AE data are not statistically analyzed but are summarized and tabulated for final reporting.

Serious AE. A serious AE is any AE that results in death, is life-threatening, results in persistent or significant disability/incapacity, or a congenital anomaly or birth defect. The Study Investigator notifies the Clinical Development Manager within 24 hours of the occurrence of any serious AE.

The schedule of events is provided in Table 17 below.

TABLE 17

	Acclimation Phase							Treatment/Observation			
	Study Day										
	Day -7	Day -6	Day -5	Day -4	Day -3	Day -2	Day -1	Day 0	Day 1	Day 2	Day 3
GHOs ¹	X	X	X	X	X	X	X	q 12	q 12	q 12	q 12

71

3. Place feces into 50 mL conical plastic centrifuge tube. Tube should be no more than half full.
4. Label with Dog ID and date.
5. Store in plastic freezer bags, grouped by date and treatment group.
6. Freeze at -80° C. until shipping to the Laboratory Investigator.
7. At the end of the collection period, ship all collected feces on dry ice to arrive overnight to the Laboratory Investigator.

Example 13

Half-Life Degradation and Lack of Antibody Interference

This in vivo study monitors degradation of Mab A v2 in vivo and evaluates the duration hemagglutination inhibition (HI) titer levels remain sufficient to neutralize vaccinal strains of CPV. The following objectives are addressed:

Characterizes the half-life degradation of Mab A v2 after subcutaneous (SC) administration

Determines the timeframe after administration that Mab A v2 will no longer inhibit active immunization against parvovirus (lack of antibody interference, or LOAI)

Serologic monitoring of HI titers after administration of Mab A v2 may characterize the degradation half-life. LOAI is determined by active immune responses against parvovirus after vaccine administration in at least 80% of dogs within each group. Active response is defined as a four-fold increase in titer over baseline (seroconversion) with achievement of antibody titers equal to or above a 1:80 dilution as determined by HI. This level of antibody response provides immunity against canine parvovirus, as shown by challenge of immunity studies (Pollock and Carmichael, JAVMA 1982 Jan. 1; 180(1):37-42).

Secondarily, safety data are collected post treatment and prior to CPV immunization which may be used contribute to the total number of dogs required to meet the 300-dog minimum requirement by the USDA-CVB to satisfy Veterinary Services Memorandum (VSM) No. 800.204.

Satisfactory completion of this study and demonstration when LOAI to CPV immunization occurs may support two label claims stating:

“This product has been shown to passively immunize healthy dogs 13 weeks of age or older for prevention of canine parvovirus (CPV) disease.”

“This product has been shown to be effective for the treatment of canine parvovirus (CPV) disease in dogs 13 weeks of age or older.”

In example 9 above, describing prior study KB-030-PK-301, Mab A v2 neutralized vaccinal CPV much like maternally derived antibody. The current study may provide information regarding the timing of CPV vaccination subsequent to administration of the Mab A v2 as a prophylactic therapy.

This is a randomized, non-blinded, open label, GCP study conducted in healthy CPV seronegative Beagle dogs. The study is up to 113 days in length with a 7-day acclimation period. During acclimation, the dogs are acclimated to feeding, housing and handling procedures; and receive an anthelmintic and antibiotic medications to eliminate common helminth and coccidia infections.

On Day 0, 5 mg/kg of Mab A v2 is administered SC in the interscapular region to Groups 1 and 2. Groups 3 and 4 do not receive Mab A v2. From Day 0 to Day 41, general health observations (GHOs), physical examinations (PEs) and

72

injection site observations (ISO) are collected at various time points to monitor the dogs for any local and systemic reactions to the Mab A v2.

On Day 42, one pre-assigned monovalent CPV vaccine is administered to Groups 1 and 3, and another pre-assigned monovalent CPV vaccine is administered to Groups 2 and 4. Each dog is administered the pre-assigned monovalent CPV vaccine (either Vaccine 1 or Vaccine 2) on Day 42 after serum has been collected for HI titers. If vaccinal seroconversion is not documented after the previous vaccine, a booster vaccination (“2nd CPV vaccine”) is administered 3 weeks after the first vaccine, on Day 63. If vaccinal seroconversion is not documented after the 2nd CPV vaccine, a third booster of the pre-assigned monovalent CPV vaccine (“3rd booster”) is administered on Day 84. Study Exit by individual dogs occurs upon receipt of HI results which document seroconversion. In the event that some dogs have not mounted an immune response to vaccination after Day 91 HI titers, a fourth and final monovalent CPV booster (“4th booster”) is administered on Day 105. Serum hemagglutination inhibition (HI) titers are measured at various time points to monitor degradation of Mab A v2 and when LOAI to CPV vaccination occurs.

From Day 43 to Study Exit, GHOs and PEs are collected at various time points to monitor the dogs for general health.

The two monovalent CPV vaccines to be preassigned and administered in the study are:

Vaccine 1	
Registered name:	Nobivac ® Canine 1-PV
Vaccine:	Canine parvovirus (modified live)
Pharmaceutical form:	Sterile injectable solution
Test dose:	1 mL (1 dose)
Route of administration:	Subcutaneous
Manufacturer:	Merck Animal Health
Packaging:	Individual vials per manufacturer packaging
Storage conditions:	2-7° C.
Accountable unit:	1 vial
Vaccine 2	
Registered name:	Recombitek ®
Vaccine:	Canine parvovirus (modified live)
Pharmaceutical form:	Sterile injectable solution
Test dose:	1 mL (1 dose)
Route of administration:	Subcutaneous
Manufacturer:	Merial
Packaging:	Individual vials per manufacturer packaging
Storage conditions:	2-7° C.
Accountable unit:	1 vial

See treatment groups in Table 18 below.

TABLE 18

Group	Number of Dogs	Treatment	Route	Concentration (mg/mL)	Dose	Monovalent Vaccine to be Administered
1	≥10	Mab A v2	SC	25 mg/mL	5 mg/kg	Vaccine 1
2	≥10	Mab A v2	SC	25 mg/mL	5 mg/kg	Vaccine 2
3*	≥5	None	NA	NA	NA	Vaccine 1
4*	≥5	None	NA	NA	NA	Vaccine 2

Vaccine 1 = Nobivac Canine 1-PV ® (Merck);
Vaccine 2 = Recombitek ® (Merial)

Randomization. Dogs (≤13 weeks old on Day 0, of any sex, and weighing ≥1.5 kg) are randomized at a 2:2:1:1 ratio to Groups 1 through 4 on or before Day 0 using a randomization list provided by the Sponsor's Study Programmer.

Treatment is randomly assigned within each block of 6 dogs. The randomization is based on random numbers generated by the PLAN procedure in SAS (SAS Institute, Cary NC, version 9.4 or later) using the randomized selection method with a seed number.

Observations

Physical Examinations (PEs). PEs are performed on all days on Days -7, 0, Study Exit and on an unscheduled basis when necessary. Physical examinations include a comprehensive assessment of all body systems, including hydration status and body weight (BW).

Injection Site Observations (ISOs). On Day -3 the injection site is clipped in preparation for dosing on Day 0. Prior to dosing, the clipped injection site is evaluated and recorded as free of any inflammation or clipper burn. Beginning on Day 0, the injection site is evaluated once daily through Day 14 for local injection site reactions including but not limited to erythema, heat, and swelling (e.g. nodule). The size of any reactions is recorded. Any injection site reaction observed is documented on the Injection Site Observation Form and is recorded as an adverse event (AE).

General Health Observations (GHOs). GHOs are conducted by the Study Investigator or a designee once daily from Day -7 through Study Termination. General health observations include, but are not limited to, observations of general physical appearance, abnormalities of food or water consumption, and/or occurrence of vomiting or diarrhea. The condition of all dogs, including those without abnormal signs, is documented on the General Health Observation Form for each dog at each observation period. On Days -7, 0 and Study Exit when PEs are conducted, the PE replaces the GHO.

Serum CPV-2 Hemagglutination (HI) Testing. Whole blood samples are collected (approximately 1.5 mL) from all dogs via a jugular, cephalic, or lateral saphenous vein and are placed into serum separator tubes and processed to serum on Days -7, 0, 1, 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, 91, 98, 105, 112. If seroconversion is documented on HI, subsequent scheduled sampling ceases, and the individual dog exits the study. Samples are stored at frozen conditions (60 to -80° C.) until they are delivered to the Analytical Laboratory at the earliest opportunity.

Clinical Pathology. On Day -7 a baseline complete blood count (CBC) and biochemical profile are evaluated to verify the dogs are healthy with no pre-existing conditions. Approximately 0.5 mL of blood is collected for hematology and approximately 1 mL is collected for clinical biochemistry from the jugular, cephalic, or lateral saphenous vein.

Adverse Events (AE). An AE is defined as any observation in dogs that is unfavorable, unexpected, and unintended, and occurs after the use of a veterinary product, whether or not considered to be product related. Once experimental treatment is initiated (Day 0 when Mab A v2 is administered), any abnormal GHO, PE, or ISO is considered an AE.

Primary Endpoint. The primary endpoint for each group is the number of days from Day 0 to the time when $\geq 80\%$ of dogs within the group have demonstrated CPV seroconversion via serum HI titers to the monovalent CPV vaccination. Seroconversion is defined as a four-fold increase in titer over baseline with achievement of antibody titers equal to or above a 1:80 dilution as determined by serum HI titers. The number of dogs per group that have demonstrated an immune response to vaccines and the study day when this occurred is not statistically analyzed but summarized and tabulated for final reporting.

See the schedule of events in Table 19 below.

TABLE 19

Study Day	Fenbendazole & Sulfadimethoxine	Randomization ¹	Body Weight	Physical Exam	General Health Observations	Mab A v2 Administration	Injection Site Observations ²	Serum CPV-2 HI Testing	CBC	Clinical Biochemistry	Clip Hair at Injection Site	1 st Monovalent CPV vaccine	2 nd Monovalent CPV vaccine ³	3 rd Monovalent CPV Vaccine ³	4 th Monovalent CPV Vaccine ³
-7	X		X	X				X	X	X					
-6	X				X										
-5	X				X										
-4	X				X										
-3	X				X										
-2					X										
-1					X										
0		X	X	X		X	X	X							
1					X		X	X							
2-6					X		X	X							
7					X		X	X							
8-13					X		X	X							
14					X		X	X							
15-20					X		X	X							
21					X		X	X							
22-27					X		X	X							
28					X		X	X							
29-34					X		X	X							
35					X		X	X							
36-41					X		X	X							
42					X		X	X				X			
43-48					X		X	X							
49					X		X	X							
50-55					X		X	X							
56					X		X	X							
57-62					X		X	X					X		
63					X		X	X							
64-69					X		X	X							
70					X		X	X							
71-76					X		X	X							
77					X		X	X							
78-83					X		X	X							
84					X		X	X						X	
85-90					X		X	X							
91					X		X	X							
92-97					X		X	X							
98					X		X	X							
99-104					X		X	X							
105					X		X	X							X
106-111					X		X	X							
112			X	X				X							

¹Randomization occurs on or before Day 0.²Injection Site Observations for clinical signs of intolerance (e.g. erythema, heat, swelling) are made daily from Day 0, prior to dosing and after dosing, through Day 14. All abnormal observations are characterized (e.g. size of swelling) and documented on the Injection Site Observation CRF³1st, 2nd, 3rd, and 4th CPV vaccines are only administered to dogs who have not demonstrated seroconversion from the previous vaccine

Multi-Center, Pivotal Safety Study in Healthy Dogs

This in vivo field safety study evaluates the local and systemic tolerance of MabA v2. Individual dogs are observed for injection site inflammation (including but not limited to erythema, heat and/or swelling) and for systemic reactions (e.g. anaphylaxis).

A minimum of 300 dogs are administered Mab A v2, either via subcutaneous (SC) (n=150) or via intravenous (IV) (n=150) routes. The safety data collected during this study is intended to meet the pivotal field safety 300-dog requirement for the United States Department of Agriculture Center for Veterinary Biologics (USDA-CVB).

Satisfactory completion of this study and demonstration of safety may support two label claims stating:

“This product has been shown to be effective for passive immunity of healthy dogs 10 weeks of age or older for prevention of canine parvovirus (CPV) disease.”

“This product has been shown to be effective for the treatment of canine parvovirus (CPV) disease in dogs 10 weeks of age or older.”

Dose volume is calculated in mL by the EDC system based on Day 0 body weight (BW). The dose is calculated by the EDC System as follows:

$$\{(\text{Body weight in kg}) \times (5 \text{ mg/kg})\} + \{\text{TBD mg/mL}\} = \text{volume in mL to be injected}$$

Subcutaneous doses are administered into the subcutis of the interscapular region. Intravenous doses re administered in a cephalic vein.

At scheduled times outlined in Table 20, individual dog health is closely monitored via physical examinations (PEs), injection site observations (ISO) and general health observations (GHOs).

Safety evaluation is through collation of all adverse events (AEs). Specifically, dogs are monitored for injection site reactions such as erythema, heat or swelling (via daily ISO) and for systemic reactions, including anaphylaxis, post administration of Mab A v2. The AE data are not statistically analyzed but are summarized and tabulated.

See treatment groups in Table 20 below.

TABLE 20

Group	Number of Dogs of Minimum Age (≤ 10 weeks)	Number of Dogs of Older Age (> 10 weeks)	Mab A v2 Treatment	Route	Concentration (mg/mL)	Dose
1	≥ 25	≥ 50	PLS No. 1	SC	TBD	5 mg/kg
2	≥ 25	≥ 50	PLS No. 2	SC	TBD	5 mg/kg
3	≥ 25	≥ 50	PLS No. 1	IV	TBD	5 mg/kg
4	≥ 25	≥ 50	PLS No. 2	IV	TBD	5 mg/kg

This is a randomized, non-masked, GCP field safety study conducted in young, healthy, purpose-bred dogs. This study is 16 days in length at each study site. At least 100 minimum age dogs (≤ 10 weeks of age) of any gender, and at least 200 dogs older than the minimum age (> 10 weeks of age) of any gender, are enrolled into the study.

Exclusion criteria. The dog:

Is currently enrolled in another study.

Has been enrolled in another study within the previous 30 days.

Has received chimeric biologics ≤ 90 days prior to Day -2. Is not cooperative for study procedures.

Is not reasonably expected to survive the duration of the study.

Has any pre-existing skin conditions which may confound ISOs.

Has current medical history suggesting a significant comorbidity (e.g., polyuria, polydipsia, polyphagia, vomiting or diarrhea, unexplained weight loss, etc.).

The minimum age dogs are located in one site (Ridgland Farms (Mount Horeb, WI)) whereas the remaining dogs are divided among Test Facilities in three geographic regions. A minimum of three Test Facilities located in three separate geographical regions and unique breeding colonies to diversify genetics as much as possible are used. At each site, dogs are randomized 1:1:1:1 to Groups 1, 2, 3 and 4 as per Table 20.

On Day 0, all dogs are administered Mab A v2 either SC or IV. Two different pre-licensing serials (PLS) of Mab A v2 are administered in this study. Within each age group and treatment method group, approximately half of the dogs randomly receive PLS No. 1 and the other half randomly receive PLS No. 2.

Randomization. Due to challenges with finding Test Facilities with appropriately aged dogs, the approximately 100 minimum age dogs is sourced from one location. Approximately 200 remaining dogs are divided among Test Facilities in at least three geographic regions and are divided as evenly as possible amongst the Test Facilities. Randomization is stratified by the Test Facility site into blocks containing four dogs. Within each site, dogs are randomized at a 1:1:1:1 ratio to Groups 1 (PLS No. 1, SC), 2 (PLS No. 2, SC), 3 (PLS No. 1, IV), and 4 (PLS No. 2, IV). The randomization is based on random numbers generated by the PLAN procedure in SAS (SAS Institute, Cary NC, version 9.4 or later) using the randomized selection method with a seed number.

Acclimation. This study is conducted in Test Facilities where the dogs are previously acclimated to diet, housing, handling procedures, etc. Therefore, only a 2 day acclimation phase is implemented to ensure dogs meet the enrollment criteria and to clip the injection sites on Day -2.

Physical Examination (PE). A PE is performed by on site veterinarian on Days -2, 0 (pre-dose), 7, and 14. PEs include a subjective assessment of general appearance and attitude, otic, ocular, oral, mucous membranes, respiratory, cardiovascular, gastrointestinal, neurologic, musculoskeletal, integumentary and genitourinary. VeDDRA low level terminology is utilized to describe specific PE abnormalities. Abnormalities (excluding pre-existing conditions and worsening abnormalities) recorded after Day 0 are considered as AEs.

Injection Site Observations (ISO). On Day -2 the injection site is clipped in preparation for dosing on Day 0. Prior to treatment, the clipped injection site is evaluated and

recorded as free of any inflammation or clipper burn. Beginning on Day 0 through Day 14, the injection site location is evaluated once daily for local injection site reactions including but not limited to erythema, heat, swelling (e.g. nodule). The size of any reactions is recorded. Any injection site reaction observed is recorded as an AE.

General Health Observations (GHOs). GHOs are conducted once daily beginning on acclimation through study end with one exception: On Days -2, 0, 7 and 14 when PE are performed, the PE replaces the GHO. General health observations include, but are not limited to, observations of general physical appearance, abnormalities of food or water consumption, and/or occurrence of vomiting or diarrhea. VEDDRA low level terminology is utilized to describe specific observations. The condition of all dogs, including those without abnormal signs, is documented.

Body weights. On Day 0, BW (recorded in kilograms (kg)) is measured using a calibrated scale and recorded. Dogs may be weighed fed or fasted. Body weights are documented.

Safety Population. The safety population consists of all dogs who receive Mab A v2.

Safety Outcomes. Safety evaluation is through collation of all AEs. Specifically, dogs are monitored for injection site reactions such as erythema, heat or swelling (via daily ISO) and systemic reactions, including anaphylaxis post administration of Mab A v2. The AE data are not statistically analyzed but are summarized and tabulated.

Adverse Events (AE). An AE is defined as any observation in dogs that is unfavorable, unexpected, and unintended, and occurs after the use of a veterinary product, whether or not considered to be product related. All AEs are documented and recorded using low level VEDDRA terms.

Serious AE. An AE that is fatal or life-threatening or requires professional intervention (e.g., the dog's health is such that the Investigator determines that the dog should either be euthanized or receive medical treatment).

Non-Serious AE. An AE that is not severe enough to require professional intervention or removal of the dog from the study.

See the schedule of events in Table 21 below.

TABLE 21

Event	Day -2	Day 0	Days 1-6	Day 7	Days 8-13	Day 14	Unscheduled
Clip Hair Over Injection Site	✓						
Inclusion/Exclusion Criteria	✓						
Signalment	✓						
Physical Exam	✓	✓		✓		✓	✓
with Body Weight							
Mab A v2 Administration		✓					
General Health Observations (GHOs)*			✓		✓		
Injection Site Observation (ISO)**		✓	✓	✓	✓	✓	✓
Record Adverse Events		✓	✓	✓	✓	✓	✓
Study Exit						✓	✓

*GHOs include, but are not limited to, observations of general physical appearance, abnormalities of food or water consumption, and/or occurrence of vomiting or diarrhea.

**The injection site is evaluated once daily from Days 0 to 14 for clinical signs of local inflammation (including but not limited to erythema, heat, swelling, nodules). Size of reactions are recorded.

Identification of Variant Canine IgG-B Fc Polypeptides for Enhanced ADCC Activity

Structural models of canine CD16 and canine IgG-B Fc were prepared using the structural coordinates of human IgG1 Fc/CD16 complex (1e4k1.pdb) as template. See Sondermann P., "The 3.2-A Crystal Structure of the Human IgG1 Fc Fragment-Fc gammaRIII complex," *Nature* 406 (6793):267-73 (2000). Amino acid residues that appear to interact between wild-type canine IgG-B Fc (SEQ ID NO: 91) and canine CD16 (SEQ ID NO: 90) were identified. Both Ser10 and Ile103 of canine IgG-B Fc SEQ ID NO: 91 are in close proximity to Lys148 of canine CD16 SEQ ID NO: 90. To gain electrostatic interaction, Ser10 and/or Ile103 of canine IgG-B Fc SEQ ID NO: 90 may be substituted with acidic amino acids Asp or Glu to introduce a salt bridge to K148 of canine CD16 SEQ ID NO: 90. The distance between Ser10Asp and Lys148; between Ser10Glu and Lys148; between Ile103Asp and Lys148; and between Ile103Glu and Lys148 are each less than 4 Å (FIG. 5). Exemplary variant canine IgG-B Fc polypeptide sequences for enhanced ADCC activity include SEQ ID NO: 92, SEQ ID NO: 93, SEQ ID NO: 94, SEQ ID NO: 95, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 98, and SEQ ID NO: 99. Such variant canine IgG-B Fc polypeptides may be used to in place of wild-type canine IgG-B Fc to enhance the killing of parvovirus infected cells.

REFERENCES

1. S. Nandi, Manoj Kumar. Canine Parvovirus: Current Perspective. *Indian J. Virol.* 2010; 21(1):31-44.
2. Carla Miranda, Gertrude Thompson. Canine parvovirus: the worldwide occurrence of antigenic variants. *Journal of General Virology.* 2016; 97, 2043-2057.
3. Melissa Kennedy, Adesola Odunayo. Canine Parvovirus. *Clinician's Brief.* 2017.
4. Brindhalakshmi B, Mukhopadhyay H K, Antony P X, Thanislass J, Vijayalakshmi P, Mangadevi N. Isolation and molecular characterization of canine and feline parvovirus strains—an updated review. *Journal of Dairy, Veterinary & Animal Research.* 2016; 3(5):164-169.
5. Emilee Venn, Karolina Preisner, Pedro Boscan, David Twedt, Lauren A. Sullivan. Evaluation of an outpatient protocol in the treatment of canine parvoviral enteritis. *Journal of Veterinary Emergency and Critical Care.* 2017; 27(1):52-65.

SEQUENCE LISTING

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-continued

20	25	30
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 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable heavy chain framework
 region HC-FR4 of Mab A variant 2 (v2) , S11A

<400> SEQUENCE: 12

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ala
1 5 10

<210> SEQ ID NO 13
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain CDRL1 of Mab A

<400> SEQUENCE: 13

Lys Ala Ser Gln Asn Val Asp Ser Asn Val Asp
1 5 10

<210> SEQ ID NO 14
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain CDRL2 of Mab A

<400> SEQUENCE: 14

Lys Ala Ser Asn Arg Asn Thr
1 5

<210> SEQ ID NO 15
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain CDRL3 of Mab A

<400> SEQUENCE: 15

Met Gln Ser Thr Ser Tyr Pro Leu Thr Phe
1 5 10

<210> SEQ ID NO 16
 <211> LENGTH: 23
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain framework

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region LC-FR1 of Mab A

<400> SEQUENCE: 16

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Met Ser Ile Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Met Asn Cys
 20

<210> SEQ ID NO 17

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable light chain framework
region LC-FR2 of Mab A

<400> SEQUENCE: 17

Trp Tyr Gln Gln Lys Thr Gly Gln Ser Pro Asn Leu Leu Ile Tyr
 1 5 10 15

<210> SEQ ID NO 18

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable light chain framework
region LC-FR3 of Mab A

<400> SEQUENCE: 18

Gly Val Pro Asp Arg Phe Thr Gly Ser Gly Ser Gly Thr Asp Phe Thr
 1 5 10 15

Phe Thr Ile Ser Asn Met Gln Ala Glu Asp Leu Ala Val Tyr Tyr Cys
 20 25 30

<210> SEQ ID NO 19

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable light chain framework
region LC-FR4 of Mab A

<400> SEQUENCE: 19

Gly Ser Gly Thr Lys Leu Glu Ile Lys
 1 5

<210> SEQ ID NO 20

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable heavy chain of Mab A, (Mab
A HC)

<400> SEQUENCE: 20

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Gln
 1 5 10 15

Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr
 20 25 30

His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45

Gly Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu Ala Leu Asn
 50 55 60

Ser Arg Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val Phe Phe

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65	70	75	80
Lys Met Ser Ser Leu Gln Thr Glu Asp Thr Ala Thr Tyr Tyr Cys Ala			
	85	90	95
Arg Pro Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe Pro Tyr Trp			
	100	105	110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser			
	115	120	

<210> SEQ ID NO 21
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable heavy chain of Mab A
 variant 2, HC-FR1 Q13A, HC-FR4 S11A, (Mab A HC v2)

<400> SEQUENCE: 21

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Ala Pro Ser Gln			
1	5	10	15
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr			
	20	25	30
His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu			
	35	40	45
Gly Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu Ala Leu Asn			
	50	55	60
Ser Arg Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val Phe Phe			
65	70	75	80
Lys Met Ser Ser Leu Gln Thr Glu Asp Thr Ala Thr Tyr Tyr Cys Ala			
	85	90	95
Arg Pro Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe Pro Tyr Trp			
	100	105	110
Gly Gln Gly Thr Leu Val Thr Val Ser Ala			
	115	120	

<210> SEQ ID NO 22
 <211> LENGTH: 107
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain of Mab A, (Mab
 A LC)

<400> SEQUENCE: 22

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Met Ser Ile Ser Val Gly			
1	5	10	15
Asp Arg Val Thr Met Asn Cys Lys Ala Ser Gln Asn Val Asp Ser Asn			
	20	25	30
Val Asp Trp Tyr Gln Gln Lys Thr Gly Gln Ser Pro Asn Leu Leu Ile			
	35	40	45
Tyr Lys Ala Ser Asn Arg Asn Thr Gly Val Pro Asp Arg Phe Thr Gly			
	50	55	60
Ser Gly Ser Gly Thr Asp Phe Thr Phe Thr Ile Ser Asn Met Gln Ala			
65	70	75	80
Glu Asp Leu Ala Val Tyr Tyr Cys Met Gln Ser Thr Ser Tyr Pro Leu			
	85	90	95
Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys			
	100	105	

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<210> SEQ ID NO 23
<211> LENGTH: 457
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic: Chimeric variable heavy chain of Mab
      A and canine IgG-B , (Chimeric A HC IgG-B)

<400> SEQUENCE: 23

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Gln
1          5          10          15

Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr
20          25          30

His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
35          40          45

Gly Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu Ala Leu Asn
50          55          60

Ser Arg Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val Phe Phe
65          70          75          80

Lys Met Ser Ser Leu Gln Thr Glu Asp Thr Ala Thr Tyr Tyr Cys Ala
85          90          95

Arg Pro Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe Pro Tyr Trp
100         105         110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Thr Ala Pro
115         120         125

Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser Gly Ser Thr
130         135         140

Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu Pro Val Thr
145         150         155         160

Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His Thr Phe Pro
165         170         175

Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Met Val Thr
180         185         190

Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys Asn Val Ala
195         200         205

His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro Lys Arg Glu
210         215         220

Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys Pro Ala Pro
225         230         235         240

Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys Pro Lys
245         250         255

Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys Val Val Val
260         265         270

Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp Phe Val Asp
275         280         285

Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu Glu Gln Phe
290         295         300

Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly His Gln Asp
305         310         315         320

Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn Lys Ala Leu
325         330         335

Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly Gln Ala His
340         345         350

Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu Leu Ser Lys
355         360         365

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Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe Pro Pro Asp
 370                               375                               380

Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro Glu Ser Lys
385                               390                               395                               400

Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser Tyr Phe Leu
                               405                               410                               415

Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg Gly Asp Thr
                               420                               425                               430

Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His Tyr Thr Gln
435                               440                               445

Glu Ser Leu Ser His Ser Pro Gly Lys
450                               455

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<210> SEQ ID NO 24
<211> LENGTH: 457
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic: Chimeric variable heavy chain of Mab
        A v2 and canine IgG-B , HC-FR1 Q13A, HC-FR4 S11A, (Chimeric A HC
        v2 IgG-B),

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<400> SEQUENCE: 24

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Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Ala Pro Ser Gln
 1                               5                               10                               15

Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr
                               20                               25                               30

His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
35                               40                               45

Gly Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu Ala Leu Asn
50                               55                               60

Ser Arg Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val Phe Phe
65                               70                               75                               80

Lys Met Ser Ser Leu Gln Thr Glu Asp Thr Ala Thr Tyr Tyr Cys Ala
85                               90                               95

Arg Pro Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe Pro Tyr Trp
100                              105                              110

Gly Gln Gly Thr Leu Val Thr Val Ser Ala Ala Ser Thr Thr Ala Pro
115                              120                              125

Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser Gly Ser Thr
130                              135                              140

Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu Pro Val Thr
145                              150                              155                              160

Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His Thr Phe Pro
165                              170                              175

Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Met Val Thr
180                              185                              190

Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys Asn Val Ala
195                              200                              205

His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro Lys Arg Glu
210                              215                              220

Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys Pro Ala Pro
225                              230                              235                              240

Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys Pro Lys
245                              250                              255

Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys Val Val Val

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260				265				270							
Asp	Leu	Asp	Pro	Glu	Asp	Pro	Glu	Val	Gln	Ile	Ser	Trp	Phe	Val	Asp
	275						280						285		
Gly	Lys	Gln	Met	Gln	Thr	Ala	Lys	Thr	Gln	Pro	Arg	Glu	Glu	Gln	Phe
	290					295					300				
Asn	Gly	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Pro	Ile	Gly	His	Gln	Asp
	305				310					315					320
Trp	Leu	Lys	Gly	Lys	Gln	Phe	Thr	Cys	Lys	Val	Asn	Asn	Lys	Ala	Leu
			325						330					335	
Pro	Ser	Pro	Ile	Glu	Arg	Thr	Ile	Ser	Lys	Ala	Arg	Gly	Gln	Ala	His
			340						345					350	
Gln	Pro	Ser	Val	Tyr	Val	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Leu	Ser	Lys
		355					360						365		
Asn	Thr	Val	Ser	Leu	Thr	Cys	Leu	Ile	Lys	Asp	Phe	Phe	Pro	Pro	Asp
	370					375					380				
Ile	Asp	Val	Glu	Trp	Gln	Ser	Asn	Gly	Gln	Gln	Glu	Pro	Glu	Ser	Lys
	385				390					395					400
Tyr	Arg	Thr	Thr	Pro	Pro	Gln	Leu	Asp	Glu	Asp	Gly	Ser	Tyr	Phe	Leu
			405						410					415	
Tyr	Ser	Lys	Leu	Ser	Val	Asp	Lys	Ser	Arg	Trp	Gln	Arg	Gly	Asp	Thr
			420						425					430	
Phe	Ile	Cys	Ala	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln
		435					440						445		
Glu	Ser	Leu	Ser	His	Ser	Pro	Gly	Lys							
	450					455									

<210> SEQ ID NO 25

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab
A and canine kappa light constant region, (Chimeric A LC kappa)

<400> SEQUENCE: 25

Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ala	Ser	Met	Ser	Ile	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Met	Asn	Cys	Lys	Ala	Ser	Gln	Asn	Val	Asp	Ser	Asn
		20						25					30		
Val	Asp	Trp	Tyr	Gln	Gln	Lys	Thr	Gly	Gln	Ser	Pro	Asn	Leu	Leu	Ile
		35				40						45			
Tyr	Lys	Ala	Ser	Asn	Arg	Asn	Thr	Gly	Val	Pro	Asp	Arg	Phe	Thr	Gly
	50					55					60				
Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Phe	Thr	Ile	Ser	Asn	Met	Gln	Ala
	65				70					75				80	
Glu	Asp	Leu	Ala	Val	Tyr	Tyr	Cys	Met	Gln	Ser	Thr	Ser	Tyr	Pro	Leu
			85						90					95	
Thr	Phe	Gly	Ser	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Arg	Asn	Asp	Ala	Gln
		100						105					110		
Pro	Ala	Val	Tyr	Leu	Phe	Gln	Pro	Ser	Pro	Asp	Gln	Leu	His	Thr	Gly
		115					120					125			
Ser	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Ser	Phe	Tyr	Pro	Lys	Asp	Ile
	130					135					140				
Asn	Val	Lys	Trp	Lys	Val	Asp	Gly	Val	Ile	Gln	Asp	Thr	Gly	Ile	Gln
	145				150					155				160	

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Pro Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr
 130                135                140

Thr Ala Pro Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser
 145                150                155                160

Gly Ser Thr Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu
                165                170                175

Pro Val Thr Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His
 180                185                190

Thr Phe Pro Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
 195                200                205

Met Val Thr Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys
 210                215                220

Asn Val Ala His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro
 225                230                235                240

Lys Arg Glu Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys
                245                250                255

Pro Ala Pro Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro
                260                265                270

Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
 275                280                285

Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
 290                295                300

Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
 305                310                315                320

Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
                325                330                335

His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
 340                345                350

Lys Ala Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly
 355                360                365

Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu
 370                375                380

Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe
 385                390                395                400

Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro
                405                410                415

Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser
 420                425                430

Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
 435                440                445

Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
 450                455                460

Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
 465                470                475

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<210> SEQ ID NO 29

<211> LENGTH: 476

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable heavy chain of Mab
 A v2 and canine IgG-B with leader sequence, HC-FR1 Q13A, HC-FR4
 S11A, (Chimeric A HC v2 IgG-B with leader)

<400> SEQUENCE: 29

Met Ala Val Leu Gly Leu Leu Leu Cys Leu Val Thr Phe Pro Ser Cys

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1	5	10	15
Val Leu Ser Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Ala	20	25	30
Pro Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu	35	40	45
Ser Ser Tyr His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu	50	55	60
Glu Trp Leu Gly Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu	65	70	75
Ala Leu Asn Ser Arg Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln	85	90	95
Val Phe Phe Lys Met Ser Ser Leu Gln Thr Glu Asp Thr Ala Thr Tyr	100	105	110
Tyr Cys Ala Arg Pro Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe	115	120	125
Pro Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ala Ala Ser Thr	130	135	140
Thr Ala Pro Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser	145	150	155
Gly Ser Thr Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu	165	170	175
Pro Val Thr Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His	180	185	190
Thr Phe Pro Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser	195	200	205
Met Val Thr Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys	210	215	220
Asn Val Ala His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro	225	230	235
Lys Arg Glu Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys	245	250	255
Pro Ala Pro Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro	260	265	270
Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys	275	280	285
Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp	290	295	300
Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu	305	310	315
Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly	325	330	335
His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn	340	345	350
Lys Ala Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly	355	360	365
Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu	370	375	380
Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe	385	390	395
Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro	405	410	415
Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser	420	425	430

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Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
 435 440 445

Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
 450 455 460

Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
 465 470 475

<210> SEQ ID NO 30

<211> LENGTH: 237

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab
 A and canine kappa light constant region with leader sequence,
 (Chimeric A LC kappa with leader)

<400> SEQUENCE: 30

Met Glu Thr Asp Thr Leu Leu Leu Trp Val Leu Leu Leu Trp Val Pro
 1 5 10 15

Gly Ser Thr Gly Asp Ile Val Met Thr Gln Thr Pro Ala Ser Met Ser
 20 25 30

Ile Ser Val Gly Asp Arg Val Thr Met Asn Cys Lys Ala Ser Gln Asn
 35 40 45

Val Asp Ser Asn Val Asp Trp Tyr Gln Gln Lys Thr Gly Gln Ser Pro
 50 55 60

Asn Leu Leu Ile Tyr Lys Ala Ser Asn Arg Asn Thr Gly Val Pro Asp
 65 70 75 80

Arg Phe Thr Gly Ser Gly Ser Gly Thr Asp Phe Thr Phe Thr Ile Ser
 85 90 95

Asn Met Gln Ala Glu Asp Leu Ala Val Tyr Tyr Cys Met Gln Ser Thr
 100 105 110

Ser Tyr Pro Leu Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys Arg
 115 120 125

Asn Asp Ala Gln Pro Ala Val Tyr Leu Phe Gln Pro Ser Pro Asp Gln
 130 135 140

Leu His Thr Gly Ser Ala Ser Val Val Cys Leu Leu Asn Ser Phe Tyr
 145 150 155 160

Pro Lys Asp Ile Asn Val Lys Trp Lys Val Asp Gly Val Ile Gln Asp
 165 170 175

Thr Gly Ile Gln Glu Ser Val Thr Glu Gln Asp Lys Asp Ser Thr Tyr
 180 185 190

Ser Leu Ser Ser Thr Leu Thr Met Ser Ser Thr Glu Tyr Leu Ser His
 195 200 205

Glu Leu Tyr Ser Cys Glu Ile Thr His Lys Ser Leu Pro Ser Thr Leu
 210 215 220

Ile Lys Ser Phe Gln Arg Ser Glu Cys Gln Arg Val Asp
 225 230 235

<210> SEQ ID NO 31

<211> LENGTH: 457

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable heavy chain of Mab
 A and feline IgG-1, (Chimeric A HC IgG-1)

<400> SEQUENCE: 31

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Gln

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1	5	10	15
Thr Leu Ser	Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr		
	20	25	30
His Val His	Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu		
	35	40	45
Gly Val Met	Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu Ala Leu Asn		
	50	55	60
Ser Arg Leu	Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val Phe Phe		
65	70	75	80
Lys Met Ser	Ser Leu Gln Thr Glu Asp Thr Ala Thr Tyr Tyr Cys Ala		
	85	90	95
Arg Pro Glu	Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe Pro Tyr Trp		
	100	105	110
Gly Gln Gly	Thr Leu Val Thr Val Ser Ser Ala Ser Thr Thr Ala Pro		
	115	120	125
Ser Val Phe	Pro Leu Ala Pro Ser Cys Gly Thr Thr Ser Gly Ala Thr		
130	135	140	
Val Ala Leu	Ala Cys Leu Val Leu Gly Tyr Phe Pro Glu Pro Val Thr		
145	150	155	160
Val Ser Trp	Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro		
	165	170	175
Ala Val Leu	Gln Ala Ser Gly Leu Tyr Ser Leu Ser Ser Met Val Thr		
	180	185	190
Val Pro Ser	Ser Arg Trp Leu Ser Asp Thr Phe Thr Cys Asn Val Ala		
	195	200	205
His Pro Pro	Ser Asn Thr Lys Val Asp Lys Thr Val Arg Lys Thr Asp		
210	215	220	
His Pro Pro	Gly Pro Lys Pro Cys Asp Cys Pro Lys Cys Pro Pro Pro		
225	230	235	240
Glu Met Leu	Gly Gly Pro Ser Ile Phe Ile Phe Pro Pro Lys Pro Lys		
	245	250	255
Asp Thr Leu	Ser Ile Ser Arg Thr Pro Glu Val Thr Cys Leu Val Val		
	260	265	270
Asp Leu Gly	Pro Asp Asp Ser Asp Val Gln Ile Thr Trp Phe Val Asp		
	275	280	285
Asn Thr Gln	Val Tyr Thr Ala Lys Thr Ser Pro Arg Glu Glu Gln Phe		
290	295	300	
Asn Ser Thr	Tyr Arg Val Val Ser Val Leu Pro Ile Leu His Gln Asp		
305	310	315	320
Trp Leu Lys	Gly Lys Glu Phe Lys Cys Lys Val Asn Ser Lys Ser Leu		
	325	330	335
Pro Ser Pro	Ile Glu Arg Thr Ile Ser Lys Ala Lys Gly Gln Pro His		
	340	345	350
Glu Pro Gln	Val Tyr Val Leu Pro Pro Ala Gln Glu Glu Leu Ser Arg		
	355	360	365
Asn Lys Val	Ser Val Thr Cys Leu Ile Lys Ser Phe His Pro Pro Asp		
	370	375	380
Ile Ala Val	Glu Trp Glu Ile Thr Gly Gln Pro Glu Pro Glu Asn Asn		
385	390	395	400
Tyr Arg Thr	Thr Pro Pro Gln Leu Asp Ser Asp Gly Thr Tyr Phe Val		
	405	410	415
Tyr Ser Lys	Leu Ser Val Asp Arg Ser His Trp Gln Arg Gly Asn Thr		
	420	425	430

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Tyr Thr Cys Ser Val Ser His Glu Ala Leu His Ser His His Thr Gln
 435 440 445

Lys Ser Leu Thr Gln Ser Pro Gly Lys
 450 455

<210> SEQ ID NO 32

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab
 A and feline kappa light constant region , (Chimeric A LC kappa)

<400> SEQUENCE: 32

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Met Ser Ile Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Met Asn Cys Lys Ala Ser Gln Asn Val Asp Ser Asn
 20 25 30

Val Asp Trp Tyr Gln Gln Lys Thr Gly Gln Ser Pro Asn Leu Leu Ile
 35 40 45

Tyr Lys Ala Ser Asn Arg Asn Thr Gly Val Pro Asp Arg Phe Thr Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Phe Thr Ile Ser Asn Met Gln Ala
 65 70 75 80

Glu Asp Leu Ala Val Tyr Tyr Cys Met Gln Ser Thr Ser Tyr Pro Leu
 85 90 95

Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys Arg Ser Asp Ala Gln
 100 105 110

Pro Ser Val Phe Leu Phe Gln Pro Ser Leu Asp Glu Leu His Thr Gly
 115 120 125

Ser Ala Ser Ile Val Cys Ile Leu Asn Asp Phe Tyr Pro Lys Glu Val
 130 135 140

Asn Val Lys Trp Lys Val Asp Gly Val Val Gln Asn Lys Gly Ile Gln
 145 150 155 160

Glu Ser Thr Thr Glu Gln Asn Ser Lys Asp Ser Thr Tyr Ser Leu Ser
 165 170 175

Ser Thr Leu Thr Met Ser Ser Thr Glu Tyr Gln Ser His Glu Lys Phe
 180 185 190

Ser Cys Glu Val Thr His Lys Ser Leu Ala Ser Thr Leu Val Lys Ser
 195 200 205

Phe Asn Arg Ser Glu Cys Gln Arg Glu
 210 215

<210> SEQ ID NO 33

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab
 A and feline kappa light constant region with no N-linked
 glycosylation site, (Chimeric A LC kappa aglycos)

<400> SEQUENCE: 33

Asp Ile Val Met Thr Gln Thr Pro Ala Ser Met Ser Ile Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Met Asn Cys Lys Ala Ser Gln Asn Val Asp Ser Asn
 20 25 30

Val Asp Trp Tyr Gln Gln Lys Thr Gly Gln Ser Pro Asn Leu Leu Ile

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35	40	45
Tyr Lys Ala Ser Asn Arg	Asn Thr Gly Val Pro	Asp Arg Phe Thr Gly
50	55	60
Ser Gly Ser Gly Thr Asp	Phe Thr Phe Thr Ile	Ser Asn Met Gln Ala
65	70	75
Glu Asp Leu Ala Val Tyr	Tyr Cys Met Gln Ser	Thr Ser Tyr Pro Leu
85	90	95
Thr Phe Gly Ser Gly Thr	Lys Leu Glu Ile Lys	Arg Ser Asp Ala Gln
100	105	110
Pro Ser Val Phe Leu Phe	Gln Pro Ser Leu Asp	Glu Leu His Thr Gly
115	120	125
Ser Ala Ser Ile Val Cys	Ile Leu Asn Asp Phe	Tyr Pro Lys Glu Val
130	135	140
Asn Val Lys Trp Lys Val	Asp Gly Val Val Gln	Asn Lys Gly Ile Gln
145	150	155
Glu Ser Thr Thr Glu Gln	Asn Ser Lys Asp Ser	Thr Tyr Ser Leu Ser
165	170	175
Ser Thr Leu Thr Met Ser	Ser Thr Glu Tyr Gln	Ser His Glu Lys Phe
180	185	190
Ser Cys Glu Val Thr His	Lys Ser Leu Ala Ser	Thr Leu Val Lys Ser
195	200	205
Phe Gln Arg Ser Glu Cys	Gln Arg Glu	
210	215	

<210> SEQ ID NO 34

<211> LENGTH: 476

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Caninized variable heavy chain of
 Mab A v3 and canine IgG-B with leader sequence, (Chimeric A HC v3
 IgG-B with leader)

<400> SEQUENCE: 34

Met Ala Val Leu Gly Leu Leu Leu Cys Leu Val Thr Phe Pro Ser Cys
1 5 10 15
Val Leu Ser Gln Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys
20 25 30
Pro Gly Gly Thr Leu Arg Leu Ser Cys Thr Val Ser Gly Phe Ser Leu
35 40 45
Ser Ser Tyr His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu
50 55 60
Glu Trp Val Ala Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu
65 70 75 80
Ala Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr
85 90 95
Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr
100 105 110
Tyr Cys Ala Arg Pro Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe
115 120 125
Pro Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr
130 135 140
Thr Ala Pro Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser
145 150 155 160
Gly Ser Thr Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu
165 170 175

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Pro Val Thr Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His
    180                      185                      190

Thr Phe Pro Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
    195                      200                      205

Met Val Thr Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys
    210                      215                      220

Asn Val Ala His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro
    225                      230                      235                      240

Lys Arg Glu Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys
    245                      250                      255

Pro Ala Pro Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro
    260                      265                      270

Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
    275                      280                      285

Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
    290                      295                      300

Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
    305                      310                      315                      320

Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
    325                      330                      335

His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
    340                      345                      350

Lys Ala Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly
    355                      360                      365

Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu
    370                      375                      380

Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe
    385                      390                      395                      400

Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro
    405                      410                      415

Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser
    420                      425                      430

Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
    435                      440                      445

Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
    450                      455                      460

Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
    465                      470                      475

```

<210> SEQ ID NO 35

<211> LENGTH: 476

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic; Caninized variable heavy chain of
Mab A v4 and canine IgG-B with leader sequence, (Caninized A HC v4
IgG-B with leader)

<400> SEQUENCE: 35

```

Met Ala Val Leu Gly Leu Leu Leu Cys Leu Val Thr Phe Pro Ser Cys
1          5          10          15

Val Leu Ser Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys
20          25          30

Pro Gly Gly Thr Leu Arg Leu Ser Cys Thr Val Ser Gly Phe Ser Leu
35          40          45

```

Ser 50	Tyr	His	Val	His	Trp 55	Val	Arg	Gln	Pro 60	Gly	Lys	Gly	Leu		
Glu 65	Trp	Leu	Gly	Val 70	Met	Trp	Asn	Asp	Gly 75	Asp	Thr	Ser	Tyr	Asn	Leu 80
Ala	Val	Lys	Gly	Arg 85	Phe	Thr	Ile	Ser	Arg 90	Asp	Asn	Ala	Lys	Asn 95	Thr
Leu	Tyr	Phe	Gln 100	Met	Asn	Ser	Leu	Arg 105	Ala	Glu	Asp	Thr	Ala	Thr	Tyr
Tyr	Cys	Ala 115	Arg	Pro	Glu	Leu	Pro 120	Gly	Leu	Thr	Tyr	Gly 125	Val	Trp	Phe
Pro	Tyr	Trp	Gly 130	Gln	Gly	Thr 135	Leu	Val	Thr	Val	Ser 140	Ser	Ala	Ser	Thr
Thr 145	Ala	Pro	Ser	Val 150	Phe	Pro	Leu	Ala	Pro	Ser 155	Cys	Gly	Ser	Thr	Ser 160
Gly	Ser	Thr	Val 165	Ala	Leu	Ala	Cys	Leu	Val 170	Ser	Gly	Tyr	Phe	Pro 175	Glu
Pro	Val	Thr	Val 180	Ser	Trp	Asn	Ser	Gly 185	Ser	Leu	Thr	Ser	Gly 190	Val	His
Thr	Phe 195	Pro	Ser	Val	Leu	Gln 200	Ser	Ser	Gly	Leu	Tyr	Ser 205	Leu	Ser	Ser
Met 210	Val	Thr	Val	Pro	Ser	Ser 215	Arg	Trp	Pro	Ser	Glu 220	Thr	Phe	Thr	Cys
Asn 225	Val	Ala	His	Pro	Ala 230	Ser	Lys	Thr	Lys	Val 235	Asp	Lys	Pro	Val	Pro 240
Lys	Arg	Glu	Asn 245	Gly	Arg	Val	Pro	Arg	Pro 250	Pro	Asp	Cys	Pro	Lys 255	Cys
Pro	Ala	Pro	Glu 260	Met	Leu	Gly	Gly	Pro 265	Ser	Val	Phe	Ile	Phe 270	Pro	Pro
Lys	Pro 275	Lys	Asp	Thr	Leu	Leu	Ile 280	Ala	Arg	Thr	Pro	Glu 285	Val	Thr	Cys
Val 290	Val	Val	Asp	Leu	Asp	Pro 295	Glu	Asp	Pro	Glu	Val 300	Gln	Ile	Ser	Trp
Phe 305	Val	Asp	Gly	Lys	Gln 310	Met	Gln	Thr	Ala	Lys 315	Thr	Gln	Pro	Arg	Glu 320
Glu	Gln	Phe	Asn 325	Gly	Thr	Tyr	Arg	Val	Val 330	Ser	Val	Leu	Pro	Ile 335	Gly
His	Gln	Asp	Trp 340	Leu	Lys	Gly	Lys	Gln	Phe 345	Thr	Cys	Lys	Val 350	Asn	Asn
Lys	Ala 355	Leu	Pro	Ser	Pro	Ile	Glu 360	Arg	Thr	Ile	Ser	Lys 365	Ala	Arg	Gly
Gln 370	Ala	His	Gln	Pro	Ser	Val 375	Tyr	Val	Leu	Pro	Pro 380	Ser	Arg	Glu	Glu
Leu 385	Ser	Lys	Asn	Thr 390	Val	Ser	Leu	Thr	Cys 395	Leu	Ile	Lys	Asp	Phe	Phe 400
Pro	Pro	Asp	Ile 405	Val	Glu	Trp	Gln	Ser 410	Asn	Gly	Gln	Gln	Glu	Pro 415	
Glu	Ser	Lys	Tyr 420	Arg	Thr	Thr	Pro	Pro 425	Gln	Leu	Asp	Glu	Asp 430	Gly	Ser
Tyr	Phe 435	Leu	Tyr	Ser	Lys	Leu	Ser	Val 440	Asp	Lys	Ser	Arg 445	Trp	Gln	Arg
Gly 450	Asp	Thr	Phe	Ile	Cys	Ala 455	Val	Met	His	Glu	Ala 460	Leu	His	Asn	His
Tyr	Thr	Gln	Glu	Ser	Leu	Ser	His	Ser	Pro	Gly	Lys				

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465

470

475

<210> SEQ ID NO 36
 <211> LENGTH: 237
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Caninized variable light chain of
 Mab A and canine kappa light constant region with leader sequence,
 (Caninized A LC kappa with leader)

<400> SEQUENCE: 36

```

Met Glu Thr Asp Thr Leu Leu Leu Trp Val Leu Leu Leu Trp Val Pro
1           5           10           15

Gly Ser Thr Gly Glu Ile Val Met Thr Gln Ser Pro Ala Ser Leu Ser
20          25          30

Leu Ser Gln Glu Glu Lys Val Thr Ile Thr Cys Lys Ala Ser Gln Asn
35          40          45

Val Asp Ser Asn Val Asp Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro
50          55          60

Lys Leu Leu Ile Tyr Lys Ala Ser Asn Arg Asn Thr Gly Val Pro Ser
65          70          75          80

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Ser Phe Thr Ile Ser
85          90          95

Ser Leu Glu Pro Glu Asp Val Ala Val Tyr Tyr Cys Met Gln Ser Thr
100         105         110

Ser Tyr Pro Leu Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg
115         120         125

Asn Asp Ala Gln Pro Ala Val Tyr Leu Phe Gln Pro Ser Pro Asp Gln
130         135         140

Leu His Thr Gly Ser Ala Ser Val Val Cys Leu Leu Asn Ser Phe Tyr
145         150         155         160

Pro Lys Asp Ile Asn Val Lys Trp Lys Val Asp Gly Val Ile Gln Asp
165         170         175

Thr Gly Ile Gln Glu Ser Val Thr Glu Gln Asp Lys Asp Ser Thr Tyr
180         185         190

Ser Leu Ser Ser Thr Leu Thr Met Ser Ser Thr Glu Tyr Leu Ser His
195         200         205

Glu Leu Tyr Ser Cys Glu Ile Thr His Lys Ser Leu Pro Ser Thr Leu
210         215         220

Ile Lys Ser Phe Gln Arg Ser Glu Cys Gln Arg Val Asp
225         230         235
  
```

<210> SEQ ID NO 37
 <211> LENGTH: 457
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Caninized variable heavy chain of
 Mab A v3 and canine IgG-B , (Caninized A HC v3 IgG-B)

<400> SEQUENCE: 37

```

Gln Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys Pro Gly Gly
1           5           10           15

Thr Leu Arg Leu Ser Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr
20          25          30

His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
  
```

Ala	Val	Met	Trp	Asn	Asp	Gly	Asp	Thr	Ser	Tyr	Asn	Leu	Ala	Val	Lys
50															
Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Asn	Thr	Leu	Tyr	Leu
65															
Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala
85															
Arg	Pro	Glu	Leu	Pro	Gly	Leu	Thr	Tyr	Gly	Val	Trp	Phe	Pro	Tyr	Trp
100															
Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Thr	Ala	Pro
115															
Ser	Val	Phe	Pro	Leu	Ala	Pro	Ser	Cys	Gly	Ser	Thr	Ser	Gly	Ser	Thr
130															
Val	Ala	Leu	Ala	Cys	Leu	Val	Ser	Gly	Tyr	Phe	Pro	Glu	Pro	Val	Thr
145															
Val	Ser	Trp	Asn	Ser	Gly	Ser	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro
165															
Ser	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Met	Val	Thr
180															
Val	Pro	Ser	Ser	Arg	Trp	Pro	Ser	Glu	Thr	Phe	Thr	Cys	Asn	Val	Ala
195															
His	Pro	Ala	Ser	Lys	Thr	Lys	Val	Asp	Lys	Pro	Val	Pro	Lys	Arg	Glu
210															
Asn	Gly	Arg	Val	Pro	Arg	Pro	Pro	Asp	Cys	Pro	Lys	Cys	Pro	Ala	Pro
225															
Glu	Met	Leu	Gly	Gly	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Lys	Pro	Lys
245															
Asp	Thr	Leu	Leu	Ile	Ala	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val
260															
Asp	Leu	Asp	Pro	Glu	Asp	Pro	Glu	Val	Gln	Ile	Ser	Trp	Phe	Val	Asp
275															
Gly	Lys	Gln	Met	Gln	Thr	Ala	Lys	Thr	Gln	Pro	Arg	Glu	Glu	Gln	Phe
290															
Asn	Gly	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Pro	Ile	Gly	His	Gln	Asp
305															
Trp	Leu	Lys	Gly	Lys	Gln	Phe	Thr	Cys	Lys	Val	Asn	Asn	Lys	Ala	Leu
325															
Pro	Ser	Pro	Ile	Glu	Arg	Thr	Ile	Ser	Lys	Ala	Arg	Gly	Gln	Ala	His
340															
Gln	Pro	Ser	Val	Tyr	Val	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Leu	Ser	Lys
355															
Asn	Thr	Val	Ser	Leu	Thr	Cys	Leu	Ile	Lys	Asp	Phe	Phe	Pro	Pro	Asp
370															
Ile	Asp	Val	Glu	Trp	Gln	Ser	Asn	Gly	Gln	Gln	Glu	Pro	Glu	Ser	Lys
385															
Tyr	Arg	Thr	Thr	Pro	Pro	Gln	Leu	Asp	Glu	Asp	Gly	Ser	Tyr	Phe	Leu
405															
Tyr	Ser	Lys	Leu	Ser	Val	Asp	Lys	Ser	Arg	Trp	Gln	Arg	Gly	Asp	Thr
420															
Phe	Ile	Cys	Ala	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln
435															
Glu	Ser	Leu	Ser	His	Ser	Pro	Gly	Lys							
450															
455															

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<210> SEQ ID NO 38
 <211> LENGTH: 457
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Caninized variable heavy chain of
 Mab A v4 and canine IgG-B , (Caninized A HC v4 IgG-B)

<400> SEQUENCE: 38

Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys Pro Gly Gly
 1 5 10 15
 Thr Leu Arg Leu Ser Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr
 20 25 30
 His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45
 Gly Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu Ala Val Lys
 50 55 60
 Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr Phe
 65 70 75 80
 Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Thr Tyr Tyr Cys Ala
 85 90 95
 Arg Pro Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe Pro Tyr Trp
 100 105 110
 Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Thr Ala Pro
 115 120 125
 Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser Gly Ser Thr
 130 135 140
 Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu Pro Val Thr
 145 150 155 160
 Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His Thr Phe Pro
 165 170 175
 Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Met Val Thr
 180 185 190
 Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys Asn Val Ala
 195 200 205
 His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro Lys Arg Glu
 210 215 220
 Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys Pro Ala Pro
 225 230 235 240
 Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys Pro Lys
 245 250 255
 Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys Val Val Val
 260 265 270
 Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp Phe Val Asp
 275 280 285
 Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu Glu Gln Phe
 290 295 300
 Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly His Gln Asp
 305 310 315 320
 Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn Lys Ala Leu
 325 330 335
 Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly Gln Ala His
 340 345 350
 Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu Leu Ser Lys
 355 360 365

-continued

Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe Pro Pro Asp
370 375 380

Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro Glu Ser Lys
385 390 395 400

Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser Tyr Phe Leu
405 410 415

Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg Gly Asp Thr
420 425 430

Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His Tyr Thr Gln
435 440 445

Glu Ser Leu Ser His Ser Pro Gly Lys
450 455

<210> SEQ ID NO 39

<211> LENGTH: 217

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Caninized variable light chain of
Mab A and canine kappa light constant region , (Caninized A LC
kappa)

<400> SEQUENCE: 39

Glu Ile Val Met Thr Gln Ser Pro Ala Ser Leu Ser Leu Ser Gln Glu
1 5 10 15

Glu Lys Val Thr Ile Thr Cys Lys Ala Ser Gln Asn Val Asp Ser Asn
20 25 30

Val Asp Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Lys Ala Ser Asn Arg Asn Thr Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Ser Phe Thr Ile Ser Ser Leu Glu Pro
65 70 75 80

Glu Asp Val Ala Val Tyr Tyr Cys Met Gln Ser Thr Ser Tyr Pro Leu
85 90 95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg Asn Asp Ala Gln
100 105 110

Pro Ala Val Tyr Leu Phe Gln Pro Ser Pro Asp Gln Leu His Thr Gly
115 120 125

Ser Ala Ser Val Val Cys Leu Leu Asn Ser Phe Tyr Pro Lys Asp Ile
130 135 140

Asn Val Lys Trp Lys Val Asp Gly Val Ile Gln Asp Thr Gly Ile Gln
145 150 155 160

Glu Ser Val Thr Glu Gln Asp Lys Asp Ser Thr Tyr Ser Leu Ser Ser
165 170 175

Thr Leu Thr Met Ser Ser Thr Glu Tyr Leu Ser His Glu Leu Tyr Ser
180 185 190

Cys Glu Ile Thr His Lys Ser Leu Pro Ser Thr Leu Ile Lys Ser Phe
195 200 205

Gln Arg Ser Glu Cys Gln Arg Val Asp
210 215

<210> SEQ ID NO 40

<211> LENGTH: 119

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Felinized variable heavy chain of

-continued

Mab A , (Felinized A HC)

<400> SEQUENCE: 40

Gln Leu Thr Leu Arg Glu Ser Gly Pro Gly Leu Val Lys Pro Ser Gln
 1 5 10 15

Ser Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr
 20 25 30

His Val His Trp Ile Arg Gln Arg Pro Gly Arg Gly Leu Glu Trp Leu
 35 40 45

Gly Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Ala Phe Gln Gly Arg
 50 55 60

Ile Ser Ile Thr Ala Asp Thr Ala Gln Asn Gln Phe Ser Leu Gln Leu
 65 70 75 80

Ser Ser Met Thr Thr Asp Asp Thr Ala Val Tyr Tyr Cys Ala Arg Pro
 85 90 95

Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe Pro Tyr Trp Gly Gln
 100 105 110

Gly Ala Leu Val Thr Val Ser
 115

<210> SEQ ID NO 41

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Felinized variable light chain of Mab A , (Felinized A LC)

<400> SEQUENCE: 41

Ala Ile Thr Met Thr Gln Ser Pro Gly Ser Leu Ala Gly Ser Pro Gly
 1 5 10 15

Gln Gln Val Thr Met Asn Cys Lys Ala Ser Gln Asn Val Asp Ser Asn
 20 25 30

Val Asp Trp Tyr Gln Gln Lys Pro Gly Gln His Pro Lys Leu Leu Ile
 35 40 45

Tyr Lys Ala Ser Asn Arg Asn Thr Gly Val Pro Asp Arg Phe Ser Gly
 50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asn Leu Gln Ala
 65 70 75 80

Glu Asp Val Ala Ser Tyr Tyr Cys Met Gln Ser Thr Ser Tyr Pro Leu
 85 90 95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 42

<211> LENGTH: 10

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable heavy chain CDRH1 of Mab B

<400> SEQUENCE: 42

Gly Phe Ser Leu Thr Ser Tyr Gly Val Ser
 1 5 10

<210> SEQ ID NO 43

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic: Variable heavy chain CDRH2 of Mab B

<400> SEQUENCE: 43

Thr Met Trp Asn Asp Gly Asp Thr Asp
1 5

<210> SEQ ID NO 44

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable heavy chain CDRH3 of Mab B

<400> SEQUENCE: 44

Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe Val Tyr
1 5 10

<210> SEQ ID NO 45

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable heavy chain framework
region HC-FR1 of Mab B

<400> SEQUENCE: 45

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Gln
1 5 10 15

Thr Leu Ser Leu Thr Cys Thr Val Ser
20 25

<210> SEQ ID NO 46

<211> LENGTH: 25

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable heavy chain framework
region HC-FR1 of Mab B variant 2 (v2) , Q16E

<400> SEQUENCE: 46

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Glu
1 5 10 15

Thr Leu Ser Leu Thr Cys Thr Val Ser
20 25

<210> SEQ ID NO 47

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable region heavy chain
framework region HC-FR2 of Mab B

<400> SEQUENCE: 47

Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Ile Gly
1 5 10

<210> SEQ ID NO 48

<211> LENGTH: 14

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable region heavy chain
framework region HC-FR2 of Mab B variant 2 (v2), I13M

<400> SEQUENCE: 48

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Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Met Gly
1 5 10

<210> SEQ ID NO 49
 <211> LENGTH: 39
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable heavy chain framework
 region HC-FR3 of Mab B

<400> SEQUENCE: 49

Tyr His Ser Ala Leu Arg Ser Arg Leu Ser Ile Ser Arg Asp Ser Ser
1 5 10 15

Lys Ser Gln Val Leu Leu Lys Met Asn Ser Leu Gln Thr Glu Asp Thr
20 25 30

Ala Met Tyr Phe Cys Ala Arg
35

<210> SEQ ID NO 50
 <211> LENGTH: 39
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable region heavy chain
 framework region HC-FR3 of Mab B variant 2 (v2), H2N, R8K, S15T,
 L21F, M34I, F36Y

<400> SEQUENCE: 50

Tyr Asn Ser Ala Leu Arg Ser Lys Leu Ser Ile Ser Arg Asp Thr Ser
1 5 10 15

Lys Ser Gln Val Phe Leu Lys Met Asn Ser Leu Gln Thr Glu Asp Thr
20 25 30

Ala Ile Tyr Tyr Cys Ala Arg
35

<210> SEQ ID NO 51
 <211> LENGTH: 10
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable heavy chain framework
 region HC-FR4 of Mab B

<400> SEQUENCE: 51

Trp Gly Gln Gly Thr Leu Val Ile Val Ser
1 5 10

<210> SEQ ID NO 52
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain CDRL1 of Mab B

<400> SEQUENCE: 52

Lys Ala Ser His Asn Ile Asn Lys Asn Leu Glu
1 5 10

<210> SEQ ID NO 53
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain CDRL2 of Mab B

-continued

<400> SEQUENCE: 53

Tyr Ala Asn Asn Leu Gln Thr
 1 5

<210> SEQ ID NO 54

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable light chain CDRL3 of Mab B

<400> SEQUENCE: 54

Tyr Gln Tyr Asn Ser Gly His Thr Phe
 1 5

<210> SEQ ID NO 55

<211> LENGTH: 23

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable light chain framework
region LC-FR1 of Mab B

<400> SEQUENCE: 55

Asp Ile Val Met Thr Gln Thr Pro Ser Leu Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Leu Asn Cys
 20

<210> SEQ ID NO 56

<211> LENGTH: 23

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable light chain framework
region LC-FR1 of Mab B variant 2 (v2), V3Q, T7S, S9P, L10V, N22S

<400> SEQUENCE: 56

Asp Ile Gln Met Thr Gln Ser Pro Pro Val Leu Ser Ala Ser Val Gly
 1 5 10 15

Asp Arg Val Thr Leu Ser Cys
 20

<210> SEQ ID NO 57

<211> LENGTH: 15

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable light chain framework
region LC-FR2 of Mab B

<400> SEQUENCE: 57

Trp Tyr Gln Gln Lys Leu Gly Glu Ala Pro Lys Leu Leu Ile Tyr
 1 5 10 15

<210> SEQ ID NO 58

<211> LENGTH: 32

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Variable light chain framework
region LC-FR3 of Mab B

<400> SEQUENCE: 58

Gly Ile Ser Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Tyr Thr
 1 5 10 15

-continued

Leu Thr Ile Ser Ser Leu Gln Pro Glu Asp Val Ala Thr Tyr Tyr Cys
 20 25 30

<210> SEQ ID NO 59
 <211> LENGTH: 32
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain framework
 region LC-FR3 of Mab B variant 2 (v2), S3P

<400> SEQUENCE: 59

Gly Ile Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Tyr Thr
 1 5 10 15

Leu Thr Ile Ser Ser Leu Gln Pro Glu Asp Val Ala Thr Tyr Tyr Cys
 20 25 30

<210> SEQ ID NO 60
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain framework
 region LC-FR4 of Mab B

<400> SEQUENCE: 60

Gly Ala Gly Thr Lys Leu Glu Leu Lys
 1 5

<210> SEQ ID NO 61
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable light chain framework
 region LC-FR4 of Mab B variant 2 (v2), L8I

<400> SEQUENCE: 61

Gly Ala Gly Thr Lys Leu Glu Ile Lys
 1 5

<210> SEQ ID NO 62
 <211> LENGTH: 121
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Variable heavy chain of Mab B, (Mab
 B HC)

<400> SEQUENCE: 62

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Gln
 1 5 10 15

Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
 20 25 30

Gly Val Ser Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45

Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr His Ser Ala Leu Arg
 50 55 60

Ser Arg Leu Ser Ile Ser Arg Asp Ser Ser Lys Ser Gln Val Leu Leu
 65 70 75 80

Lys Met Asn Ser Leu Gln Thr Glu Asp Thr Ala Met Tyr Phe Cys Ala
 85 90 95

Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe Val Tyr Trp

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100	105	110
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Gly Gln Gly Thr Leu Val Ile Val Ser
115 120

<210> SEQ ID NO 63
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic: Variable heavy chain of Mab B
variant 2, HC-FR1 Q16E, HC-FR2 I13M, HC-FR3 H2N, HC-FR3 R8K,
HC-FR3 S15T, HC-FR3 L21F, HC-FR3 M34I, HC-FR3 F36Y, (Mab B HC v2),

<400> SEQUENCE: 63

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Glu	
1 5 10 15	
Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr	
20 25 30	
Gly Val Ser Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Met	
35 40 45	
Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr Asn Ser Ala Leu Arg	
50 55 60	
Ser Lys Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val Phe Leu	
65 70 75 80	
Lys Met Asn Ser Leu Gln Thr Glu Asp Thr Ala Ile Tyr Tyr Cys Ala	
85 90 95	
Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe Val Tyr Trp	
100 105 110	
Gly Gln Gly Thr Leu Val Ile Val Ser	
115 120	

<210> SEQ ID NO 64
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic: Variable light chain of Mab B, (Mab
B LC)

<400> SEQUENCE: 64

Asp Ile Val Met Thr Gln Thr Pro Ser Leu Leu Ser Ala Ser Val Gly	
1 5 10 15	
Asp Arg Val Thr Leu Asn Cys Lys Ala Ser His Asn Ile Asn Lys Asn	
20 25 30	
Leu Glu Trp Tyr Gln Gln Lys Leu Gly Glu Ala Pro Lys Leu Leu Ile	
35 40 45	
Tyr Tyr Ala Asn Asn Leu Gln Thr Gly Ile Ser Ser Arg Phe Ser Gly	
50 55 60	
Ser Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro	
65 70 75 80	
Glu Asp Val Ala Thr Tyr Tyr Cys Tyr Gln Tyr Asn Ser Gly His Thr	
85 90 95	
Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys	
100 105	

<210> SEQ ID NO 65
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic: Variable light chain of Mab B
variant 2, LC-FR1 V3Q, LC-FR1 T7S, LC-FR1 S9P, LC-FR1 L10V, LC-FR1
N22S , LC-FR3 S3P, LC-FR4 L8I, (Mab B LC v2),

<400> SEQUENCE: 65

```

Asp Ile Gln Met Thr Gln Ser Pro Pro Val Leu Ser Ala Ser Val Gly
1             5             10             15

Asp Arg Val Thr Leu Ser Cys Lys Ala Ser His Asn Ile Asn Lys Asn
                20             25             30

Leu Glu Trp Tyr Gln Gln Lys Leu Gly Glu Ala Pro Lys Leu Leu Ile
            35             40             45

Tyr Tyr Ala Asn Asn Leu Gln Thr Gly Ile Pro Ser Arg Phe Ser Gly
            50             55             60

Ser Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro
65             70             75             80

Glu Asp Val Ala Thr Tyr Tyr Cys Tyr Gln Tyr Asn Ser Gly His Thr
            85             90             95

Phe Gly Ala Gly Thr Lys Leu Glu Ile Lys
            100             105

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<210> SEQ ID NO 66

<211> LENGTH: 456

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable heavy chain of Mab
B and canine IgG-B , (Chimeric B HC IgG-B)

<400> SEQUENCE: 66

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Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Gln
1             5             10             15

Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
            20             25             30

Gly Val Ser Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Ile
            35             40             45

Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr His Ser Ala Leu Arg
50             55             60

Ser Arg Leu Ser Ile Ser Arg Asp Ser Ser Lys Ser Gln Val Leu Leu
65             70             75             80

Lys Met Asn Ser Leu Gln Thr Glu Asp Thr Ala Met Tyr Phe Cys Ala
            85             90             95

Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe Val Tyr Trp
            100             105             110

Gly Gln Gly Thr Leu Val Ile Val Ser Ala Ser Thr Thr Ala Pro Ser
115             120             125

Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser Gly Ser Thr Val
130             135             140

Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu Pro Val Thr Val
145             150             155             160

Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His Thr Phe Pro Ser
            165             170             175

Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Met Val Thr Val
            180             185             190

Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys Asn Val Ala His
            195             200             205

Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro Lys Arg Glu Asn
210             215             220

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Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys Pro Ala Pro Glu
 225 230 235 240
 Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys Pro Lys Asp
 245 250 255
 Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys Val Val Val Asp
 260 265 270
 Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp Phe Val Asp Gly
 275 280 285
 Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu Glu Gln Phe Asn
 290 295 300
 Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly His Gln Asp Trp
 305 310 315 320
 Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn Lys Ala Leu Pro
 325 330 335
 Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly Gln Ala His Gln
 340 345 350
 Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu Leu Ser Lys Asn
 355 360 365
 Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe Pro Pro Asp Ile
 370 375 380
 Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro Glu Ser Lys Tyr
 385 390 395 400
 Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser Tyr Phe Leu Tyr
 405 410 415
 Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg Gly Asp Thr Phe
 420 425 430
 Ile Cys Ala Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Glu
 435 440 445
 Ser Leu Ser His Ser Pro Gly Lys
 450 455

<210> SEQ ID NO 67

<211> LENGTH: 457

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable heavy chain of Mab
 B v2 and canine IgG-B , HC-FR1 Q16E, HC-FR2 I13M, HC-FR3 H2N,
 HC-FR3 R8K, HC-FR3 S15T, HC-FR3 L21F, HC-FR3 M34I, HC-FR3 F36Y,
 (Chimeric B HC v2 IgG-B)

<400> SEQUENCE: 67

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Glu
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
 20 25 30
 Gly Val Ser Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Met
 35 40 45
 Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr Asn Ser Ala Leu Arg
 50 55 60
 Ser Lys Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val Phe Leu
 65 70 75 80
 Lys Met Asn Ser Leu Gln Thr Glu Asp Thr Ala Ile Tyr Tyr Cys Ala
 85 90 95
 Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe Val Tyr Trp
 100 105 110

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Gly Gln Gly Thr Leu Val Ile Val Ser Ala Ala Ser Thr Thr Ala Pro
 115 120 125
 Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser Gly Ser Thr
 130 135 140
 Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu Pro Val Thr
 145 150 155 160
 Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His Thr Phe Pro
 165 170 175
 Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Met Val Thr
 180 185 190
 Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys Asn Val Ala
 195 200 205
 His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro Lys Arg Glu
 210 215 220
 Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys Pro Ala Pro
 225 230 235 240
 Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys Pro Lys
 245 250 255
 Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys Val Val Val
 260 265 270
 Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp Phe Val Asp
 275 280 285
 Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu Glu Gln Phe
 290 295 300
 Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly His Gln Asp
 305 310 315 320
 Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn Lys Ala Leu
 325 330 335
 Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly Gln Ala His
 340 345 350
 Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu Leu Ser Lys
 355 360 365
 Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe Pro Pro Asp
 370 375 380
 Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro Glu Ser Lys
 385 390 395 400
 Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser Tyr Phe Leu
 405 410 415
 Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg Gly Asp Thr
 420 425 430
 Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His Tyr Thr Gln
 435 440 445
 Glu Ser Leu Ser His Ser Pro Gly Lys
 450 455

<210> SEQ ID NO 68

<211> LENGTH: 216

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

 <223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab
 B and canine kappa light constant region, (Chimeric B LC kappa)

<400> SEQUENCE: 68

Asp Ile Val Met Thr Gln Thr Pro Ser Leu Leu Ser Ala Ser Val Gly

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1	5	10	15
Asp Arg Val Thr Leu Asn Cys Lys Ala Ser His Asn Ile Asn Lys Asn	20	25	30
Leu Glu Trp Tyr Gln Gln Lys Leu Gly Glu Ala Pro Lys Leu Leu Ile	35	40	45
Tyr Tyr Ala Asn Asn Leu Gln Thr Gly Ile Ser Ser Arg Phe Ser Gly	50	55	60
Ser Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro	65	70	75
Glu Asp Val Ala Thr Tyr Tyr Cys Tyr Gln Tyr Asn Ser Gly His Thr	85	90	95
Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys Arg Asn Asp Ala Gln Pro	100	105	110
Ala Val Tyr Leu Phe Gln Pro Ser Pro Asp Gln Leu His Thr Gly Ser	115	120	125
Ala Ser Val Val Cys Leu Leu Asn Ser Phe Tyr Pro Lys Asp Ile Asn	130	135	140
Val Lys Trp Lys Val Asp Gly Val Ile Gln Asp Thr Gly Ile Gln Glu	145	150	155
Ser Val Thr Glu Gln Asp Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr	165	170	175
Leu Thr Met Ser Ser Thr Glu Tyr Leu Ser His Glu Leu Tyr Ser Cys	180	185	190
Glu Ile Thr His Lys Ser Leu Pro Ser Thr Leu Ile Lys Ser Phe Gln	195	200	205
Arg Ser Glu Cys Gln Arg Val Asp	210	215	

<210> SEQ ID NO 69

<211> LENGTH: 216

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab
 B v2 and canine kappa light constant region, , LC-FR1 V3Q, LC-FR1
 T7S, LC-FR1 S9P, LC-FR1 L10V, LC-FR1 N22S , LC-FR3 S3P, LC-FR4
 L8I, (Chimeric B LC v2 kappa)

<400> SEQUENCE: 69

Asp Ile Gln Met Thr Gln Ser Pro Pro Val Leu Ser Ala Ser Val Gly	1	5	10	15
Asp Arg Val Thr Leu Ser Cys Lys Ala Ser His Asn Ile Asn Lys Asn	20	25	30	
Leu Glu Trp Tyr Gln Gln Lys Leu Gly Glu Ala Pro Lys Leu Leu Ile	35	40	45	
Tyr Tyr Ala Asn Asn Leu Gln Thr Gly Ile Pro Ser Arg Phe Ser Gly	50	55	60	
Ser Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro	65	70	75	80
Glu Asp Val Ala Thr Tyr Tyr Cys Tyr Gln Tyr Asn Ser Gly His Thr	85	90	95	
Phe Gly Ala Gly Thr Lys Leu Glu Ile Lys Arg Asn Asp Ala Gln Pro	100	105	110	
Ala Val Tyr Leu Phe Gln Pro Ser Pro Asp Gln Leu His Thr Gly Ser	115	120	125	
Ala Ser Val Val Cys Leu Leu Asn Ser Phe Tyr Pro Lys Asp Ile Asn				

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130	135	140
Val Lys Trp Lys Val Asp Gly Val Ile Gln Asp Thr Gly Ile Gln Glu		
145	150	155 160
Ser Val Thr Glu Gln Asp Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr		
	165	170 175
Leu Thr Met Ser Ser Thr Glu Tyr Leu Ser His Glu Leu Tyr Ser Cys		
	180	185 190
Glu Ile Thr His Lys Ser Leu Pro Ser Thr Leu Ile Lys Ser Phe Gln		
	195	200 205
Arg Ser Glu Cys Gln Arg Val Asp		
	210	215

<210> SEQ ID NO 70
 <211> LENGTH: 475
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Chimeric variable heavy chain of Mab
 B and canine IgG-B with leader sequence, (Canine chimeric B HC
 with leader)

 <400> SEQUENCE: 70

Met Ala Val Leu Gly Leu Leu Leu Cys Leu Val Thr Phe Pro Ser Cys		
1	5	10 15
Val Leu Ser Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln		
	20	25 30
Pro Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu		
	35	40 45
Thr Ser Tyr Gly Val Ser Trp Val Arg Gln Pro Pro Gly Lys Gly Leu		
	50	55 60
Glu Trp Ile Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr His Ser		
	65	70 75 80
Ala Leu Arg Ser Arg Leu Ser Ile Ser Arg Asp Ser Ser Lys Ser Gln		
	85	90 95
Val Leu Leu Lys Met Asn Ser Leu Gln Thr Glu Asp Thr Ala Met Tyr		
	100	105 110
Phe Cys Ala Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe		
	115	120 125
Val Tyr Trp Gly Gln Gly Thr Leu Val Ile Val Ser Ala Ser Thr Thr		
	130	135 140
Ala Pro Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser Gly		
	145	150 155 160
Ser Thr Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu Pro		
	165	170 175
Val Thr Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His Thr		
	180	185 190
Phe Pro Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Met		
	195	200 205
Val Thr Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys Asn		
	210	215 220
Val Ala His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro Lys		
	225	230 235 240
Arg Glu Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys Pro		
	245	250 255
Ala Pro Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys		
	260	265 270

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Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys Val
  275                      280                      285

Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp Phe
  290                      295                      300

Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu Glu
  305                      310                      315                      320

Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly His
  325                      330                      335

Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn Lys
  340                      345                      350

Ala Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly Gln
  355                      360                      365

Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu Leu
  370                      375                      380

Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe Pro
  385                      390                      395                      400

Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro Glu
  405                      410                      415

Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser Tyr
  420                      425                      430

Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg Gly
  435                      440                      445

Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His Tyr
  450                      455                      460

Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
  465                      470                      475

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<210> SEQ ID NO 71
<211> LENGTH: 476
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic: Chimeric variable heavy chain of Mab
      B v2 and canine IgG-B with leader sequence, HC-FR1 Q16E, HC-FR2
      I13M, HC-FR3 H2N, HC-FR3 R8K, HC-FR3 S15T, HC-FR3 L21F, HC-FR3
      M34I, HC-FR3 F36Y, (Canine chimeric B HC v2 IgG-B with leader),

<400> SEQUENCE: 71

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Met Ala Val Leu Gly Leu Leu Leu Cys Leu Val Thr Phe Pro Ser Cys
  1      5      10      15

Val Leu Ser Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln
  20      25      30

Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu
  35      40      45

Thr Ser Tyr Gly Val Ser Trp Val Arg Gln Pro Pro Gly Lys Gly Leu
  50      55      60

Glu Trp Met Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr Asn Ser
  65      70      75      80

Ala Leu Arg Ser Lys Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln
  85      90      95

Val Phe Leu Lys Met Asn Ser Leu Gln Thr Glu Asp Thr Ala Ile Tyr
  100     105     110

Tyr Cys Ala Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe
  115     120     125

Val Tyr Trp Gly Gln Gly Thr Leu Val Ile Val Ser Ala Ala Ser Thr
  130     135     140

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Thr Ala Pro Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser
 145 150 155 160
 Gly Ser Thr Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu
 165 170 175
 Pro Val Thr Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His
 180 185 190
 Thr Phe Pro Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
 195 200 205
 Met Val Thr Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys
 210 215 220
 Asn Val Ala His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro
 225 230 235 240
 Lys Arg Glu Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys
 245 250 255
 Pro Ala Pro Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro
 260 265 270
 Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
 275 280 285
 Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
 290 295 300
 Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
 305 310 315 320
 Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
 325 330 335
 His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
 340 345 350
 Lys Ala Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly
 355 360 365
 Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu
 370 375 380
 Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe
 385 390 395 400
 Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro
 405 410 415
 Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser
 420 425 430
 Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
 435 440 445
 Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
 450 455 460
 Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
 465 470 475

<210> SEQ ID NO 72

<211> LENGTH: 236

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab
 B and canine kappa light constant region with leader sequence,
 (Canine chimeric B LC kappa with leader)

<400> SEQUENCE: 72

Met Glu Thr Asp Thr Leu Leu Leu Trp Val Leu Leu Leu Trp Val Pro
 1 5 10 15

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Gly	Ser	Thr	Gly	Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ser	Leu	Leu	Ser
			20					25					30		
Ala	Ser	Val	Gly	Asp	Arg	Val	Thr	Leu	Asn	Cys	Lys	Ala	Ser	His	Asn
		35					40					45			
Ile	Asn	Lys	Asn	Leu	Glu	Trp	Tyr	Gln	Gln	Lys	Leu	Gly	Glu	Ala	Pro
	50					55					60				
Lys	Leu	Leu	Ile	Tyr	Tyr	Ala	Asn	Asn	Leu	Gln	Thr	Gly	Ile	Ser	Ser
65				70						75				80	
Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser
			85					90						95	
Ser	Leu	Gln	Pro	Glu	Asp	Val	Ala	Thr	Tyr	Tyr	Cys	Tyr	Gln	Tyr	Asn
		100						105					110		
Ser	Gly	His	Thr	Phe	Gly	Ala	Gly	Thr	Lys	Leu	Glu	Leu	Lys	Arg	Asn
	115						120					125			
Asp	Ala	Gln	Pro	Ala	Val	Tyr	Leu	Phe	Gln	Pro	Ser	Pro	Asp	Gln	Leu
	130					135					140				
His	Thr	Gly	Ser	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Ser	Phe	Tyr	Pro
145					150					155					160
Lys	Asp	Ile	Asn	Val	Lys	Trp	Lys	Val	Asp	Gly	Val	Ile	Gln	Asp	Thr
			165					170						175	
Gly	Ile	Gln	Glu	Ser	Val	Thr	Glu	Gln	Asp	Lys	Asp	Ser	Thr	Tyr	Ser
		180						185					190		
Leu	Ser	Ser	Thr	Leu	Thr	Met	Ser	Ser	Thr	Glu	Tyr	Leu	Ser	His	Glu
	195						200					205			
Leu	Tyr	Ser	Cys	Glu	Ile	Thr	His	Lys	Ser	Leu	Pro	Ser	Thr	Leu	Ile
	210					215					220				
Lys	Ser	Phe	Gln	Arg	Ser	Glu	Cys	Gln	Arg	Val	Asp				
225					230					235					

<210> SEQ ID NO 73

<211> LENGTH: 236

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab B v2 and canine kappa light constant region with leader sequence, LC-FR1 V3Q, LC-FR1 T7S, LC-FR1 S9P, LC-FR1 L10V, LC-FR1 N22S, LC-FR3 S3P, LC-FR4 L8I, (Canine chimeric B LC v2 kappa with leader)

<400> SEQUENCE: 73

Met	Glu	Thr	Asp	Thr	Leu	Leu	Leu	Trp	Val	Leu	Leu	Leu	Trp	Val	Pro
1				5					10					15	
Gly	Ser	Thr	Gly	Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Pro	Val	Leu	Ser
			20				25					30			
Ala	Ser	Val	Gly	Asp	Arg	Val	Thr	Leu	Ser	Cys	Lys	Ala	Ser	His	Asn
		35					40					45			
Ile	Asn	Lys	Asn	Leu	Glu	Trp	Tyr	Gln	Gln	Lys	Leu	Gly	Glu	Ala	Pro
	50					55					60				
Lys	Leu	Leu	Ile	Tyr	Tyr	Ala	Asn	Asn	Leu	Gln	Thr	Gly	Ile	Pro	Ser
65				70						75				80	
Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser
			85					90						95	
Ser	Leu	Gln	Pro	Glu	Asp	Val	Ala	Thr	Tyr	Tyr	Cys	Tyr	Gln	Tyr	Asn
		100						105					110		
Ser	Gly	His	Thr	Phe	Gly	Ala	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Arg	Asn
	115						120					125			

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Asp Ala Gln Pro Ala Val Tyr Leu Phe Gln Pro Ser Pro Asp Gln Leu
 130 135 140
 His Thr Gly Ser Ala Ser Val Val Cys Leu Leu Asn Ser Phe Tyr Pro
 145 150 155 160
 Lys Asp Ile Asn Val Lys Trp Lys Val Asp Gly Val Ile Gln Asp Thr
 165 170 175
 Gly Ile Gln Glu Ser Val Thr Glu Gln Asp Lys Asp Ser Thr Tyr Ser
 180 185 190
 Leu Ser Ser Thr Leu Thr Met Ser Ser Thr Glu Tyr Leu Ser His Glu
 195 200 205
 Leu Tyr Ser Cys Glu Ile Thr His Lys Ser Leu Pro Ser Thr Leu Ile
 210 215 220
 Lys Ser Phe Gln Arg Ser Glu Cys Gln Arg Val Asp
 225 230 235

<210> SEQ ID NO 74

<211> LENGTH: 456

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable heavy chain of Mab
 B and feline IgG-1, (Feline chimeric B HC IgG-1)

<400> SEQUENCE: 74

Gln Val Gln Leu Lys Glu Ser Gly Pro Gly Leu Val Gln Pro Ser Gln
 1 5 10 15
 Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
 20 25 30
 Gly Val Ser Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45
 Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr His Ser Ala Leu Arg
 50 55 60
 Ser Arg Leu Ser Ile Ser Arg Asp Ser Ser Lys Ser Gln Val Leu Leu
 65 70 75 80
 Lys Met Asn Ser Leu Gln Thr Glu Asp Thr Ala Met Tyr Phe Cys Ala
 85 90 95
 Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe Val Tyr Trp
 100 105 110
 Gly Gln Gly Thr Leu Val Ile Val Ser Ala Ser Thr Thr Ala Pro Ser
 115 120 125
 Val Phe Pro Leu Ala Pro Ser Cys Gly Thr Thr Ser Gly Ala Thr Val
 130 135 140
 Ala Leu Ala Cys Leu Val Leu Gly Tyr Phe Pro Glu Pro Val Thr Val
 145 150 155 160
 Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala
 165 170 175
 Val Leu Gln Ala Ser Gly Leu Tyr Ser Leu Ser Ser Met Val Thr Val
 180 185 190
 Pro Ser Ser Arg Trp Leu Ser Asp Thr Phe Thr Cys Asn Val Ala His
 195 200 205
 Pro Pro Ser Asn Thr Lys Val Asp Lys Thr Val Arg Lys Thr Asp His
 210 215 220
 Pro Pro Gly Pro Lys Pro Cys Asp Cys Pro Lys Cys Pro Pro Pro Glu
 225 230 235 240
 Met Leu Gly Gly Pro Ser Ile Phe Ile Phe Pro Pro Lys Pro Lys Asp

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245					250					255					
Thr	Leu	Ser	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Leu	Val	Val	Asp
			260					265					270		
Leu	Gly	Pro	Asp	Asp	Ser	Asp	Val	Gln	Ile	Thr	Trp	Phe	Val	Asp	Asn
		275					280					285			
Thr	Gln	Val	Tyr	Thr	Ala	Lys	Thr	Ser	Pro	Arg	Glu	Glu	Gln	Phe	Asn
	290					295					300				
Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Pro	Ile	Leu	His	Gln	Asp	Trp
305					310					315				320	
Leu	Lys	Gly	Lys	Glu	Phe	Lys	Cys	Lys	Val	Asn	Ser	Lys	Ser	Leu	Pro
			325						330					335	
Ser	Pro	Ile	Glu	Arg	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	His	Glu
			340					345					350		
Pro	Gln	Val	Tyr	Val	Leu	Pro	Pro	Ala	Gln	Glu	Glu	Leu	Ser	Arg	Asn
		355					360					365			
Lys	Val	Ser	Val	Thr	Cys	Leu	Ile	Lys	Ser	Phe	His	Pro	Pro	Asp	Ile
	370					375					380				
Ala	Val	Glu	Trp	Glu	Ile	Thr	Gly	Gln	Pro	Glu	Pro	Glu	Asn	Asn	Tyr
385					390					395					400
Arg	Thr	Thr	Pro	Pro	Gln	Leu	Asp	Ser	Asp	Gly	Thr	Tyr	Phe	Val	Tyr
			405						410					415	
Ser	Lys	Leu	Ser	Val	Asp	Arg	Ser	His	Trp	Gln	Arg	Gly	Asn	Thr	Tyr
		420						425					430		
Thr	Cys	Ser	Val	Ser	His	Glu	Ala	Leu	His	Ser	His	His	Thr	Gln	Lys
		435				440						445			
Ser	Leu	Thr	Gln	Ser	Pro	Gly	Lys								
	450					455									

<210> SEQ ID NO 75

<211> LENGTH: 216

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab
B and feline kappa light constant region , (Feline chimeric B LC
kappa)

<400> SEQUENCE: 75

Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Ser	Leu	Leu	Ser	Ala	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Leu	Asn	Cys	Lys	Ala	Ser	His	Asn	Ile	Asn	Lys	Asn
		20					25						30		
Leu	Glu	Trp	Tyr	Gln	Gln	Lys	Leu	Gly	Glu	Ala	Pro	Lys	Leu	Leu	Ile
	35					40						45			
Tyr	Tyr	Ala	Asn	Asn	Leu	Gln	Thr	Gly	Ile	Ser	Ser	Arg	Phe	Ser	Gly
	50				55					60					
Ser	Gly	Ser	Gly	Thr	Asp	Tyr	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro
65				70					75					80	
Glu	Asp	Val	Ala	Thr	Tyr	Tyr	Cys	Tyr	Gln	Tyr	Asn	Ser	Gly	His	Thr
			85						90					95	
Phe	Gly	Ala	Gly	Thr	Lys	Leu	Glu	Leu	Lys	Arg	Ser	Asp	Ala	Gln	Pro
		100						105					110		
Ser	Val	Phe	Leu	Phe	Gln	Pro	Ser	Leu	Asp	Glu	Leu	His	Thr	Gly	Ser
		115				120						125			
Ala	Ser	Ile	Val	Cys	Ile	Leu	Asn	Asp	Phe	Tyr	Pro	Lys	Glu	Val	Asn
	130					135					140				

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Val Lys Trp Lys Val Asp Gly Val Val Gln Asn Lys Gly Ile Gln Glu
145                150                155                160

Ser Thr Thr Glu Gln Asn Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser
                165                170                175

Thr Leu Thr Met Ser Ser Thr Glu Tyr Gln Ser His Glu Lys Phe Ser
                180                185                190

Cys Glu Val Thr His Lys Ser Leu Ala Ser Thr Leu Val Lys Ser Phe
                195                200                205

Asn Arg Ser Glu Cys Gln Arg Glu
                210                215

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<210> SEQ ID NO 76
<211> LENGTH: 216
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic: Chimeric variable light chain of Mab
      B and feline kappa light constant region with no N-linked
      glycosylation site, (Feline chimeric B LC kappa aglycos)

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<400> SEQUENCE: 76

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Asp Ile Val Met Thr Gln Thr Pro Ser Leu Leu Ser Ala Ser Val Gly
1             5             10             15

Asp Arg Val Thr Leu Asn Cys Lys Ala Ser His Asn Ile Asn Lys Asn
20            25            30

Leu Glu Trp Tyr Gln Gln Lys Leu Gly Glu Ala Pro Lys Leu Leu Ile
35            40            45

Tyr Tyr Ala Asn Asn Leu Gln Thr Gly Ile Ser Ser Arg Phe Ser Gly
50            55            60

Ser Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro
65            70            75            80

Glu Asp Val Ala Thr Tyr Tyr Cys Tyr Gln Tyr Asn Ser Gly His Thr
85            90            95

Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys Arg Ser Asp Ala Gln Pro
100           105           110

Ser Val Phe Leu Phe Gln Pro Ser Leu Asp Glu Leu His Thr Gly Ser
115           120           125

Ala Ser Ile Val Cys Ile Leu Asn Asp Phe Tyr Pro Lys Glu Val Asn
130           135           140

Val Lys Trp Lys Val Asp Gly Val Val Gln Asn Lys Gly Ile Gln Glu
145           150           155           160

Ser Thr Thr Glu Gln Asn Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser
165           170           175

Thr Leu Thr Met Ser Ser Thr Glu Tyr Gln Ser His Glu Lys Phe Ser
180           185           190

Cys Glu Val Thr His Lys Ser Leu Ala Ser Thr Leu Val Lys Ser Phe
195           200           205

Gln Arg Ser Glu Cys Gln Arg Glu
210           215

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<210> SEQ ID NO 77
<211> LENGTH: 476
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic: Caninized variable heavy chain of
      Mab B v3 and canine IgG-B with leader sequence, (Caninized B HC v3
      IgG-B with leader)

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<400> SEQUENCE: 77

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Met Ala Val Leu Gly Leu Leu Leu Cys Leu Val Thr Phe Pro Ser Cys
1      5      10      15
Val Leu Ser Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys
20      25      30
Pro Gly Gly Thr Leu Arg Leu Ser Cys Thr Val Ser Gly Phe Ser Leu
35      40      45
Thr Ser Tyr Gly Val Ser Trp Val Arg Gln Ser Pro Gly Lys Gly Leu
50      55      60
Glu Trp Ile Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr His Ser
65      70      75      80
Ala Val Lys Gly Gln Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln
85      90      95
Val Phe Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Met Tyr
100     105     110
Tyr Cys Ala Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe
115     120     125
Val Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr
130     135     140
Thr Ala Pro Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser
145     150     155     160
Gly Ser Thr Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu
165     170     175
Pro Val Thr Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His
180     185     190
Thr Phe Pro Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
195     200     205
Met Val Thr Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys
210     215     220
Asn Val Ala His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro
225     230     235     240
Lys Arg Glu Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys
245     250     255
Pro Ala Pro Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro
260     265     270
Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
275     280     285
Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
290     295     300
Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
305     310     315     320
Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
325     330     335
His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
340     345     350
Lys Ala Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly
355     360     365
Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu
370     375     380
Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe
385     390     395     400
Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro

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405	410	415
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Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser
 420 425 430

Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
 435 440 445

Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
 450 455 460

Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
 465 470 475

<210> SEQ ID NO 78
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Caninized variable light chain of
 Mab B v3 and canine kappa light constant region with leader
 sequence, (Caninized B LC kappa with leader)

<400> SEQUENCE: 78

Met Glu Thr Asp Thr Leu Leu Leu Trp Val Leu Leu Leu Trp Val Pro
 1 5 10 15

Gly Ser Thr Gly Glu Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala
 20 25 30

Val Ser Leu Gly Gln Arg Ala Thr Ile Ser Cys Lys Ala Ser His Asn
 35 40 45

Ile Asn Lys Asn Leu Glu Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro
 50 55 60

Lys Leu Leu Ile Tyr Tyr Ala Asn Asn Leu Gln Thr Gly Val Pro Ala
 65 70 75 80

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Tyr Ser Leu Asn Ile His
 85 90 95

Pro Met Glu Glu Asp Asp Thr Ala Met Tyr Tyr Cys Tyr Gln Tyr Asn
 100 105 110

Ser Gly His Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Asn
 115 120 125

Asp Ala Gln Pro Ala Val Tyr Leu Phe Gln Pro Ser Pro Asp Gln Leu
 130 135 140

His Thr Gly Ser Ala Ser Val Val Cys Leu Leu Asn Ser Phe Tyr Pro
 145 150 155 160

Lys Asp Ile Asn Val Lys Trp Lys Val Asp Gly Val Ile Gln Asp Thr
 165 170 175

Gly Ile Gln Glu Ser Val Thr Glu Gln Asp Lys Asp Ser Thr Tyr Ser
 180 185 190

Leu Ser Ser Thr Leu Thr Met Ser Ser Thr Glu Tyr Leu Ser His Glu
 195 200 205

Leu Tyr Ser Cys Glu Ile Thr His Lys Ser Leu Pro Ser Thr Leu Ile
 210 215 220

Lys Ser Phe Gln Arg Ser Glu Cys Gln Arg Val Asp
 225 230 235

<210> SEQ ID NO 79
 <211> LENGTH: 457
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Caninized variable heavy chain of
 Mab B v3 and canine IgG-B , (Caninized B HC v3 IgG-B)

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<400> SEQUENCE: 79

Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys Pro Gly Gly
 1 5 10 15
 Thr Leu Arg Leu Ser Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
 20 25 30
 Gly Val Ser Trp Val Arg Gln Ser Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45
 Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr His Ser Ala Val Lys
 50 55 60
 Gly Gln Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val Phe Leu
 65 70 75 80
 Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Met Tyr Tyr Cys Ala
 85 90 95
 Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe Val Tyr Trp
 100 105 110
 Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Thr Ala Pro
 115 120 125
 Ser Val Phe Pro Leu Ala Pro Ser Cys Gly Ser Thr Ser Gly Ser Thr
 130 135 140
 Val Ala Leu Ala Cys Leu Val Ser Gly Tyr Phe Pro Glu Pro Val Thr
 145 150 155 160
 Val Ser Trp Asn Ser Gly Ser Leu Thr Ser Gly Val His Thr Phe Pro
 165 170 175
 Ser Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Met Val Thr
 180 185 190
 Val Pro Ser Ser Arg Trp Pro Ser Glu Thr Phe Thr Cys Asn Val Ala
 195 200 205
 His Pro Ala Ser Lys Thr Lys Val Asp Lys Pro Val Pro Lys Arg Glu
 210 215 220
 Asn Gly Arg Val Pro Arg Pro Pro Asp Cys Pro Lys Cys Pro Ala Pro
 225 230 235 240
 Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro Lys Pro Lys
 245 250 255
 Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys Val Val Val
 260 265 270
 Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp Phe Val Asp
 275 280 285
 Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu Glu Gln Phe
 290 295 300
 Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly His Gln Asp
 305 310 315 320
 Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn Lys Ala Leu
 325 330 335
 Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly Gln Ala His
 340 345 350
 Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu Leu Ser Lys
 355 360 365
 Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe Pro Pro Asp
 370 375 380
 Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro Glu Ser Lys
 385 390 395 400
 Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser Tyr Phe Leu

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	405		410		415
Tyr Ser Lys	Leu Ser Val Asp Lys Ser Arg Trp Gln Arg Gly Asp Thr				
	420		425		430
Phe Ile Cys	Ala Val Met His Glu Ala Leu His Asn His Tyr Thr Gln				
	435		440		445
Glu Ser Leu Ser His Ser Pro Gly Lys					
	450		455		

<210> SEQ ID NO 80
 <211> LENGTH: 216
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Caninized variable light chain of
 Mab B v3 and canine kappa light constant region, (Caninized B LC
 v3 kappa)

<400> SEQUENCE: 80

Glu Ile Val	Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Leu Gly
1	5 10 15
Gln Arg Ala	Thr Ile Ser Cys Lys Ala Ser His Asn Ile Asn Lys Asn
	20 25 30
Leu Glu Trp	Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
	35 40 45
Tyr Tyr Ala	Asn Asn Leu Gln Thr Gly Val Pro Ala Arg Phe Ser Gly
	50 55 60
Ser Gly Ser	Gly Thr Asp Tyr Ser Leu Asn Ile His Pro Met Glu Glu
65	70 75 80
Asp Asp Thr	Ala Met Tyr Tyr Cys Tyr Gln Tyr Asn Ser Gly His Thr
	85 90 95
Phe Gly Gly	Gly Thr Lys Leu Glu Ile Lys Arg Asn Asp Ala Gln Pro
	100 105 110
Ala Val Tyr	Leu Phe Gln Pro Ser Pro Asp Gln Leu His Thr Gly Ser
	115 120 125
Ala Ser Val	Val Cys Leu Leu Asn Ser Phe Tyr Pro Lys Asp Ile Asn
	130 135 140
Val Lys Trp	Lys Val Asp Gly Val Ile Gln Asp Thr Gly Ile Gln Glu
145	150 155 160
Ser Val Thr	Glu Gln Asp Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr
	165 170 175
Leu Thr Met	Ser Ser Thr Glu Tyr Leu Ser His Glu Leu Tyr Ser Cys
	180 185 190
Glu Ile Thr	His Lys Ser Leu Pro Ser Thr Leu Ile Lys Ser Phe Gln
	195 200 205
Arg Ser Glu	Cys Gln Arg Val Asp
	210 215

<210> SEQ ID NO 81
 <211> LENGTH: 114
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Mab E variable HC
 <400> SEQUENCE: 81

Gly Pro Gly	Leu Val Gln Pro Ser Gln Thr Leu Ser Leu Thr Cys Thr
1	5 10 15
Val Ser Gly	Phe Ser Leu Ser Ser Tyr His Val His Trp Val Arg Gln

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Pro	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Leu	Gly	Val	Met	Trp	Asn	Asp	Gly
		35					40					45			
Asp	Thr	Ser	Tyr	Asn	Leu	Ala	Leu	Asn	Ser	Arg	Leu	Ser	Ile	Ser	Arg
	50				55					60					
Asp	Thr	Ser	Lys	Ser	Gln	Val	Phe	Phe	Lys	Met	Ser	Ser	Leu	Gln	Thr
65					70					75				80	
Glu	Asp	Thr	Ala	Thr	Tyr	Tyr	Cys	Ala	Arg	Pro	Glu	Leu	Pro	Gly	Leu
			85						90					95	
Thr	Tyr	Gly	Val	Trp	Phe	Pro	Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr
		100						105					110		

Val Ser

<210> SEQ ID NO 82
 <211> LENGTH: 101
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Mab E variable LC

<400> SEQUENCE: 82

Ala	Ser	Met	Ser	Ile	Ser	Val	Gly	Asp	Arg	Val	Thr	Met	Asn	Cys	Lys
1			5						10					15	
Ala	Ser	Gln	Asn	Val	Asp	Ser	Asn	Val	Asp	Trp	Tyr	Gln	Gln	Lys	Thr
		20					25					30			
Gly	Gln	Ser	Pro	Asn	Leu	Leu	Ile	Tyr	Lys	Ala	Ser	Asn	Arg	Asn	Thr
	35						40					45			
Gly	Val	Pro	Asp	Arg	Phe	Thr	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr
	50				55						60				
Phe	Thr	Ile	Ser	Asn	Met	Gln	Ala	Glu	Asp	Leu	Ala	Val	Tyr	Tyr	Cys
65				70					75					80	
Met	Gln	Ser	Thr	Ser	Tyr	Pro	Leu	Thr	Phe	Gly	Ser	Gly	Thr	Lys	Leu
			85					90						95	

Glu Ile Lys Arg Ala
100

<210> SEQ ID NO 83
 <211> LENGTH: 114
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Mab F variable HC

<400> SEQUENCE: 83

Gly	Pro	Gly	Leu	Val	Gln	Pro	Ser	Gln	Thr	Leu	Ser	Leu	Thr	Cys	Thr
1			5						10					15	
Val	Ser	Gly	Phe	Ser	Leu	Thr	Ser	Tyr	Gly	Val	Ser	Trp	Val	Arg	Gln
		20						25					30		
Pro	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Ile	Gly	Thr	Met	Trp	Asn	Asp	Gly
		35					40					45			
Asp	Thr	Asp	Tyr	His	Ser	Ala	Leu	Arg	Ser	Arg	Leu	Ser	Ile	Ser	Arg
	50					55					60				
Asp	Ser	Ser	Lys	Ser	Gln	Val	Leu	Leu	Lys	Met	Asn	Ser	Leu	Gln	Thr
65					70					75				80	
Glu	Asp	Thr	Ala	Met	Tyr	Phe	Cys	Ala	Arg	Ser	Gln	Leu	Pro	Gly	Tyr
			85						90					95	

Asn Leu Arg Gly Trp Phe Val Tyr Trp Gly Gln Gly Thr Leu Val Ile

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100	105	110
Val Ser		
<210> SEQ ID NO 84 <211> LENGTH: 100 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic: Mab F variable LC <400> SEQUENCE: 84		
Ser Leu Leu Ser Ala Ser Val Gly Asp Arg Val Thr Leu Asn Cys Lys		
1 5 10 15		
Ala Ser His Asn Ile Asn Lys Asn Leu Glu Trp Tyr Gln Gln Lys Leu		
20 25 30		
Gly Glu Ala Pro Lys Leu Leu Ile Tyr Tyr Ala Asn Asn Leu Gln Thr		
35 40 45		
Gly Ile Ser Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Tyr Thr		
50 55 60		
Leu Thr Ile Ser Ser Leu Gln Pro Glu Asp Val Ala Thr Tyr Tyr Cys		
65 70 75 80		
Tyr Gln Tyr Asn Ser Gly His Thr Phe Gly Ala Gly Thr Lys Leu Glu		
85 90 95		
Leu Lys Arg Ala		
100		
<210> SEQ ID NO 85 <211> LENGTH: 122 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic: Caninized variable heavy chain of Mab A v3 <400> SEQUENCE: 85		
Gln Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys Pro Gly Gly		
1 5 10 15		
Thr Leu Arg Leu Ser Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr		
20 25 30		
His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Val		
35 40 45		
Ala Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu Ala Val Lys		
50 55 60		
Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr Leu		
65 70 75 80		
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala		
85 90 95		
Arg Pro Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe Pro Tyr Trp		
100 105 110		
Gly Gln Gly Thr Leu Val Thr Val Ser Ser		
115 120		
<210> SEQ ID NO 86 <211> LENGTH: 122 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: Synthetic: Caninized variable heavy chain of Mab A v4		

-continued

<400> SEQUENCE: 86

Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys Pro Gly Gly
 1 5 10 15
 Thr Leu Arg Leu Ser Cys Thr Val Ser Gly Phe Ser Leu Ser Ser Tyr
 20 25 30
 His Val His Trp Val Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu
 35 40 45
 Gly Val Met Trp Asn Asp Gly Asp Thr Ser Tyr Asn Leu Ala Val Lys
 50 55 60
 Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr Phe
 65 70 75 80
 Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Thr Tyr Tyr Cys Ala
 85 90 95
 Arg Pro Glu Leu Pro Gly Leu Thr Tyr Gly Val Trp Phe Pro Tyr Trp
 100 105 110
 Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 115 120

<210> SEQ ID NO 87

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Caninized variable light chain of Mab A

<400> SEQUENCE: 87

Glu Ile Val Met Thr Gln Ser Pro Ala Ser Leu Ser Leu Ser Gln Glu
 1 5 10 15
 Glu Lys Val Thr Ile Thr Cys Lys Ala Ser Gln Asn Val Asp Ser Asn
 20 25 30
 Val Asp Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Lys Leu Leu Ile
 35 40 45
 Tyr Lys Ala Ser Asn Arg Asn Thr Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Ser Phe Thr Ile Ser Ser Leu Glu Pro
 65 70 75 80
 Glu Asp Val Ala Val Tyr Tyr Cys Met Gln Ser Thr Ser Tyr Pro Leu
 85 90 95
 Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
 100 105

<210> SEQ ID NO 88

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Caninized variable heavy chain of Mab B v3

<400> SEQUENCE: 88

Glu Val Gln Leu Val Glu Ser Gly Gly Asp Leu Val Lys Pro Gly Gly
 1 5 10 15
 Thr Leu Arg Leu Ser Cys Thr Val Ser Gly Phe Ser Leu Thr Ser Tyr
 20 25 30
 Gly Val Ser Trp Val Arg Gln Ser Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45
 Gly Thr Met Trp Asn Asp Gly Asp Thr Asp Tyr His Ser Ala Val Lys

-continued

50	55	60
Gly Gln Leu Ser Ile Ser Arg Asp Thr Ser Lys Ser Gln Val Phe Leu		
65	70	75 80
Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Met Tyr Tyr Cys Ala		
	85	90 95
Arg Ser Gln Leu Pro Gly Tyr Asn Leu Arg Gly Trp Phe Val Tyr Trp		
	100	105 110
Gly Gln Gly Thr Leu Val Thr Val Ser Ser		
	115	120

<210> SEQ ID NO 89
 <211> LENGTH: 106
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Caninized variable light chain of
 Mab B v3

<400> SEQUENCE: 89

Glu Ile Val Leu Thr Gln Ser Pro Ala Ser Leu Ala Val Ser Leu Gly
1 5 10 15
Gln Arg Ala Thr Ile Ser Cys Lys Ala Ser His Asn Ile Asn Lys Asn
20 25 30
Leu Glu Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile
35 40 45
Tyr Tyr Ala Asn Asn Leu Gln Thr Gly Val Pro Ala Arg Phe Ser Gly
50 55 60
Ser Gly Ser Gly Thr Asp Tyr Ser Leu Asn Ile His Pro Met Glu Glu
65 70 75 80
Asp Asp Thr Ala Met Tyr Tyr Cys Tyr Gln Tyr Asn Ser Gly His Thr
85 90 95
Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys
100 105

<210> SEQ ID NO 90
 <211> LENGTH: 216
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Exemplary canine CD16 with linker,
 and poly-His

<400> SEQUENCE: 90

Met Trp Gln Leu Val Ser Ser Thr Ala Leu Leu Leu Val Ser Ala
1 5 10 15
Gly Thr Gln Ala Ala Asp Val Pro Lys Ala Val Val Val Leu Glu Pro
20 25 30
Lys Trp Asn Arg Val Leu Thr Met Asp Ser Val Thr Leu Lys Cys Gln
35 40 45
Gly Asp His Leu Leu Arg Asp Asn Tyr Thr Trp Leu His Asn Gly Arg
50 55 60
Pro Ile Ser Asn Gln Ile Ser Thr Tyr Ile Ile Lys Asn Ala Ser Ile
65 70 75 80
Lys Asn Ser Gly Glu Tyr Arg Cys Gln Thr Asp Gln Ser Lys Leu Ser
85 90 95
Asp Pro Val Gln Leu Glu Val His Thr Gly Trp Leu Leu Gln Val
100 105 110
Pro Arg Leu Val Phe Gln Glu Gly Glu Leu Ile Gln Leu Lys Cys His

-continued

115	120	125
Ser Trp Lys Asn Thr Pro Val Arg Asn Val Gln Tyr Phe Gln Asn Gly		
130	135	140
Arg Gly Lys Lys Phe Phe Tyr Asn Asn Ser Glu Tyr His Ile Pro Ala		
145	150	155
Ala Thr Ser Glu His Asn Gly Ser Tyr Phe Cys Arg Gly Ile Ile Gly		
	165	170
		175
Lys Lys Asn Glu Ser Ser Glu Ala Val Asn Ile Ile Ile Gln Gly Ser		
	180	185
		190
Ser Leu Pro Ser Thr Ser Leu Leu Leu Ser His Trp Pro Gln Gly Ser		
	195	200
		205
Gly Ser His His His His His His		
	210	215

<210> SEQ ID NO 91
 <211> LENGTH: 220
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Exemplary wild-type canine IgG-B Fc

<400> SEQUENCE: 91

Pro Ala Pro Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro		
1	5	10
		15
Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys		
	20	25
		30
Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp		
	35	40
		45
Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu		
	50	55
		60
Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly		
	65	70
		75
		80
His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn		
	85	90
		95
Lys Ala Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly		
	100	105
		110
Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu		
	115	120
		125
Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe		
	130	135
		140
Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro		
	145	150
		155
		160
Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser		
	165	170
		175
Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg		
	180	185
		190
Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His		
	195	200
		205
Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys		
	210	215
		220

<210> SEQ ID NO 92
 <211> LENGTH: 220
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Exemplary variant canine IgG-B Fc,

-continued

CD16 enhancing binding mutant, S10D

<400> SEQUENCE: 92

```

Pro Ala Pro Glu Met Leu Gly Gly Pro Asp Val Phe Ile Phe Pro Pro
1           5           10           15

Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
20           25           30

Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
35           40           45

Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
50           55           60

Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
65           70           75           80

His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
85           90           95

Lys Ala Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly
100          105          110

Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu
115          120          125

Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe
130          135          140

Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro
145          150          155          160

Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser
165          170          175

Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
180          185          190

Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
195          200          205

Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
210          215          220

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<210> SEQ ID NO 93

<211> LENGTH: 220

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Exemplary variant canine IgG-B Fc,
CD16 enhancing binding mutant, S10E

<400> SEQUENCE: 93

```

Pro Ala Pro Glu Met Leu Gly Gly Pro Glu Val Phe Ile Phe Pro Pro
1           5           10           15

Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
20           25           30

Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
35           40           45

Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
50           55           60

Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
65           70           75           80

His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
85           90           95

Lys Ala Leu Pro Ser Pro Ile Glu Arg Thr Ile Ser Lys Ala Arg Gly
100          105          110

Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu

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115					120					125					
Leu	Ser	Lys	Asn	Thr	Val	Ser	Leu	Thr	Cys	Leu	Ile	Lys	Asp	Phe	Phe
130						135					140				
Pro	Pro	Asp	Ile	Asp	Val	Glu	Trp	Gln	Ser	Asn	Gly	Gln	Gln	Glu	Pro
145					150					155					160
Glu	Ser	Lys	Tyr	Arg	Thr	Thr	Pro	Pro	Gln	Leu	Asp	Glu	Asp	Gly	Ser
				165					170					175	
Tyr	Phe	Leu	Tyr	Ser	Lys	Leu	Ser	Val	Asp	Lys	Ser	Arg	Trp	Gln	Arg
		180						185					190		
Gly	Asp	Thr	Phe	Ile	Cys	Ala	Val	Met	His	Glu	Ala	Leu	His	Asn	His
		195					200					205			
Tyr	Thr	Gln	Glu	Ser	Leu	Ser	His	Ser	Pro	Gly	Lys				
	210					215					220				

<210> SEQ ID NO 94

<211> LENGTH: 220

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Exemplary variant canine IgG-B Fc, CD16 enhancing binding mutant, I103D

<400> SEQUENCE: 94

Pro	Ala	Pro	Glu	Met	Leu	Gly	Gly	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro
1				5				10					15		
Lys	Pro	Lys	Asp	Thr	Leu	Leu	Ile	Ala	Arg	Thr	Pro	Glu	Val	Thr	Cys
			20					25					30		
Val	Val	Val	Asp	Leu	Asp	Pro	Glu	Asp	Pro	Glu	Val	Gln	Ile	Ser	Trp
		35				40					45				
Phe	Val	Asp	Gly	Lys	Gln	Met	Gln	Thr	Ala	Lys	Thr	Gln	Pro	Arg	Glu
	50					55					60				
Glu	Gln	Phe	Asn	Gly	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Pro	Ile	Gly
65					70					75					80
His	Gln	Asp	Trp	Leu	Lys	Gly	Lys	Gln	Phe	Thr	Cys	Lys	Val	Asn	Asn
			85						90					95	
Lys	Ala	Leu	Pro	Ser	Pro	Asp	Glu	Arg	Thr	Ile	Ser	Lys	Ala	Arg	Gly
		100						105					110		
Gln	Ala	His	Gln	Pro	Ser	Val	Tyr	Val	Leu	Pro	Pro	Ser	Arg	Glu	Glu
		115					120					125			
Leu	Ser	Lys	Asn	Thr	Val	Ser	Leu	Thr	Cys	Leu	Ile	Lys	Asp	Phe	Phe
	130						135				140				
Pro	Pro	Asp	Ile	Asp	Val	Glu	Trp	Gln	Ser	Asn	Gly	Gln	Gln	Glu	Pro
145					150					155					160
Glu	Ser	Lys	Tyr	Arg	Thr	Thr	Pro	Pro	Gln	Leu	Asp	Glu	Asp	Gly	Ser
				165					170					175	
Tyr	Phe	Leu	Tyr	Ser	Lys	Leu	Ser	Val	Asp	Lys	Ser	Arg	Trp	Gln	Arg
		180						185					190		
Gly	Asp	Thr	Phe	Ile	Cys	Ala	Val	Met	His	Glu	Ala	Leu	His	Asn	His
		195					200					205			
Tyr	Thr	Gln	Glu	Ser	Leu	Ser	His	Ser	Pro	Gly	Lys				
	210					215					220				

<210> SEQ ID NO 95

<211> LENGTH: 220

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

-continued

<223> OTHER INFORMATION: Synthetic: Exemplary variant canine IgG-B Fc, CD16 enhancing binding mutant, I103E

<400> SEQUENCE: 95

```

Pro Ala Pro Glu Met Leu Gly Gly Pro Ser Val Phe Ile Phe Pro Pro
1           5           10           15
Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
20           25           30
Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
35           40           45
Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
50           55           60
Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
65           70           75           80
His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
85           90           95
Lys Ala Leu Pro Ser Pro Glu Glu Arg Thr Ile Ser Lys Ala Arg Gly
100          105          110
Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu
115          120          125
Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe
130          135          140
Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro
145          150          155          160
Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser
165          170          175
Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
180          185          190
Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
195          200          205
Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
210          215          220

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<210> SEQ ID NO 96

<211> LENGTH: 220

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Exemplary variant canine IgG-B Fc, CD16 enhancing binding mutant, S10D, I103D

<400> SEQUENCE: 96

```

Pro Ala Pro Glu Met Leu Gly Gly Pro Asp Val Phe Ile Phe Pro Pro
1           5           10           15
Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
20           25           30
Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
35           40           45
Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
50           55           60
Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
65           70           75           80
His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
85           90           95
Lys Ala Leu Pro Ser Pro Asp Glu Arg Thr Ile Ser Lys Ala Arg Gly
100          105          110

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-continued

Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu
 115 120 125

Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe
 130 135 140

Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro
 145 150 155 160

Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser
 165 170 175

Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
 180 185 190

Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
 195 200 205

Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
 210 215 220

<210> SEQ ID NO 97
 <211> LENGTH: 220
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic: Exemplary variant canine IgG-B Fc,
 CD16 enhancing binding mutant, S10D, I103E

<400> SEQUENCE: 97

Pro Ala Pro Glu Met Leu Gly Gly Pro Asp Val Phe Ile Phe Pro Pro
 1 5 10 15

Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
 20 25 30

Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
 35 40 45

Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
 50 55 60

Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
 65 70 75 80

His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
 85 90 95

Lys Ala Leu Pro Ser Pro Glu Glu Arg Thr Ile Ser Lys Ala Arg Gly
 100 105 110

Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu
 115 120 125

Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe
 130 135 140

Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro
 145 150 155 160

Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser
 165 170 175

Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
 180 185 190

Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
 195 200 205

Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
 210 215 220

<210> SEQ ID NO 98
 <211> LENGTH: 220
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Exemplary variant canine IgG-B Fc, CD16 enhancing binding mutant, S10E, I103D

<400> SEQUENCE: 98

```

Pro Ala Pro Glu Met Leu Gly Gly Pro Glu Val Phe Ile Phe Pro Pro
1           5           10           15
Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
20           25           30
Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
35           40           45
Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
50           55           60
Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
65           70           75           80
His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
85           90           95
Lys Ala Leu Pro Ser Pro Asp Glu Arg Thr Ile Ser Lys Ala Arg Gly
100          105          110
Gln Ala His Gln Pro Ser Val Tyr Val Leu Pro Pro Ser Arg Glu Glu
115          120          125
Leu Ser Lys Asn Thr Val Ser Leu Thr Cys Leu Ile Lys Asp Phe Phe
130          135          140
Pro Pro Asp Ile Asp Val Glu Trp Gln Ser Asn Gly Gln Gln Glu Pro
145          150          155          160
Glu Ser Lys Tyr Arg Thr Thr Pro Pro Gln Leu Asp Glu Asp Gly Ser
165          170          175
Tyr Phe Leu Tyr Ser Lys Leu Ser Val Asp Lys Ser Arg Trp Gln Arg
180          185          190
Gly Asp Thr Phe Ile Cys Ala Val Met His Glu Ala Leu His Asn His
195          200          205
Tyr Thr Gln Glu Ser Leu Ser His Ser Pro Gly Lys
210          215          220

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<210> SEQ ID NO 99

<211> LENGTH: 220

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic: Exemplary variant canine IgG-B Fc, CD16 enhancing binding mutant, S10E, I103E

<400> SEQUENCE: 99

```

Pro Ala Pro Glu Met Leu Gly Gly Pro Glu Val Phe Ile Phe Pro Pro
1           5           10           15
Lys Pro Lys Asp Thr Leu Leu Ile Ala Arg Thr Pro Glu Val Thr Cys
20           25           30
Val Val Val Asp Leu Asp Pro Glu Asp Pro Glu Val Gln Ile Ser Trp
35           40           45
Phe Val Asp Gly Lys Gln Met Gln Thr Ala Lys Thr Gln Pro Arg Glu
50           55           60
Glu Gln Phe Asn Gly Thr Tyr Arg Val Val Ser Val Leu Pro Ile Gly
65           70           75           80
His Gln Asp Trp Leu Lys Gly Lys Gln Phe Thr Cys Lys Val Asn Asn
85           90           95
Lys Ala Leu Pro Ser Pro Glu Glu Arg Thr Ile Ser Lys Ala Arg Gly
100          105          110

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Gln	Ala	His	Gln	Pro	Ser	Val	Tyr	Val	Leu	Pro	Pro	Ser	Arg	Glu	Glu
	115						120					125			
Leu	Ser	Lys	Asn	Thr	Val	Ser	Leu	Thr	Cys	Leu	Ile	Lys	Asp	Phe	Phe
	130					135				140					
Pro	Pro	Asp	Ile	Asp	Val	Glu	Trp	Gln	Ser	Asn	Gly	Gln	Gln	Glu	Pro
	145				150				155					160	
Glu	Ser	Lys	Tyr	Arg	Thr	Thr	Pro	Pro	Gln	Leu	Asp	Glu	Asp	Gly	Ser
			165						170					175	
Tyr	Phe	Leu	Tyr	Ser	Lys	Leu	Ser	Val	Asp	Lys	Ser	Arg	Trp	Gln	Arg
		180					185					190			
Gly	Asp	Thr	Phe	Ile	Cys	Ala	Val	Met	His	Glu	Ala	Leu	His	Asn	His
	195					200						205			
Tyr	Thr	Gln	Glu	Ser	Leu	Ser	His	Ser	Pro	Gly	Lys				
	210				215					220					

The invention claimed is:

1. An antibody that binds to canine parvovirus and/or feline parvovirus, wherein the antibody comprises:

- (a) (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 4,
- (ii) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 5,
- (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 6, and
- (b) (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 13,
- (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 14,
- (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 15, and

wherein the heavy chain sequence is at least 95% identical to SEQ ID NO: 24 and the light chain sequence is at least 95% identical to SEQ ID NO: 25.

2. The antibody of claim 1, wherein the heavy chain sequence is at least 98% identical to SEQ ID NO: 24 and the light chain sequence is at least 98% identical to SEQ ID NO: 25.

3. The antibody of claim 1, wherein the heavy chain sequence is at least 99% identical to SEQ ID NO: 24 and the light chain sequence is at least 99% identical to SEQ ID NO: 25.

4. The antibody of claim 1, wherein said antibody is isolated.

5. The antibody of claim 1, wherein said antibody is a chimeric antibody.

6. The antibody of claim 1, wherein the antibody binds to an epitope comprising the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, and/or SEQ ID NO: 3.

7. The antibody of claim 1, wherein the antibody binds to canine parvovirus or feline parvovirus as determined by immunoblot analysis and/or biolayer interferometry.

8. The antibody of claim 1, wherein the antibody at a concentration of 200 µg/mL has an hemagglutination inhibition value of at least 8000, of at least 16000, or of at least 32000.

9. The antibody of claim 1, wherein the antibody is a monoclonal antibody.

10. An isolated nucleic acid encoding the antibody of claim 1.

11. A host cell comprising the nucleic acid of claim 10.

12. A method of producing an antibody comprising culturing the host cell of claim 11 and isolating the antibody.

13. A pharmaceutical composition comprising the antibody of claim 1 and a pharmaceutically acceptable carrier.

14. The pharmaceutical composition of claim 13, wherein the pharmaceutically acceptable carrier is phosphate buffered saline.

15. A method of providing passive immunity in a canine or a feline subject against infection with a canine or feline parvovirus comprising administering to the subject a therapeutically effective amount of the antibody of claim 1 that binds to the canine or feline parvovirus.

16. The method of claim 15, wherein the antibody is administered before or after exposure to the canine or feline parvovirus or after infection with the canine or feline parvovirus.

17. The method of claim 16, wherein the antibody is administered after the subject has exhibited at least one symptoms selected from the group consisting of fever, vomiting, diarrhea, lymphopenia, and septicemia.

18. An antibody that binds to canine parvovirus and/or feline parvovirus, wherein the antibody comprises:

- (a) (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 4,
- (ii) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 5,
- (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 6, and
- (b) (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 13,
- (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 14,
- (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 15, and

wherein the heavy chain sequence comprises SEQ ID NO: 24 with optionally 1, 2, 3, 4, or 5 amino acid substitutions and the light chain sequence comprises SEQ ID NO: 25 with optionally 1, 2, 3, 4, or 5 amino acid substitutions.

19. The antibody of claim 18, wherein said amino acid substitutions are conservative substitutions.

20. A method of treating a canine or feline parvoviral infection in a canine or a feline subject comprising administering to the subject a therapeutically effective amount of the antibody of claim 1 that binds to the canine or feline parvovirus.

187

21. A method for detecting a parvoviral infection in a sample from a canine or a feline subject comprising contacting the sample with the antibody of claim 1 under conditions permissive for the binding of the antibody to a parvovirus, and detecting whether a complex is formed between the antibody and the parvovirus in the sample.

22. A variant IgG Fc polypeptide comprising:

- a) an aspartic acid or a glutamic acid at a position corresponding to position 10 of SEQ ID NO: 91;
- b) an aspartic acid or a glutamic acid at position 10 of SEQ ID NO: 91;
- c) an aspartic acid or a glutamic acid at a position corresponding to position 103 of SEQ ID NO: 91;
- d) an aspartic acid or a glutamic acid at position 103 of SEQ ID NO: 91;
- e) an aspartic acid or a glutamic acid at a position corresponding to position 10 and/or position 103 of SEQ ID NO: 91; or
- f) an aspartic acid or a glutamic acid at position 10 and/or position 103 of SEQ ID NO: 91.

23. The antibody of claim 1, wherein the heavy chain comprises a variable heavy chain framework region (HC-FR1) comprising the amino acid sequence of SEQ ID NO: 8.

24. The antibody of claim 2, wherein the heavy chain comprises a HC-FR1 comprising the amino acid sequence of SEQ ID NO: 8.

188

25. The antibody of claim 19, wherein the heavy chain comprises a HC-FR1 comprising the amino acid sequence of SEQ ID NO: 8.

26. The antibody of claim 18, wherein the antibody comprises:

- (a) (i) a CDR-H1 comprising the amino acid sequence of SEQ ID NO: 4,
- (ii) a CDR-H2 comprising the amino acid sequence of SEQ ID NO: 5,
- (iii) a CDR-H3 comprising the amino acid sequence of SEQ ID NO: 6, and
- (b) (i) a CDR-L1 comprising the amino acid sequence of SEQ ID NO: 13,
- (ii) a CDR-L2 comprising the amino acid sequence of SEQ ID NO: 14,
- (iii) a CDR-L3 comprising the amino acid sequence of SEQ ID NO: 15, and

wherein the heavy chain sequence comprises SEQ ID NO: 24 and the light chain sequence comprises SEQ ID NO: 25.

27. The antibody of claim 1, wherein the antibody binds to canine parvovirus.

28. The antibody of claim 19, wherein the antibody binds to canine parvovirus.

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